

UEFI boot: how does that actually work, then?

UEFI 启动：它究竟是如何工作的？

Adam Williamson — [2014-01-25 21:07](#)

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It's AdamW Essay Time again! If you're looking for something short and snappy, look elsewhere.

又到了 AdamW 的文章时间！如果你想找的是短小精悍的文章，那就别看这个了。

[Kamil Paral](#) kindly informs me I'm a chronic sufferer of [Graphomania](#). Always nice to know what's wrong with you.

[Kamil Paral](#) 好心地告诉我，我患有慢性[书写癖](#)。很高兴知道你有什么毛病。

IMPORTANT NOTE TO INDUSTRY FOLKS: This blog post is aimed at regular everyday folks; it's intended to dispel a few common myths and help regular people understand UEFI a bit better. It is not a low-level fully detailed and 100% technically accurate explanation, and I'm not a professional firmware engineer or anything like that. If you're actually building an operating system or hardware or something, please don't rely on my simplified explanations or ask me for help; I'm just an idiot on the internet. If you're doing that kind of thing and you have money, join the UEFI Forum or ask your suppliers or check your reference implementation or whatever. If you don't have money, try asking your peers with more experience, nicely. **END**

IMPORTANT NOTE

致业内人士的重要提示：这篇博文面向普通大众，旨在消除一些常见的误解，帮助大家更好地理解 UEFI。它并非详尽的底层技术解释，也并非 100% 精确的技术说明，我也不是专业的固件工程师。如果您正在开发操作系统、硬件或其他相关产品，请不要依赖我简化的解释或向我寻求帮助；我只是个网络小白。如果您有这方面的预算，可以加入 UEFI 论坛、咨询供应商、查阅参考实现等等。如果您预算有限，可以礼貌地向更有经验的同行请教。**重要提示结束**

You've probably read a lot of stuff on the internet about UEFI. Here is something important you should understand: 95% of it was probably garbage. If you think you know about UEFI, and you derived your knowledge anywhere other than the UEFI specifications, [mjb59's blog](#) or one of a few other vaguely reliable locations/people - [Rod Smith](#), or [Peter Jones](#), or Chris Murphy, or the documentation of the relatively few OSes whose developers actually know what the hell they're doing with UEFI - what you think you know is likely a toxic mix of misunderstandings, misconceptions, half-truths, propaganda and downright lies. So you should probably forget it all.

你可能在网上看过很多关于 UEFI 的资料。但有一点你必须明白：其中 95% 的内容可能都是垃圾。如果你自认为了解 UEFI，而你的知识来源并非 UEFI 规范、[mjb59 的博客](#)，或是其他少数几个勉强可靠的来源/人士——比如 [Rod Smith](#)、[Peter Jones](#)、Chris Murphy，又或是少数几个真正懂 UEFI 的操作系统的官方文档——那么你所谓的了解很可能只是一堆误解、错误观念、半真半假的信息、宣传和彻头彻尾的谎言的混合体。所以，你最好把这些都忘掉。

Good, now we've got that out of the way. What I mostly want to talk about is bootloading, because that's the bit of firmware that matters most to most people, and the bit news sites are always banging on about and wildly misunderstanding.

好了，现在我们已经解决了这个问题。我主要想谈的是引导加载，因为这部分固件对大多数人来说最为重要，也是新闻网站总是大肆渲染却又严重误解的部分。

Terminology 术语

First, let's get some terminology out of the way. Both [BIOS](#) and [UEFI](#) are types of [firmware](#) for computers. BIOS-style firmware is (mostly) only ever found on [IBM PC compatible computers](#). UEFI is meant to be more generic, and can be found on systems which are not in the 'IBM PC compatible' class.

首先，[我们](#)先来明确一些术语。BIOS 和 [UEFI](#) 都是计算机[固件](#)的类型。BIOS 类型的固件（通常）只存在于 [IBM PC 兼容计算机](#)上。UEFI 则更加通用，可以用于非 IBM PC 兼容的系统中。

You do not have a 'UEFI BIOS'. No-one has a 'UEFI BIOS'. Please don't ever say 'UEFI BIOS'. BIOS is not a generic term for all PC firmware, it is a particular type of PC firmware. Your computer has a firmware. If it's an IBM PC compatible computer, it's almost certainly either a BIOS or a UEFI firmware. If you're running [Coreboot](#), congratulations, Mr./Ms. Exception. You may be proud of yourself.

你没有“UEFI BIOS”。没有人会用“UEFI BIOS”。请永远不要再说“UEFI BIOS”这个词。BIOS 不是所有电脑固件的通用术语，它指的是一种特定的电脑固件。你的电脑有固件。如果是 IBM PC 兼容机，那么它几乎肯定是 BIOS 或 UEFI 固件。如果你正在运行 [Coreboot](#)，恭喜你，特例先生/女士。你可以为此感到自豪。

[Secure Boot](#) is not the same thing as UEFI. Do not ever use those terms interchangeably. Secure Boot is a single effectively optional element of the UEFI specification, which was added in version 2.2 of the UEFI specification. We will talk about precisely what it is later, but for now, just remember it is not the same thing about UEFI. You need to understand what Secure Boot is, and what UEFI is, and which of the two you are actually talking about at any given time. We'll talk about UEFI first, and then we'll talk about Secure Boot as an 'extension' to UEFI, because that's basically what it is.

[安全启动 \(Secure Boot\)](#) 和 UEFI 并非同一概念。切勿将这两个术语混用。安全启动是 UEFI 规范中一个可选的独立组件，它是在 UEFI 规范 2.2 版本中添加的。我们稍后会详细讨论它的具体内容，但现在，您只需记住它与 UEFI 不同即可。您需要理解安全启动和 UEFI 的定义，并明确您在讨论时实际指的是哪一方面。我们将首先讨论 UEFI，然后再讨论作为 UEFI“扩展”的安全启动，因为它本质上就是 UEFI 的扩展。

Bonus Historical Note: UEFI was not invented by, is not controlled by, and has never been controlled by Microsoft. Its predecessor and basis, EFI, was developed and published by Intel. UEFI is managed by the [UEFI Forum](#). Microsoft is a member of the UEFI forum. So is Red Hat, and so is Apple, and so is just about every major PC manufacturer, Intel (obviously), AMD, and a [laundry list of other major and minor hardware, software and firmware companies and organizations](#). It is a broad consensus specification, with all the messiness that entails, some of which we'll talk about specifically later. It is no one company's Evil Vehicle Of Evilness.

补充历史背景：UEFI 并非由微软发明、控制，也从未被微软控制过。它的前身和基础 EFI 是由英特尔开发和发布的。UEFI 由 [UEFI 论坛](#) 管理。微软是 UEFI 论坛的成员，红帽、苹果以及几乎所有主流 PC 制造商、英特尔（当然）、AMD，以及[其他众多大大小小的硬件、软件和固件公司及组织也都是其成员](#)。它是一个广泛共识的规范，因此也难免存在一些复杂性，我们稍后会详细讨论。它并非任何一家公司的“邪恶工具”。

References 参考

If you really want to understand UEFI, it's a really good idea to go and read the UEFI specification. You can do this. It's very easy. You don't have to pay anyone any money. I am not going to tell you that reading it will be the most fun you've ever had, because it won't. But it won't be a waste of your time. You can find it [right here on the official UEFI site](#). You have to check a couple of boxes, but you are not signing your soul away to Satan, or anything. It's fine. As I write this, the current version of the spec is [2.4 Errata A](#), and that's the version this post is written with regard to.

如果你真的想了解 UEFI，最好去阅读一下 UEFI 规范。你可以做到，这很简单，而且完全免费。我不会说阅读规范会是你人生中最有趣的经历，因为它不会。但这绝对不会浪费你的时间。你可以在[在 UEFI 官方网站上](#)找到它。你需要勾选几个选项，但你不需要为此付出任何代价。没关系。在我写这篇文章的时候，规范的最新版本是 [2.4 Errata A](#)，本文也是基于这个版本撰写的。

There is no BIOS specification. BIOS is a *de facto* standard - it works the way it worked on actual IBM PCs, in the 1980s. That's kind of one of the reasons UEFI exists.

没有 BIOS 规范。BIOS 是事实上的标准——它的工作方式与 20 世纪 80 年代 IBM PC 的实际工作方式相同。这在某种程度上也是 UEFI 存在的原因之一。

Now, to keep things simple, let's consider two worlds. One is the world of IBM PC compatible computers - hereafter referred to just as PCs - before UEFI and GPT (we'll come to GPT) existed. This is the world a lot of you are probably familiar with and may understand quite well. Let's talk about how booting works on PCs with BIOS firmware.

为了简化说明，我们不妨考虑两个不同的世界。一个是 IBM PC 兼容机的世界——以下简称 PC——在 UEFI 和 GPT（我们稍后会讲到 GPT）出现之前的世界。很多人可能对这个世界比较熟悉，也比较了解。接下来，我们来谈谈使用 BIOS 固件的 PC 是如何启动的。

BIOS booting BIOS 启动

It works, in fact, in a very, very simple way. On your bog-standard old-skool BIOS PC, you have one or more disks which have an [MBR](#). The MBR is another de facto standard; basically, the very start of the disk describes the partitions on the disk in a particular format, and contains a 'boot loader', a very small piece of code that a BIOS firmware knows how to execute, whose job it is to boot the operating system(s). (Modern bootloaders frequently are much bigger than can be contained in the MBR space and have to use a multi-stage design where the bit in the MBR just knows how to load the next stage from somewhere else, but that's not important to us right now).

实际上，它的工作原理非常简单。在你那台普通的旧式 BIOS 电脑上，会有一个或多个带有 [MBR](#)（主引导记录）的磁盘。MBR 是另一个事实上的标准；基本上，磁盘的起始部分以特定格式描述了磁盘上的分区，并包含一个“引导加载程序”，这是一小段 BIOS 固件能够执行的代码，其作用是启动操作系统。（现代的引导加载程序通常比 MBR 空间所能容纳的要大得多，因此必须采用多阶段设计，MBR 中的相应位只需知道如何从其他地方加载下一阶段的代码即可，但这与我们目前的实际情况无关。）

All a BIOS firmware knows, in the context of booting the system, is what disks the system contains. You, the owner of this BIOS-based computer, can tell the BIOS firmware which disk you want it to boot the system from. The firmware has no knowledge of anything beyond that. It executes the bootloader it finds in the MBR of the specified disk, and that's it. The firmware is no longer involved in booting.

在系统启动过程中，BIOS 固件只知道系统包含哪些磁盘。作为这台基于 BIOS 的计算机的所有者，您可以告诉 BIOS 固件您希望从哪个磁盘启动系统。除此之外，固件对其他任何信息都一无所知。它会执行在指定磁盘的 MBR（主引导记录）中找到的引导加载程序，仅此而已。固件不再参与系统启动过程。

In the BIOS world, absolutely all forms of multi-booting are handled above the firmware layer. The firmware layer doesn't really know what a bootloader is, or what an operating system is. Hell, it doesn't know what a partition is. All it can do is run the boot loader from a disk's MBR. You also cannot configure the boot process from outside of the firmware.

在 BIOS 世界里，所有形式的多系统启动都是在固件层之上处理的。固件层实际上并不了解什么是引导加载程序，什么是操作系统。它甚至不知道什么是分区。它所能做的只是从磁盘的主引导记录（MBR）运行引导加载程序。你也无法在固件之外配置启动过程。

UEFI booting: background

UEFI 启动：后台

OK, so we have our background, the BIOS world. Now let's look at how booting works on a UEFI system. Even if you don't grasp the details of this post, grasp this: *it is completely different*. Completely and utterly different from how BIOS booting works. You cannot apply any of your understanding of BIOS booting to native UEFI booting. You cannot make a little tweak to a system designed for the world of BIOS booting and apply it to *native* UEFI booting. You need to understand that it is a completely different world.

好了，我们已经了解了 BIOS 的背景知识。现在让我们来看看 UEFI 系统的启动机制。即使你没能完全理解本文的细节，也请记住一点：它与 BIOS 启动完全不同。你无法将任何关于 BIOS 启动的理解直接套用到原生 UEFI 启动上。你不能对一个为 BIOS 启动设计的系统进行一些简单的调整，然后将其应用到原生 UEFI 启动中。你需要明白，这是一个完全不同的世界。

Here's another important thing to understand: many UEFI firmwares implement some kind of *BIOS compatibility mode*, sometimes referred to as a *CSM*. Many UEFI firmwares can boot a system just like a BIOS firmware would - they can look for an MBR on a disk, and execute the boot loader from that MBR, and leave everything subsequently up to that bootloader. People sometimes *incorrectly* refer to using this feature as 'disabling UEFI', which is *linguistically* nonsensical. You cannot 'disable' your system's firmware. It's just a stupid term. Don't use it, but understand what people really mean when they say it. They are talking about using a UEFI firmware's ability to boot the system 'BIOS-style' rather than native UEFI style.

还有一点需要理解：许多 UEFI 固件都实现了某种 BIOS 兼容模式，有时也称为 CSM。许多 UEFI 固件可以像 BIOS 固件一样启动系统——它们可以在磁盘上查找 MBR，并从该 MBR 执行引导加载程序，之后的所有操作都由该引导加载程序完成。人们有时会错误地将使用此功能称为“禁用 UEFI”，这在语言上是不通顺的。你无法“禁用”系统的固件。这只是一个愚蠢的说法。不要使用它，但要理解人们说这个词时的真正含义。他们指的是使用 UEFI 固件以“BIOS 方式”而非原生 UEFI 方式启动系统。

What I'm going to describe is *native* UEFI booting. If you have a UEFI-based system whose firmware has the BIOS compatibility feature, and you decide to use it, and you apply this decision consistently, then as far as booting is concerned, you can pretend your system is BIOS-based, and just do everything the way you did with BIOS-style booting. If you're going to do this, though, just make sure you *do* apply it consistently. I really can't recommend strongly enough that you do *not* attempt to mix UEFI-native and BIOS-compatible booting of permanently-installed operating systems on the same computer, and *especially* not on the same disk. It is a terrible terrible idea and will cause you heartache and pain. If you decide to do it, don't come crying to me.

我要描述的是原生 UEFI 启动。如果你的系统基于 UEFI，并且固件支持 BIOS 兼容功能，你决定使用 BIOS 启动，并且始终如一地执行此操作，那么在启动方面，你可以假装你的系统是基于 BIOS 的，并像使用 BIOS 启动一样进行所有操作。但是，如果你要这样做，请务必确保始终如一地执行此操作。我强烈建议你不要在同一台电脑上，尤其是在同一块硬盘上，尝试混合使用 UEFI 原生启动和 BIOS 兼容启动来启动永久安装的操作系统。这是一个极其糟糕的主意，会让你痛苦不堪。如果你执意要这样做，到时候可别来找我哭诉。

For the sake of sanity, I am going to assume the use of disks with a [GPT](#) partition table, and EFI FAT32 [EFI system partitions](#). Depending on how deep you're going to dive into this stuff you *may* find out that it's not strictly speaking the case that you can *always* assume you'll be dealing with GPT disks and EFI FAT32 ESPs when dealing with UEFI native boot, but the UEFI specification is quite strongly tied to GPT disks and EFI FAT32 ESPs, and this is what you'll be dealing with in 99% of cases. Unless you're dealing with Macs, and quite frankly, *screw* Macs.

为了方便起见，我假设你使用的是 [GPT](#) 分区表磁盘和 EFI FAT32 [系统分区](#)。根据你深入研究的程度，你可能会发现，严格来说，在处理 UEFI 原生启动时，并非总是能假设你遇到的都是 GPT 磁盘和 EFI FAT32 ESP 分区。但 UEFI 规范与 GPT 磁盘和 EFI FAT32 ESP 分区紧密相关，99% 的情况下你都会遇到这种情况。除非你用的是 Mac，说实话，Mac 什么的都见鬼去吧。

Edit note: the following sections (up to *Implications and Complications*) were heavily revised on 2014-01-26, a few hours after the initial version of this post went up, based on feedback from Peter Jones. Consider this to be v2.0 of the post. An earlier version was written in a somewhat less accurate and more confusing way.

编辑说明： 以下章节（直至“影响与复杂情况”部分）已于 2014 年 1 月 26 日，即本文初稿发布数小时后，根据 Peter Jones 的反馈进行了大幅修订。请将此版本视为本文的 2.0 版本。早期版本措辞不够准确，也较为晦涩难懂。

UEFI native booting: how it actually works - background

UEFI 原生启动：其工作原理 - 背景

OK, with that out of the way, let's get to the meat. This is how native UEFI booting actually works. It's probably helpful to go into this with a bit of high-level background.

好了，说完这些，我们进入正题。这就是原生 UEFI 启动的实际工作原理。事先了解一些背景知识可能会有所帮助。

UEFI provides *much* more infrastructure at the firmware level for handling system boot. It's nowhere near as simple as BIOS. Unlike BIOS, UEFI certainly does understand, to varying degrees, the concepts of 'disk partitions' and 'bootloaders' and 'operating systems'.

UEFI 在固件层面提供了更完善的系统启动处理机制，远比 BIOS 复杂得多。与 BIOS 不同的是，UEFI 在不同程度上理解“磁盘分区”、“引导加载程序”和“操作系统”等概念。

You can sort of look at the BIOS boot process, and look at the UEFI process, and see how the UEFI process extends various bits to address specific problems.

你可以查看 BIOS 启动过程，再查看 UEFI 过程，看看 UEFI 过程是如何扩展各个部分来解决特定问题的。

The BIOS/MBR approach to finding the bootloader is pretty janky, when you think about it. It's very 'special sauce': this particular tiny space at the front of the disk contains magic code that only really makes much sense to the system firmware and special utilities for writing it. There are several problems with this approach.

仔细想想，BIOS/MBR 查找引导加载程序的方式其实相当不靠谱。它非常“特殊”：磁盘前端这块小小的空间里包含着只有系统固件和专门用于写入它的工具才能理解的“魔法代码”。这种方法存在诸多问题。

- It's inconvenient to deal with - you need special utilities to write the MBR, and just about the only way to find out what's in one is to dd the contents out and examine them.

处理起来很不方便——你需要特殊的工具来写入 MBR，而且几乎唯一的办法就是把 MBR 的内容导出来检查。

- As noted above, the MBR itself is not big enough for many modern bootloaders. What they do is install a small part of themselves to the MBR proper, and the rest to the empty space on the disk between where the conventional MBR ends and the first partition begins. There's a rather big problem with this (well, the whole design is a big problem, but never mind), which is that there's no reliable convention for where the first partition should begin, so it's difficult to be sure there'll be enough space. One thing you usually can rely on is that there won't be *enough* space for some bootloader configurations.

如上所述，MBR 本身对于许多现代引导加载程序来说容量不足。它们的做法是将自身的一小部分安装到 MBR 本身，其余部分则安装到传统 MBR 结束位置和第一个分区开始位置之间的磁盘空白区域。这种方法存在一个相当大的问题（实际上，整个设计本身就是一个大问题，不过暂且不谈），那就是第一个分区的起始位置并没有可靠的约定，因此很难确保有足够的空间。通常可以肯定的是，某些引导加载程序配置肯定无法获得 *足够的* 空间。

- The design doesn't provide any standardized layer or mechanism for selecting boot targets other than disks...but people *want* to select boot targets other than disks. i.e. they want to have multiple bootable 'things' - usually operating systems - per disk. The only way to do this, in the BIOS/MBR world, is for the bootloaders to handle it; but there's no widely accepted convention for the right way to do this. There are many many different approaches, none of which is particularly interoperable with any of the others, none of which is a widely accepted standard or convention, and it's very difficult to write tooling at the OS / OS installation layer that handles multiboot cleanly. It's just a very messy design.

这种设计没有提供任何标准化的层或机制来选择磁盘以外的启动目标.....但人们 *希望* 选择磁盘以外的启动目标。也就是说，他们希望每个磁盘上都有多个可启动的“东西”（通常是操作系统）。在 BIOS/MBR 的世界里，唯一的方法是让引导加载程序来处理；但目前还没有被广泛接受的正确方法。存在许多不同的方法，这些方法彼此之间缺乏兼容性，也没有任何一种是被广泛接受的标准或惯例，而且在操作系统/操作系统安装层编写能够干净利落地处理多重启动的工具非常困难。这真是一个非常混乱的设计。

- The design doesn't provide a standard way of booting from anything *except* disks. We're not going to really talk about that in this article, but just be aware it's another advantage of UEFI booting: it provides a standard way for booting from, for instance, a remote server.

这种设计没有提供除磁盘以外的任何标准启动方式。本文不会详细讨论这一点，但需要注意的是，这是 UEFI 启动的另一个优势：它提供了一种从例如远程服务器启动的标准方式。

- There's no mechanism for levels above the firmware to configure the firmware's boot behaviour.

目前没有机制允许固件上层系统配置固件的启动行为。

So you can imagine the UEFI Elves sitting around and considering this problem, and coming up with a solution. Instead of the firmware only knowing about disks and one 'magic' location per disk where bootloader code might reside, UEFI has much more infrastructure at the firmware layer for handling boot loading. Let's look at all the things it defines that are relevant here.

你可以想象 UEFI 的开发者们围坐在一起，思考这个问题，并最终找到了解决方案。与固件只了解磁盘以及每个磁盘上一个用于存放引导加载程序代码的“神奇”位置不同，UEFI 在固件层拥有更完善的基础设施来处理引导加载。让我们来看看它定义了哪些与此相关的内容。

EFI executables EFI 可执行文件

The UEFI spec defines an executable format and requires all UEFI firmwares be capable of executing code in this format. When you write a bootloader for native UEFI, you write in this format. This is pretty simple and straightforward, and doesn't need any further explanation: it's just a Good Thing that we now have a firmware specification which actually defines a common format for code the firmware can execute.

UEFI 规范定义了一种可执行文件格式，并要求所有 UEFI 固件都必须能够执行这种格式的代码。编写原生 UEFI 引导加载程序时，也需要使用这种格式。这非常简单明了，无需赘述：我们现在有了一个固件规范，它定义了固件可以执行的通用代码格式，这真是一件好事。

The GPT (GUID partition table) format

GPT（GUID 分区表）格式

The [GUID Partition Table](#) format is very much tied in with the UEFI specification, and again, this isn't something particularly complex or in need of much explanation, it's just a good bit of groundwork the spec provides. GPT is just a standard for doing partition tables - the information at the start of a disk that defines what partitions that disk contains. It's a better standard for doing this than MBR/'MS-DOS' partition tables were in many ways, and the UEFI spec requires that UEFI-compliant firmwares be capable of interpreting GPT (it also requires them to be capable of interpreting MBR, for backwards compatibility). All of this is useful groundwork: what's going on here is the spec is establishing certain capabilities that everything above the firmware layer can rely on the firmware to have.

[GUID 分区表](#)格式与 UEFI 规范紧密相关，而且这本身并不复杂，也不需要过多解释，它只是规范提供的一项重要基础。GPT 只是分区表的标准——磁盘开头的信息定义了该磁盘包含哪些分区。在很多方面，它比 MBR/“MS-DOS”分区表更胜一筹，UEFI 规范要求符合 UEFI 标准的固件能够解析 GPT（为了向后兼容，也要求固件能够解析 MBR）。所有这些都是有用的基础：规范的作用在于建立某些功能，固件层之上的所有组件都可以依赖固件具备这些功能。

EFI system partitions EFI 系统分区

I actually really wrapped my head around the EFI system partition concept while revising this post, and it was a great 'aha!' moment. Really, the concept of 'EFI system partitions' is just an answer to the problem of the 'special sauce' MBR space. The concept of some undefined amount of empty space at the start of a disk being 'where bootloader code lives' is a pretty crappy design, as we saw above. EFI system partitions are just UEFI's solution to that.¹

在修改这篇文章的过程中，我真正理解了 EFI 系统分区概念，这真是一个令人恍然大悟的时刻。实际上，“EFI 系统分区”的概念是为了解决 MBR 空间这个“特殊空间”的问题。正如我们前面看到的，将磁盘开头的一段不确定大小的空白空间作为“引导加载程序代码的存放位置”的设计相当糟糕。EFI 系统分区正是 UEFI 针对这个问题提出的解决方案。

The solution is this: we require the firmware layer to be capable of reading some specific types of filesystem. The UEFI spec requires that compliant firmwares be capable of reading the FAT12, FAT16 and FAT32 variants of the [FAT](#) format, in essence. In fact what it does is codify a particular interpretation of those formats as they existed at the point UEFI was accepted, and say that UEFI compliant firmwares must be capable of reading those formats. As the spec puts it:

解决方案如下：我们要求固件层能够读取某些特定类型的文件系统。UEFI 规范要求符合规范的固件能够读取 FAT12、FAT16 和 FAT32 等 [FAT](#) 格式变体。实际上，它对这些格式在 UEFI 被接受时的特定解释进行了规范化，并规定符合 UEFI 规范的固件必须能够读取这些格式。正如规范中所述：

"The file system supported by the Extensible Firmware Interface is based on the FAT file system. EFI defines a specific version of FAT that is explicitly documented and testable. Conformance to the EFI specification and its associate reference documents is the only definition of FAT that needs to be implemented to support EFI. To differentiate the EFI file system from pure FAT, a new partition file system type has been defined."

“可扩展固件接口 (EFI) 支持的文件系统基于 FAT 文件系统。EFI 定义了一个特定版本的 FAT，该版本有明确的文档说明且可测试。只需符合 EFI 规范及其相关参考文档即可支持 EFI。为了将 EFI 文件系统与纯 FAT 区分开来，我们定义了一种新的分区文件系统类型。”

An 'EFI system partition' is really just any partition formatted with one of the UEFI spec-defined variants of FAT and given a specific GPT partition type to help the firmware find it. And the purpose of this is just as described above: allow everyone to rely on the fact that the firmware layer will definitely be able to read data from a pretty 'normal' disk partition. Hopefully it's clear why this is a better design: instead of having to write bootloader code to the

'magic' space at the start of an MBR disk, operating systems and so on can just create, format and mount partitions in a widely understood format and put bootloader code and anything else that they might want the firmware to read there.

“EFI 系统分区”实际上就是任何使用 UEFI 规范定义的 FAT 格式变体格式化的分区，并被赋予特定的 GPT 分区类型，以便固件能够找到它。其目的正如上文所述：确保固件层能够从一个相当“普通”的磁盘分区读取数据。希望大家能够明白为什么这是一个更优的设计：操作系统等程序无需将引导加载程序代码写入 MBR 磁盘开头的“魔法”空间，只需创建、格式化并挂载一个广泛理解的分区，并将引导加载程序代码以及固件可能需要读取的任何其他数据放入其中即可。

The whole ESP thing seemed a bit bizarre and confusing to me at first, so I hope this section explains why it's actually a very sensible idea and a good design - the bizarre and confusing thing is really the BIOS/MBR design, where the only way for you to write something from the OS layer that you knew the firmware layer could consume was to write it into some (but you didn't know how much) Magic Space at the start of a disk, a convention which isn't actually codified anywhere. That really *isn't* a very sensible or understandable design, if you step back and take a look at it.

一开始我觉得 ESP（扩展空间）的概念有点奇怪，也让人困惑，所以我希望这部分能解释清楚为什么它实际上是一个非常合理且设计精良的想法——真正奇怪和令人困惑的是 BIOS/MBR 的设计。在这种设计中，你唯一能从操作系统层写入固件层可以读取的数据的方法，就是将其写入磁盘开头的某个（但你不知道具体大小的）“魔法空间”。而这种做法实际上并没有在任何地方得到规范。如果你仔细审视一下，就会发现这确实是一个非常不合理且难以理解的设计。

As we'll note later, the UEFI spec tends to take a 'you must at least do these things' approach - it rarely prohibits firmwares from doing anything else. It's not against the spec to write a firmware that can execute code in other formats, read other types of partition table, and read partitions formatted with filesystems other than the UEFI variants of FAT. But a UEFI compliant firmware must *at least* do all these things, so if you are writing an OS or something else that you want to run on *any* UEFI compliant firmware, this is why the EFI system partition concept is so important: it gives you (at least in theory) 100% confidence that you can put an EFI executable on a partition formatted with the UEFI FAT implementation and the correct GPT partition type, and the system firmware will be able to read it. This is the thing you can take to the bank, like 'the firmware will be able to execute some bootloader code I put in the MBR space' was in the BIOS world.

正如我们稍后会提到的，UEFI 规范倾向于采用“至少必须做到这些”的方法——它很少禁止固件执行其他操作。编写能够执行其他格式代码、读取其他类型分区表以及读取使用除 UEFI FAT 变体之外的其他文件系统格式化的分区的固件，并不违反规范。但是，符合 UEFI 规范的固件必须至少做到所有这些。因此，如果您正在编写操作系统或其他程序，并且希望它们能在任何符合 UEFI 规范的固件上运行，那么这就是 EFI 系统分区概念如此重要的原因：它（至少在理论上）让您 100% 确信，您可以将 EFI 可执行文件放在使用 UEFI FAT 实现和正确的 GPT 分区类型格式化的分区上，并且系统固件能够读取它。这就像 BIOS 时代“固件能够执行我放在 MBR 空间中的一些引导加载程序代码”一样，您可以完全放心。

So now we have three important bits of groundwork the UEFI spec provides: thanks to these requirements, any other layer can confidently rely on the fact that the firmware:

现在我们有 UEFI 规范提供的三个重要基础：由于这些要求，任何其他层都可以确信固件具备以下特性：

- Can read a partition table
 - 可以读取分区表
- Can access files in some specific filesystems
 - 可以访问某些特定文件系统中的文件。
- Can execute code in a particular format
 - 能够执行特定格式的代码

This is much more than you can rely on a BIOS firmware being capable of. However, in order to complete the vision of a firmware layer that can handle booting multiple targets - not just disks - we need one more bit of groundwork: there needs to be a mechanism by which the firmware *finds* the various possible boot targets, and a way to configure it.

这远远超出了 BIOS 固件的能力范围。然而，为了实现能够处理多个启动目标（而不仅仅是磁盘）的固件层愿景，我们还需要做一些基础工作：需要一种机制，让固件能够找到各种可能的启动目标，以及一种配置该机制的方法。

The UEFI boot manager

UEFI 启动管理器

The UEFI spec defines something called the *UEFI boot manager*. (Linux distributions contain a tool called **efibootmgr** which is used to manipulate the configuration of the UEFI boot manager). As a sample of what you can expect to find if you do read the UEFI spec, it defines the UEFI boot manager thusly:

UEFI 规范定义了一种称为 *UEFI 启动管理器* 的组件。（Linux 发行版包含一个名为 `efibootmgr` 的工具，用于操作 UEFI 启动管理器的配置）。以下是 UEFI 规范中对 UEFI 启动管理器的定义示例，供您参考：

"The UEFI boot manager is a firmware policy engine that can be configured by modifying architecturally defined global NVRAM variables. The boot manager will attempt to load UEFI drivers and UEFI applications (including UEFI OS boot loaders) in an order defined by the global NVRAM variables."

“UEFI 启动管理器是一个固件策略引擎，可以通过修改架构定义的全局 NVRAM 变量进行配置。启动管理器将尝试按照全局 NVRAM 变量定义的顺序加载 UEFI 驱动程序和 UEFI 应用程序（包括 UEFI 操作系统启动加载程序）。”

Well, that's that cleared up, let's move on. ;) No, not really. Let's translate that to Human. With only a reasonable degree of simplification, you can think of the UEFI boot manager as being a boot menu. With a BIOS firmware, your firmware level 'boot menu' is, necessarily, the disks connected to the system at boot time - no more, no less. This is not true with a UEFI firmware.

好了，这下清楚了，我们继续吧。;) 不，其实不是。我们来用通俗易懂的方式解释一下。简单来说，你可以把 UEFI 启动管理器想象成一个启动菜单。对于 BIOS 固件来说，固件级别的“启动菜单”必然就是启动时连接到系统的磁盘——不多也不少。但 UEFI 固件并非如此。

The UEFI boot manager can be configured - simply put, you can add and remove entries from the 'boot menu'. The firmware can also (it fact the spec requires it to, in various cases) effectively 'generate' entries in this boot menu, according to the disks attached to the system and possibly some firmware configuration settings. It can also be examined - you can look at what's in it.

UEFI 启动管理器是可以配置的——简单来说，您可以添加和删除“启动菜单”中的条目。固件也可以（实际上，在某些情况下，规范要求固件必须这样做）根据连接到系统的磁盘以及可能的固件配置设置，有效地“生成”此启动菜单中的条目。此外，您还可以查看它的内容。

One rather great thing UEFI provides is a mechanism for doing this *from other layers*: you can configure the system boot behaviour from a booted operating system. You can do all this by using the `efibootmgr` tool, once you have Linux booted via UEFI somehow. There are Windows tools for it too, but I'm not terribly familiar with them. Let's have a look at some typical `efibootmgr` output, which I stole and slightly tweaked from the Fedora forums:

UEFI 的一个非常棒的功能是允许从其他层进行配置：你可以从已启动的操作系统配置系统启动行为。一旦你通过某种方式通过 UEFI 启动了 Linux，就可以使用 `efibootmgr` 工具完成所有这些操作。Windows 也有类似的工具，但我不太熟悉它们。让我们来看一些典型的 `efibootmgr` 输出，这些输出是我从 Fedora 论坛上获取并稍作修改的：

```
[root@system directory]# efibootmgr -v
BootCurrent: 0002
Timeout: 3 seconds
BootOrder: 0003,0002,0000,0004
Boot0000* CD/DVD Drive  BIOS(3,0,00)
Boot0001* Hard Drive    HD(2,0,00)
Boot0002* Fedora        HD(1,800,61800,6d98f360-cb3e-4727-8fed-5ce0c040365d)File(\EFI\fedora\grubx64.efi)
Boot0003* opensuse       HD(1,800,61800,6d98f360-cb3e-4727-8fed-5ce0c040365d)File(\EFI\opensuse\grubx64.efi)
Boot0004* Hard Drive     BIOS(2,0,00)P0: ST1500DM003-9YN16G
[root@system directory]#
```

This is a nice clean example I stole and slightly tweaked from the Fedora forums. We can see a few things going on here.

这是一个简洁明了的例子，我从 Fedora 论坛上借鉴并稍作修改。我们可以看到这里发生了一些事情。

The first line tells you which of the 'boot menu' entries you are *currently* booted from. The second is pretty obvious (if the firmware presents a boot menu-like interface to the UEFI boot manager, that's the timeout before it goes ahead and boots the default entry). The BootOrder is the order in which the entries in the list will be tried. The rest of the output shows the actual boot entries. We'll describe what they actually do later.

第一行显示当前系统是从哪个“启动菜单”条目启动的。第二行的含义显而易见（如果固件向 UEFI 启动管理器呈现类似启动菜单的界面，则表示在启动默认条目之前需要等待的超时时间）。BootOrder 表示列表中条目的尝试顺序。其余输出显示实际的启动条目。我们稍后会详细介绍它们的具体功能。

If you boot a UEFI firmware entirely normally, without doing any of the tweaks we'll discuss later, what it ought to do is try to boot from each of the 'entries' in the 'boot menu', in the order listed in *BootOrder*. So on this system it would try to boot the entry called 'opensuse', then if that failed, the one called 'Fedora', then 'CD/DVD Drive', and then the second 'Hard Drive'.

如果完全正常地启动 UEFI 固件，而不进行我们稍后将讨论的任何调整，它应该会按照 *BootOrder* 中列出的顺序，尝试从“启动菜单”中的每个“条目”启动。因此，在这个系统中，它会尝试启动名为“opensuse”的条目，如果失败，则尝试启动名为“Fedora”的条目，然后是“CD/DVD 驱动器”，最后是第二个“硬盘驱动器”。

UEFI native booting: how it actually works – boot manager entries

UEFI 原生启动：其工作原理 - 启动管理器条目

What does these entries actually *mean*, though? There's actually a *huge* range of possibilities that makes up rather a large part of the complexity of the UEFI spec all by itself. If you're reading the spec, pour yourself an extremely large shot of gin and turn to the EFI_DEVICE_PATH_PROTOCOL section, but note that this is a generic protocol that's used for other things than booting - it's UEFI's Official Way Of Identifying Devices For All Purposes, used for boot manager entries but also for all sorts of other purposes. Not every possible EFI device path makes sense as a UEFI boot manager entry, for obvious reasons (you're probably not going to get too far trying to boot from your video adapter). But you can certainly have an entry that points to, say, a PXE server, not a disk partition. The spec has lots of bits defining valid non-disk boot targets that can be added to the UEFI boot manager configuration.

这些条目究竟意味着什么呢？实际上，UEFI 规范本身就包含着极其丰富的可能性，而这本身就构成了其复杂性的很大一部分。如果你正在阅读规范，不妨给自己倒上一大杯杜松子酒，然后翻到 EFI_DEVICE_PATH_PROTOCOL 部分。但请注意，这是一个通用协议，其用途远不止于启动——它是 UEFI 官方的设备识别方式，用于各种用途，包括启动管理器条目。并非所有可能的 EFI 设备路径都适合作为 UEFI 启动管理器条目，原因显而易见（例如，你不太可能尝试从显卡启动）。但你当然可以创建一个指向 PXE 服务器而非磁盘分区的条目。规范中包含大量定义有效非磁盘启动目标的条目，这些目标可以添加到 UEFI 启动管理器配置中。

For our purposes, though, lets just consider fairly normal disks connected to the system. In this case we can consider three types of entry you're likely to come across.

不过，就我们的目的而言，我们只考虑连接到系统的普通磁盘。在这种情况下，我们可以考虑三种你可能会遇到的条目类型。

BIOS compatibility boot entries

BIOS 兼容性启动项

Boot0000 and Boot0004 in this example are actually BIOS compatibility mode entries, not UEFI native entries. They have not been added to the UEFI boot manager configuration by any external agency, but generated by the firmware itself - this is a common way for a UEFI firmware to implement BIOS compatibility booting, by generating UEFI boot manager entries that trigger a BIOS-compatible boot of a given device. How they *present this to the user* is a different question, as we'll see later. Whether you see any of these entries or not will depend on your particular firmware, and its configuration. Each of these entries just gives a name - 'CD/DVD Drive', 'Hard Drive' - and says "if this entry is selected, boot this disk (where 'this disk' is **3,0,00** for Boot0000 and **2,0,00** for Boot0004) in BIOS compatibility mode".

在本例中，Boot0000 和 Boot0004 实际上是 BIOS 兼容模式条目，而非 UEFI 原生条目。它们并非由任何外部机构添加到 UEFI 启动管理器配置中，而是由固件自身生成——这是 UEFI 固件实现 BIOS 兼容启动的常用方法，即生成 UEFI 启动管理器条目，触发指定设备的 BIOS 兼容启动。至于如何*向用户呈现这些信息*，则是另一个问题，我们稍后会看到。您是否能看到这些条目取决于您的固件及其配置。每个条目都只提供一个名称——“CD/DVD 驱动器”、“硬盘驱动器”——并表示“如果选中此条目，则以 BIOS 兼容模式启动此磁盘（其中 Boot0000 的“此磁盘”为 **3,0,00**，Boot0004 的“此磁盘”为 **2,0,00**）”。

'Fallback path' UEFI native boot entries

UEFI 原生启动项的“回退路径”

Boot0001 is an entry (fictional, and somewhat unlikely, but it's for illustrative purposes) that tells the firmware to try and boot from a particular disk, and in UEFI mode not BIOS compatibility mode, but doesn't tell it anything more. It doesn't specify a particular boot target on the disk - it just says to boot the disk.

Boot0001 是一个条目（虚构的，可能性不大，但仅用于说明目的），它指示固件尝试从特定磁盘启动，并且使用 UEFI 模式而非 BIOS 兼容模式，但除此之外没有其他信息。它没有指定磁盘上的具体启动目标——只是指示从该磁盘启动。

The UEFI spec defines a sort of 'fallback' path for booting this kind of boot manager entry, which works in principle somewhat like BIOS drive booting: it looks in a standard location for some boot loader code. The details are different, though.

UEFI 规范定义了一种用于启动此类启动管理器条目的“备用”路径，其工作原理与 BIOS 驱动器启动有些类似：它会在标准位置查找一些引导加载程序代码。不过，具体细节有所不同。

What the firmware will actually do when trying to boot in this way is reasonably simple. The firmware will look through each EFI system partition on the disk in the order they exist on the disk. Within the ESP, it will look for a file with a specific name and location. On an x86-64 PC, it will look for the file `\EFI\BOOT\BOOTx64.EFI`. What it actually looks for is `\EFI\BOOT\BOOT{machine type short-name}.EFI` - 'x64' is the "machine type short-name" for x86-64 PCs. The other possibilities are `BOOTIA32.EFI` (x86-32), `BOOTIA64.EFI` (Itanium), `BOOTARM.EFI` (AArch32 - that is, 32-bit ARM) and

BOOTAA64.EFI (AArch64 - that is, 64-bit ARM). It will then execute the first qualifying file it finds (obviously, the file needs to be in the executable format defined in the UEFI specification).

固件尝试以这种方式启动时实际执行的操作相当简单。固件会按照磁盘上的顺序遍历每个 EFI 系统分区。在 ESP 分区内，它会查找具有特定名称和位置的文件。在 x86-64 PC 上，它会查找文件 **\EFI\BOOT\BOOTx64.EFI**。它实际查找的是 **\EFI\BOOT\BOOT{machine type short-name}.EFI** ——“x64”是 x86-64 PC 的“机器类型简称”。其他可能的文件包括 **BOOTIA32.EFI** (x86-32)、**BOOTIA64.EFI** (Itanium)、**BOOTARM.EFI** (AArch32, 即 32 位 ARM) 和 **BOOTAA64.EFI** (AArch64, 即 64 位 ARM)。然后它将执行找到的第一个符合条件的文件（显然，该文件需要是 UEFI 规范中定义的可执行格式）。

This mechanism is not designed for booting permanently-installed OSes. It's more designed for booting hotpluggable, device-agnostic media, like live images and OS install media. And this is indeed what it's usually used for. If you look at a UEFI-capable live or install medium for a Linux distribution or other OS, you'll find it has a GPT partition table and contains a FAT-formatted partition at or near the start of the device, with the GPT partition type that identifies it as an EFI system partition. Within that partition there will be a **\EFI\BOOT** directory with at least one of the specially-named files above. When you boot a Fedora live or install medium in UEFI-native mode, this is the mechanism that is used. The **BOOTx64.EFI** (or whatever) file handles the rest of the boot process from there, booting the actual operating system contained on the medium.

这种机制并非设计用于启动永久安装的操作系统。它更适合启动热插拔、与设备无关的介质，例如 Live 镜像和操作系统安装介质。事实上，它通常也用于此目的。如果您查看支持 UEFI 的 Linux 发行版或其他操作系统的 Live 或安装介质，您会发现它具有 GPT 分区表，并且在设备起始位置或附近包含一个 FAT 格式的分区，该分区的 GPT 分区类型表明它是一个 EFI 系统分区。在该分区内，会有一个名为 **\EFI\BOOT** 的目录，其中包含至少一个上述特殊命名的文件。当您以 UEFI 原生模式启动 Fedora Live 或安装介质时，就会使用此机制。**BOOTx64.EFI**（或其他名称）文件会处理后续的启动过程，启动介质上实际的操作系统。

Full UEFI native boot entries

完整的 UEFI 原生启动项

Boot0002 and Boot0003 are 'typical' entries for operating systems permanently installed to permanent storage devices. These entries show us the full power of the UEFI boot mechanism, by not just saying "boot from this disk", but "boot this specific bootloader in this specific location on this specific disk", using all the 'groundwork' we talked about above.

Boot0002 和 Boot0003 是永久安装到永久存储设备上的操作系统的“典型”启动项。这些启动项展现了 UEFI 启动机制的全部功能，它不仅指定“从此磁盘启动”，而且还指定“从此磁盘上的此特定位置启动此特定引导加载程序”，并运用了我们前面讨论的所有“基础工作”。

Boot0002 is a boot entry produced by a UEFI-native Fedora installation. Boot0003 is a boot entry produced by a UEFI-native OpenSUSE installation. As you may be able to tell, all they're saying is "load this file from this partition". The partition is the **HD(1,800,61800,6d98f360-cb3e-4727-8fed-5ce0c040365d)** bit: that's referring to a specific partition (using the **EFI_DEVICE_PATH_PROTOCOL**, which I'm really not going to attempt to explain in any detail - you don't necessarily need to know it, if you interact with the boot manager via the firmware interface and **efibootmgr**). The file is the **File(\EFI\opensuse\grubx64.efi)** bit: that just means "load the file in this location on the partition we just described". The partition in question will almost always be one that qualifies as an EFI system partition, because of the considerations above: that's the type of partition we can trust the firmware to be able to access.

Boot0002 是由 UEFI 原生 Fedora 安装生成的启动项。Boot0003 是由 UEFI 原生 OpenSUSE 安装生成的启动项。正如您所见，它们的意思都是“从该分区加载此文件”。分区指的是 **HD(1,800,61800,6d98f360-cb3e-4727-8fed-5ce0c040365d)** 位：它指向一个特定的分区（使用 **EFI_DEVICE_PATH_PROTOCOL**，我并不打算详细解释它——如果您通过固件接口和 **efibootmgr** 与启动管理器交互，则不一定需要了解它）。文件指的是 **File(\EFI\opensuse\grubx64.efi)** 位：它表示“加载我们刚才描述的分区上此位置的文件”。由于上述原因，所涉及的分区几乎总是符合 EFI 系统分区的定义：这是我们可以信任固件能够访问的分区类型。

This is the mechanism the UEFI spec provides for operating systems to make themselves available for booting: the operating system is intended to install a bootloader which loads the OS kernel and so on to an EFI system partition, and add an entry to the UEFI boot manager configuration with a name - obviously, this will usually be derived from the operating system's name - and the location of the bootloader (in EFI executable format) that is intended for loading that operating system.

这是 UEFI 规范为操作系统提供的启动机制：操作系统旨在安装一个引导加载程序，该程序会将操作系统内核等加载到 EFI 系统分区，并在 UEFI 启动管理器配置中添加一个条目，该条目具有一个名称（显然，这通常会从操作系统的名称派生而来）以及用于加载该操作系统的引导加载程序（EFI 可执行格式）的位置。

Linux distributions use the **efibootmgr** tool to deal with the UEFI boot manager. What a Linux distribution actually does, so far as bootloading is concerned, when you do a UEFI native install is really pretty simple: it creates an EFI system partition if one does not already exist, installs an EFI boot loader with an appropriate configuration - often grub2-efi, but there are others - into a correct path in the EFI system partition, and calls **efibootmgr** to add an appropriately-named UEFI boot manager entry pointing to its boot loader. Most distros will use an existing EFI system partition if there is one,

though it's perfectly valid to create a new one and use that instead: as we've noted, UEFI is a permissive spec, and if you follow the design logically, there's really no problem with having just as many EFI system partitions as you want.

Linux 发行版使用 `efibootmgr` 工具来处理 UEFI 启动管理器。就启动加载而言，Linux 发行版在进行 UEFI 原生安装时实际执行的操作非常简单：它会创建一个 EFI 系统分区（如果尚不存在），将具有适当配置的 EFI 启动加载程序（通常是 `grub2-efi`，但也存在其他选择）安装到 EFI 系统分区中的正确路径，然后调用 `efibootmgr` 添加一个指向其启动加载程序的、名称合适的 UEFI 启动管理器条目。大多数发行版会使用现有的 EFI 系统分区（如果存在），但创建一个新的分区并使用它也是完全有效的：正如我们提到的，UEFI 是一个宽松的规范，如果您遵循其逻辑设计，那么拥有任意数量的 EFI 系统分区实际上没有任何问题。

Configuring the boot process (the firmware UI)

配置启动过程（固件用户界面）

The above describes the basic mechanism the UEFI spec defines that manages the UEFI boot process. It's important to realize that your firmware user interface may well not represent this mechanism very clearly. Unfortunately, the spec intentionally refrains from defining how the boot process should be represented to the user or how the user should be allowed to configure it, and what that means - since we're dealing with [firmware engineers](#) - is that every firmware does it differently, and some do it insanely.

以上描述了 UEFI 规范定义的用于管理 UEFI 启动过程的基本机制。需要注意的是，您的固件用户界面可能无法清晰地展现这一机制。遗憾的是，该规范有意避免定义如何向用户呈现启动过程，以及允许用户如何配置它。这意味着——考虑到我们面对的是[固件工程师](#)——每个固件的实现方式都不同，有些甚至非常复杂。

Many firmwares do have fairly reasonable interfaces for boot configuration. A good firmware design will at least show you the boot order, with a reasonable representation of the entries on it, and let you add or remove entries, change the order, or override the order for a specific boot (by changing it just for that boot, or directly instructing the firmware to boot a particular menu entry, or even giving you the option to simply say "boot this disk", either in BIOS compatibility mode or UEFI 'fallback' mode - my firmware does this). Such an interface will often show 'full' UEFI native boot entries (like the Fedora and openSUSE examples we saw earlier) only by their name; you have to examine the `efibootmgr -v` output to know precisely what these entries will actually try and *do* when invoked.

许多固件都提供了相当合理的启动配置界面。一个好的固件设计至少会显示启动顺序，并以合理的方式呈现启动项，同时允许用户添加或删除启动项、更改启动顺序，或者覆盖特定启动的启动顺序（例如，仅针对该启动更改启动顺序，或直接指示固件启动特定菜单项，甚至允许用户简单地指定“从此磁盘启动”，无论是在 BIOS 兼容模式还是 UEFI“回退”模式下——我的固件就支持此功能）。这样的界面通常只会显示“完整”的 UEFI 原生启动项（例如我们之前看到的 Fedora 和 openSUSE 示例），而无法显示其名称；用户需要查看 `efibootmgr -v` 输出才能确切了解这些启动项在被调用时实际会执行哪些操作。

Some firmwares try to abstract and simplify the configuration, and may do a good or a bad job of it. For instance, if you have an option to 'enable or disable' BIOS compatibility mode, what it'll really likely *do* is configure whether the firmware adds BIOS compatibility entries for attached drives to the UEFI boot manager configuration or not. If you have an option to 'enable or disable' UEFI native booting, what likely really happens when you 'disable' it is that the firmware changes the UEFI boot manager configuration to leave all UEFI-native entries out of the BootOrder.

有些固件试图抽象化和简化配置，但效果可能好也可能坏。例如，如果可以选择“启用或禁用”BIOS 兼容模式，它实际的*作用*很可能是配置固件是否将外接驱动器的 BIOS 兼容条目添加到 UEFI 启动管理器配置中。如果可以选择“启用或禁用”UEFI 原生启动，禁用后固件实际的作用很可能是更改 UEFI 启动管理器配置，将所有 UEFI 原生启动条目从启动顺序中移除。

The key point to remember is that any configuration option inside your firmware interface which is to do with booting is really, behind the scenes, configuring the behaviour of the UEFI boot manager. If you understand all the stuff we've discussed above, you may well find it easier to figure out what's *really* happening when you twiddle the knobs your firmware interface exposes.

需要记住的关键一点是，固件界面中任何与启动相关的配置选项，实际上都是在底层配置 UEFI 启动管理器的行为。如果您理解了我們上面讨论的所有内容，那么当您调整固件界面中显示的那些参数时，就更容易弄清楚究竟发生了什么。

In the BIOS world, you'll remember, you don't always find that systems are configured to try and boot from removable drives - CD, USB - before booting from permanent drives. Some are, and some aren't. Some will try CD before the hard disks, but not USB. People have got fairly used to having to check the BIOS configuration to ensure the boot order is 'correct' when trying to install a new operating system.

在 BIOS 设置方面，您可能还记得，并非所有系统都会配置为先尝试从可移动驱动器（例如光盘、U 盘）启动，然后再从固定驱动器启动。有些系统会这样做，有些则不会。有些系统会先尝试光盘启动，然后再尝试硬盘，但不会先尝试 U 盘启动。人们已经相当习惯于在安装新操作系统时检查 BIOS 配置，以确保启动顺序“正确”。

This applies to the UEFI world too, but because of the added flexibility/complexity of the UEFI boot manager mechanism, it can look unfamiliar and scary.

这也适用于 UEFI 世界，但由于 UEFI 启动管理器机制的灵活性/复杂性增加，它看起来可能既陌生又可怕。

If you want to ensure that your system tries to boot from removable devices using the 'fallback' mechanism before it tries to boot 'permanent' boot entries - as you will want to do if you want to, say, install Fedora - you need this to be the default for your firmware, or you need to be able to tell the firmware this. Depending on your firmware's interface, you may find there is a 'menu entry' for each attached removable device and you just have to adjust the boot order to put it at the top of the list, or you may find that there is the mechanism to directly request 'UEFI fallback boot of this particular disk', or you may find that the firmware tries to abstract the configuration somehow. We just don't know, and that makes writing instructions for this quite hard. But now you broadly understand how things work behind the scenes, you may find it easier to understand your firmware user interface's representation of that.

如果您希望系统在尝试启动“永久”启动项之前，先尝试使用“回退”机制从可移动设备启动（例如，安装 Fedora 时），则需要将此设置为固件的默认设置，或者需要能够告知固件此设置。根据固件的界面，您可能会发现每个连接的可移动设备都有一个“菜单项”，只需调整启动顺序将其置于列表顶部即可；或者您可能会发现存在直接请求“此特定磁盘的 UEFI 回退启动”的机制；或者您可能会发现固件以某种方式尝试抽象化此配置。我们对此并不完全了解，因此很难编写相关说明。但现在您大致了解了其背后的工作原理，您可能会发现更容易理解固件用户界面对此的表示。

Configuring the boot process (from an operating system)

配置启动过程（从操作系统）

As we've noted above, unlike in the BIOS world, you can actually configure the UEFI boot process from the operating system level. If you have an insane firmware, you may *have* to do this in order to achieve what you want.

正如我们前面提到的，与 BIOS 不同，你可以在操作系统级别配置 UEFI 启动过程。如果你的固件非常复杂，你可能需要这样做才能实现你想要的功能。

You can use the **efibootmgr** tool mentioned earlier to add, delete and modify entries in the UEFI boot manager configuration, and actually do quite a lot of other stuff with it too. You can change the boot order. You can tell it to boot some particular entry in the list on the next boot, instead of using the *BootOrder* list (if you or some other tool has configured this to happen, your **efibootmgr -v** output will include a *BootNext* item stating which menu entry will be loaded on the next boot). There are tools for Windows that can do this stuff from Windows, too. So if you're really struggling to manage to do whatever it is you want to do with UEFI boot configuration from your firmware interface, but you *can* boot a UEFI native operating system of some kind, you may want to consider doing your boot configuration from that operating system rather than from the firmware UI.

您可以使用前面提到的 **efibootmgr** 工具来添加、删除和修改 UEFI 启动管理器配置中的条目，实际上它还可以执行许多其他操作。您可以更改启动顺序。您可以指定下次启动时启动列表中的某个特定条目，而不是使用 *BootOrder* 列表（如果您或其他工具已配置此操作，则 **efibootmgr -v** 输出将包含一个 *BootNext* 项，指示下次启动时将加载哪个菜单项）。Windows 系统下也有一些工具可以执行这些操作。因此，如果您确实难以通过固件界面完成所需的 UEFI 启动配置，但您可以启动某种 UEFI 原生操作系统，则不妨考虑从该操作系统而不是固件 UI 进行启动配置。

So to recap: 总结一下：

- Your UEFI firmware contains something very like what you think of as a boot menu.
您的 UEFI 固件包含类似于您通常所说的启动菜单的内容。
- You can query its configuration with **efibootmgr -v**, from any UEFI-native boot of a Linux OS, and also *change* its configuration with **efibootmgr** (see the man page for details).
您可以使用 **efibootmgr -v** 从 Linux 操作系统的任何 UEFI 原生启动中查询其配置，也可以使用 **efibootmgr** 更改其配置（有关详细信息，请参阅手册页）。
- This 'boot menu' can contain entries that say 'boot this disk in BIOS compatibility mode', 'boot this disk in UEFI native mode via the fallback path' (which will use the 'look for BOOT(something).EFI' method described above), or 'boot the specific EFI format executable at this specific location (almost always on an EFI system partition)'.
此“启动菜单”可以包含以下条目：以 BIOS 兼容模式启动此磁盘、通过回退路径以 UEFI 本机模式启动此磁盘（这将使用上面描述的“查找 BOOT(something).EFI”方法）或“启动此特定位置的特定 EFI 格式可执行文件（几乎总是在 EFI 系统分区上）”。
- The nice, clean design that the UEFI spec is trying to imply is that all operating systems should install a bootloader of their own to an EFI system partition, add entries to this 'boot menu' that point to themselves, and butt out from trying to take control of booting anything else.
UEFI 规范试图传达的简洁设计理念是，所有操作系统都应该在 EFI 系统分区中安装自己的引导加载程序，向该“启动菜单”添加指向自身的条目，并且不试图控制其他任何程序的启动。
- Your firmware UI has free rein to represent this mechanism to you in whatever way it wants, and it may do this well, or it may do this poorly.
固件用户界面可以自由地以任何方式向您呈现此机制，它可能做得很好，也可能做得不好。

Installing operating systems to UEFI-based computers

在基于 UEFI 的计算机上安装操作系统

Let's have a quick look at some specific consequences of the above that relate to installing operating systems on UEFI computers.

让我们快速看一下上述内容与在 UEFI 计算机上安装操作系统相关的具体后果。

UEFI native and BIOS compatibility booting

UEFI 原生和 BIOS 兼容性启动

Here's a very very simple one which people sometimes miss:

这是一个非常非常简单的道理，但人们有时会忽略：

- If you boot the installation medium in 'UEFI native' mode, it will do a UEFI native install of the operating system: it will try to write an EFI-format bootloader to an EFI system partition, and attempt to add an entry to the UEFI boot manager 'boot menu' which loads that bootloader.

如果以“UEFI 本机”模式启动安装介质，它将执行 UEFI 本机操作系统安装：它会尝试将 EFI 格式的引导加载程序写入 EFI 系统分区，并尝试向 UEFI 启动管理器“启动菜单”添加一个条目，以加载该引导加载程序。

- If you boot the installation medium in 'BIOS compatibility' mode, it will do a BIOS compatible install of the operating system: it will try to write an MBR-type bootloader to the magic MBR space on a disk.

如果以“BIOS 兼容”模式启动安装介质，它将以 BIOS 兼容的方式安装操作系统：它会尝试将 MBR 类型的引导加载程序写入磁盘上的魔数 MBR 空间。

This applies (with one minor caveat I'm going to paper over for now) to all OSes of which I'm aware. So you probably want to make sure you understand how, in your firmware, you can choose to boot a removable device in UEFI native mode and how you can choose to boot it in BIOS compatibility mode, and make sure you pick whichever one you actually want to use for your installation.

这适用于我所知的所有操作系统（除了一个我暂时忽略的小问题）。因此，您可能需要确保了解如何在固件中选择以 UEFI 原生模式启动可移动设备，以及如何选择以 BIOS 兼容模式启动，并确保选择您实际想要用于安装的模式。

You really cannot do a completely successful UEFI-native installation of an OS if you boot its installation medium in BIOS compatibility mode, because the installer will not be able to configure the UEFI boot manager (this is only possible when booted UEFI-native).

如果以 BIOS 兼容模式启动安装介质，则无法完全成功地以 UEFI 原生方式安装操作系统，因为安装程序将无法配置 UEFI 启动管理器（只有在以 UEFI 原生方式启动时才有可能）。

It is theoretically possible for an OS installer to install the OS in the BIOS style - that is, write a bootloader to a disk's MBR - after being booted in UEFI native mode, but most of them *won't* do this, and that's probably sensible.

理论上，操作系统安装程序可以在以 UEFI 原生模式启动后，以 BIOS 风格安装操作系统（即将引导加载程序写入磁盘的 MBR），但大多数安装程序不会这样做，这可能是明智之举。

Finding out which mode you're booted in

找出你当前启动的模式。

It is possible that you might find yourself with your operating system installer booted, and not sure whether it's actually booted in UEFI native mode or BIOS compatibility mode. Don't panic! It's pretty easy to find out which, in a few different ways. One of the easiest is just to try and talk to the UEFI boot manager. If what you have booted is a Linux installer or environment, and you can get to a shell (ctrl-alt-f2 in the Fedora installer, for instance), run **efibootmgr -v**. If you're booted in UEFI native mode, you'll get your UEFI boot manager configuration, as shown above. If you're booted in BIOS compatibility mode, you'll get something like this:

您可能会遇到这样的情况：操作系统安装程序已经启动，但您不确定它是以 UEFI 原生模式还是 BIOS 兼容模式启动的。别担心！有几种方法可以轻松确定。其中最简单的方法之一是尝试与 UEFI 启动管理器通信。如果您启动的是 Linux 安装程序或环境，并且您可以进入 shell（例如，在 Fedora 安装程序中按 Ctrl+Alt+F2），请运行 **efibootmgr -v**。如果您以 UEFI 原生模式启动，您将看到 UEFI 启动管理器的配置，如上所示。如果您以 BIOS 兼容模式启动，您将看到类似这样的信息：

```
Fatal: Couldn't open either sysfs or procfs directories for accessing EFI variables.
Try 'modprobe efivars' as root.
```

If you've booted some other operating system, you can try running a utility native to that OS which tries to talk to the UEFI boot manager, and see if you get sensible output or a similar kind of error. Or you can examine the system logs and search for 'efi' and/or 'uefi', and you'll probably find some kind of indication.

如果你已经启动了其他操作系统，可以尝试运行该操作系统自带的、会尝试与 UEFI 启动管理器通信的实用程序，看看是否能得到合理的输出或类似的错误信息。或者，你也可以查看系统日志，搜索“efi”和/或“uefi”，或许能找到一些线索。

Enabling UEFI native boot

启用 UEFI 原生启动

To be bootable in UEFI native mode, your OS installation medium must obviously actually comply with all this stuff we've just described: it's got to have a GPT partition table, and an EFI system partition with a bootloader in the correct 'fallback' path - \EFI\BOOT\BOOTx64.EFI (or the other names for the other platforms). If you're having trouble doing a UEFI native boot of your installation medium and can't figure out why, check that this is actually the case. Notably, when using the **livecd-iso-to-disk** tool to write a Fedora image to a USB stick, you must pass the **--efi** parameter to configure the stick to be UEFI bootable.

要使操作系统安装介质能够以 UEFI 原生模式启动，显然必须满足我们刚才描述的所有条件：它必须具有 GPT 分区表，以及一个包含引导加载程序的 EFI 系统分区，并且该引导加载程序必须位于正确的“备用”路径 \EFI\BOOT\BOOTx64.EFI（或其他平台对应的名称）。如果您在安装介质上无法以 UEFI 原生模式启动，并且找不到原因，请检查是否确实如此。值得注意的是，当使用 **livecd-iso-to-disk** 工具将 Fedora 镜像写入 U 盘时，必须传递 **--efi** 参数才能将 U 盘配置为 UEFI 启动。

Forcing BIOS compatibility boot

强制 BIOS 兼容启动

If your firmware seems to make it very difficult to boot from a removable medium in BIOS compatibility mode, but you really want to do that, there's a handy trick you can use: just make the medium not UEFI native bootable at all. You can do this pretty easily by just wiping all the EFI system partitions. (Alternatively, if using **livecd-iso-to-disk** to create a USB stick from a Fedora image, you can just leave out the **--efi** parameter and it won't be UEFI bootable). If at that point your firmware refuses to boot it in BIOS compatibility mode, commence swearing at your firmware vendor (if you didn't already).

如果你的固件似乎让从可移动介质在 BIOS 兼容模式下启动变得非常困难，但你又确实想这样做，这里有一个方便的小技巧：直接让介质完全无法通过 UEFI 启动。你可以通过擦除所有 EFI 系统分区轻松做到这一点。（或者，如果你使用 **livecd-iso-to-disk** 从 Fedora 镜像创建 USB 启动盘，你可以省略 **--efi** 参数，这样它就不会是 UEFI 启动盘了）。如果这时你的固件仍然拒绝在 BIOS 兼容模式下启动，那就开始咒骂你的固件供应商吧（如果你还没这么做的话）。

Disk formats (MBR vs. GPT)

磁盘格式（MBR 与 GPT）

Here's another very important consideration:

还有一点非常重要：

- If you want to do a 'BIOS compatibility' type installation, you probably want to install to an MBR formatted disk.

如果要进行“BIOS 兼容性”类型的安装，最好安装到 MBR 格式的磁盘上。

- If you want to do a UEFI native installation, you probably want to install to a GPT formatted disk.

如果要进行 UEFI 原生安装，最好安装到 GPT 格式的磁盘上。

Of course, to make life complicated, many firmwares *can* boot BIOS-style from a GPT formatted disk. UEFI firmwares are in fact technically *required* to be able to boot UEFI-style from an MBR formatted disk (though we are not particularly confident that they all really can). But you really should avoid this if at all possible. This consideration is quite important, as it's one that trips up quite a few people. For instance, it's a bad idea to boot an OS installer in UEFI native mode and then attempt to install to an MBR formatted disk without reformatting it. This is very likely to fail. Most modern OS installers will automatically reformat the disk in the correct format if you allow them to completely wipe it, but if you try and tell the installer 'do a UEFI native

installation to this MBR formatted disk and don't reformat it because it has data on it that I care about', it's very likely to fail, even though this configuration is technically covered in the UEFI specification. Specifically, Windows and Fedora at least explicitly disallow this configuration.

当然，事情并非总是如此简单，许多固件可以从 GPT 格式的磁盘以 BIOS 方式启动。实际上，UEFI 固件在技术上必须能够从 MBR 格式的磁盘以 UEFI 方式启动（尽管我们并不确定所有固件都能做到这一点）。但您最好尽可能避免这种情况。这一点非常重要，因为它很容易让用户犯错。例如，在 UEFI 原生模式下启动操作系统安装程序，然后尝试在未重新格式化的情况下将其安装到 MBR 格式的磁盘上，这是一个糟糕的做法。这样做很可能会失败。大多数现代操作系统安装程序会在您允许其完全擦除磁盘的情况下自动将其重新格式化为正确的格式，但如果您尝试告诉安装程序“对这个 MBR 格式的磁盘进行 UEFI 原生安装，并且不要重新格式化它，因为它上面有我关心的数据”，则很可能会失败，即使这种配置在 UEFI 规范中理论上是允许的。具体来说，Windows 和 Fedora 至少明确禁止这种配置。

Checking the disk format

检查磁盘格式

You can use the **parted** utility to check the format of a given disk:

您可以使用 **parted** 工具来检查给定磁盘的格式：

```
[adamw@adam Downloads]$ sudo parted /dev/sda
GNU Parted 3.1
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) p
Model: ATA C300-CTFDDAC128M (scsi)
Disk /dev/sda: 128GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number   Start   End     Size    Type    File system  Flags
  1       1049kB  525MB   524MB   primary ext4          boot
  2       525MB   128GB   128GB   primary                lvm

(parted)
```

See that **Partition table: msdos**? This is an MBR/MS-DOS formatted disk. If it was GPT-formatted, that would say **gpt**. You can reformat the disk with the other type of partition table by doing **mklabel gpt** or **mklabel msdos** from within parted. **This will destroy the contents of the disk.**

看到 **Partition table: msdos** 吗？这是一个 MBR/MS-DOS 格式的磁盘。如果是 GPT 格式，则会显示为 **gpt**。您可以使用 parted 工具中的 **mklabel gpt** 或 **mklabel msdos** 命令重新格式化磁盘，使用另一种分区表格式。但这会清除磁盘上的所有数据。

With most OS installers, if you pick a disk configuration that blows away the entire contents of the target disk, the installer will automatically reformat it using the most appropriate configuration for the type of installation you're doing, but if you want to use an existing disk without reformatting it, you're going to have to check how it's formatted and take this into account.

对于大多数操作系统安装程序，如果您选择的磁盘配置会清除目标磁盘上的所有内容，安装程序会自动使用最适合您正在进行的安装类型的配置重新格式化该磁盘。但是，如果您想使用现有磁盘而不重新格式化，则必须检查其格式并考虑这一点。

Handling EFI system partition if doing manual partitioning

手动分区时如何处理 EFI 系统分区。

I can only give authoritative advice for Fedora here, but the gist may be useful for other distros / OSes.

我在这里只能提供针对 Fedora 的权威建议，但要点可能对其他发行版/操作系统有用。

If you allow Fedora to handle partitioning for you when doing a UEFI native installation - and you use a GPT-formatted disk, or allow it to reformat the disk (by deleting all existing partitions) - it will handle the EFI system partition stuff for you.

如果您允许 Fedora 在进行 UEFI 原生安装时为您处理分区，并且您使用 GPT 格式的磁盘，或者允许它重新格式化磁盘（通过删除所有现有分区），它将为您的 EFI 系统分区相关事宜。

If you use custom partitioning, though, it will expect you to provide an EFI system partition for the installer to use. If you don't do this, the installer will complain (with a somewhat confusing error message) and refuse to let you start the installation.

不过，如果您使用自定义分区，安装程序会要求您提供一个 EFI 系统分区。如果您不这样做，安装程序会发出警告（错误信息可能令人困惑），并拒绝启动安装。

So if you're doing a UEFI native install and using custom partitioning, you need to ensure that a partition of the 'EFI system partition' type is mounted at `/boot/efi` - this is where Fedora expects to find the EFI system partition it's using. If there is an existing EFI system partition on the system, just set its mount point to `/boot/efi`. If there is not an EFI system partition yet, create a partition, set its type to *EFI system partition*, make it at least 200MB big (500MB is good), and set its mount point to `/boot/efi`.

因此，如果您进行的是 UEFI 原生安装并使用自定义分区，则需要确保将类型为“EFI 系统分区”的分区挂载到 `/boot/efi` 目录——Fedora 期望在此处找到它使用的 EFI 系统分区。如果系统上已存在 EFI 系统分区，只需将其挂载点设置为 `/boot/efi` 即可。如果尚不存在 EFI 系统分区，请创建一个分区，将其类型设置为 *EFI 系统分区*，大小至少为 200MB（500MB 更佳），并将其挂载点设置为 `/boot/efi`。

A specific example 一个具体的例子

To boil down the above: if you bought a Windows 8 or later system, you *almost* certainly have a UEFI native install of Windows to a GPT-formatted disk. This means that if you want to install another OS alongside that Windows install, you almost certainly want to do a UEFI-native installation of your other OS. If you don't like all this UEFI nonsense and want to go back to the good old world you're familiar with, you will, I'm afraid, have to blow away the UEFI-native Windows installation, and it would be a good idea to reformat the disk to MBR.

总结以上内容：如果您购买的是 Windows 8 或更高版本的系统，那么您的系统中几乎肯定已经安装了 UEFI 原生 Windows 系统，并且磁盘格式为 GPT。这意味着，如果您想在 Windows 系统旁边安装另一个操作系统，那么您几乎肯定需要以 UEFI 原生方式安装该操作系统。如果您不喜欢 UEFI 的各种特性，想要回到您熟悉的传统系统，恐怕您只能删除 UEFI 原生 Windows 系统，并且最好将磁盘重新格式化为 MBR 格式。

Implications and Complications

影响和并发症

So, that's how UEFI booting works, at least a reasonable approximation. When I describe it like that, it almost all makes sense, right?

所以，这就是 UEFI 启动的工作原理，至少可以大致概括一下。我这么描述之后，是不是就都明白了？

However, all is not sweetness and light. There are problems. There always are.

然而，生活并非总是美好轻松。问题总是存在的。

Attentive readers may have noticed that I've talked about the UEFI spec *providing* a mechanism. This is accurate, and important. As the UEFI spec is a 'broad consensus' sort of thing, one of its major shortcomings (looked at from a particular perspective) is that it's nowhere near prescriptive enough.

细心的读者可能已经注意到，我曾提到 UEFI 规范*提供*了一种机制。这没错，而且很重要。由于 UEFI 规范是一种“广泛共识”的产物，因此（从特定角度来看）它的主要缺陷之一是缺乏足够的规范性。

If you read the UEFI spec critically, its basic approach is to define a set of functions that UEFI compliant firmwares must support. What it doesn't do a lot of at all is strictly requiring things to be done in any particular way, or not done in any particular way.

如果你仔细研读 UEFI 规范，就会发现它的基本方法是定义一组 UEFI 兼容固件必须支持的功能。但它很少严格规定某些功能必须以某种特定方式实现，或者必须避免以某种特定方式实现。

So: the spec says that a system firmware must do all the stuff I've described above, in order to be considered a UEFI-compliant firmware. The spec, however, doesn't talk about what operating systems 'should' or 'must' do at all, and it doesn't say that firmwares must *not* support (or no-one may expect them to support, or whatever)...anything at all. If you're making a UEFI firmware, in other words, you have to support GPT formatted disks, and FAT-formatted EFI system partitions, and you must read UEFI boot manager entries in the standard format, and you must do this and that and the other - but you can also do any *other* crap you like.

所以：规范规定，系统固件必须完成我上面描述的所有操作，才能被视为符合 UEFI 标准的固件。然而，规范完全没有提及操作系统“应该”或“必须”做什么，也没有规定固件*不能*支持（或者不应该要求支持，等等）任何东西。换句话说，如果你要开发 UEFI 固件，你必须支持 GPT 格式的磁盘和 FAT 格式的 EFI 系统分区，你必须以标准格式读取 UEFI 启动管理器条目，你必须做这个做那个等等——但你也可以做*任何*你想做的事情。

It's pretty easy to read certain implications from the spec - it carefully sets up this nice mechanism for handling OS (or other 'bootable thing') selection at the firmware level, for instance, with the clear implication "hey, it'd be great if all OSes were written to this mechanism". But the UEFI spec doesn't *require* that, and neither does any other widely-respected specification.

从规范中很容易看出一些含义——例如，它精心设计了一种在固件层面处理操作系统（或其他“可启动项”）选择的机制，其明显的含义是“如果所有操作系统都采用这种机制就太好了”。但 UEFI 规范并没有*强制*要求这样做，其他任何广受认可的规范也没有。

So, what happens in the real world is that we wind up with really dumb crap. Apple, for instance, ships at least *some* Macs with their bootloaders in an HFS+ partition. The spec says a UEFI-compliant firmware must support UEFI FAT partitions with the specific GPT partition type that identifies them as an "EFI system partition", but it doesn't say the firmware can't *also* recognize some other filesystem type and load a bootloader from that. (Whether you consider such a partition to be an "EFI system partition" or not is an interesting philosophical conundrum, but let's skate right over that for now).

所以，现实世界中我们最终得到的往往是一些非常愚蠢的东西。例如，苹果公司至少在某些 Mac 电脑中，引导加载程序位于 HFS+ 分区中。规范规定，符合 UEFI 标准的固件必须支持 UEFI FAT 分区，并支持特定的 GPT 分区类型，以将其标识为“EFI 系统分区”。但规范并没有规定固件不能识别其他文件系统类型并从中加载引导加载程序。（你是否认为这样的分区是“EFI 系统分区”是一个有趣的哲学难题，但我们暂且略过这个问题。）

The world would pretty clearly be a better place if everyone just damn well used the EFI system partition format the spec goes to such great pains to define, but Apple is Apple and we can't have nice things, so Apple went right ahead and wrote firmwares that *also* can read and load code from HFS+ partitions, and now everyone else has to deal with that or tell Macs to go and get boned. Apple also goes quite a long way beyond the spec in its boot process design, and if you want your alternative OS to show up on its graphical boot menu with a nice icon and things, you have to do more than what the UEFI spec would suggest.

如果大家都能乖乖使用 EFI 系统分区格式（毕竟 EFI 规范费尽心思地定义了这种格式），世界无疑会变得更美好。但苹果就是苹果，我们总是事与愿违。于是，苹果直接编写了能够读取和加载 HFS+ 分区代码的固件，现在其他厂商要么得忍受这种不兼容，要么就只能眼睁睁地看着 Mac 被淘汰。苹果在启动过程的设计上也远远超出了规范的要求。如果你想让你的第三方操作系统出现在图形化的启动菜单中，并显示漂亮的图标，那么你必须做的远不止 UEFI 规范所建议的那些。

There are various similar incredibly annoying corner cases we've come across, but let's not go into them all right now. This post is long enough.

我们还遇到过各种类似且极其恼人的特殊情况，但这里就不一一赘述了。这篇文章已经够长了。

Also, as we noted earlier, the spec makes no requirements as to how the mechanism should be represented to the user. So if a couple of software companies write OSes to behave 'nicely' according to the conventions the spec is clearly designed to back, and install EFI boot loaders and define EFI boot manager entries with nice clear names - like, oh say, "Fedora" and "Windows" - they are implicitly relying on the firmware to then give the user *some* kind of sane interface *somewhere* relatively discoverable that lets them choose to boot "Windows" or "Fedora". The more firmwares don't do a good job of this, the less willing OS engineers will be to rely on the 'proper' conventions, and the more likely they'll be to start rebuilding ugly hacks above the firmware level.

此外，正如我们之前提到的，该规范并未对如何向用户呈现该机制做出任何要求。因此，如果几家软件公司编写的操作系统能够“良好地”遵循该规范明确支持的约定，并安装 EFI 引导加载程序，以及使用清晰易懂的名称（例如“Fedora”和“Windows”）定义 EFI 引导管理器条目，那么它们实际上依赖于固件，希望固件能够为用户提供一个相对容易找到的合理界面，让用户选择启动“Windows”或“Fedora”。固件在这方面做得越差，操作系统工程师就越不愿意依赖“正确”的约定，也就越有可能在固件层之上构建丑陋的临时解决方案。

To be fair, we could do somewhat more at the OS level. We could present all those neat efibootmgr capabilities rather more obviously - we can use that 'don't respect BootOrder on the next boot, but instead boot *this*' capability, for instance, and have 'Reboot to Windows' as an option. It'd be kinda nice if someone looked at exposing all this functionality somewhere more obvious than efibootmgr. Windows 8 systems do use this, to some extent - you can reboot your system to the firmware UI from the Windows 8 settings menus, for instance. But still.

平心而论，我们在操作系统层面还可以做得更多。我们可以更清晰地展示 efibootmgr 的所有强大功能——例如，我们可以利用“下次启动时不遵循启动顺序，而是启动这个程序”的功能，并将“重启到 Windows”作为一个选项。如果有人能考虑将所有这些功能以比 efibootmgr 更直观的方式呈现出来，那就太好了。Windows 8 系统在某种程度上确实使用了这些功能——例如，您可以从 Windows 8 设置菜单重启系统进入固件界面。但即便如此，我们仍然需要改进。

All this is really incredibly frustrating, because UEFI is so *close* to making things really a lot better. The BIOS approach doesn't provide any kind of convention or standard for multibooting at all - it *has* to be handled entirely above the firmware level. We (the industry) could have come up with some sort of convention for handling multiboot, but we never did, so it just became a multiple-decade epic fail, where each operating system came up with its own approach and lots of people wrote their own bootloaders which tried to subsume all the operating systems and all the operating systems and independent bootloaders merrily fought like cats in a sack. I mean, pre-UEFI multibooting is such a clusterf**k it's not even worth going into, it's broken sixteen ways from Sunday by definition.

这一切真的令人无比沮丧，因为 UEFI 本来离真正改善现状只差一步之遥。BIOS 方案完全没有提供任何多启动的约定或标准——它必须完全在固件层之上处理。我们（业界）本可以制定某种处理多启动的约定，但我们从未这样做，所以它就变成了一场持续数十年的惨败，每个操作系统都提出了自己的方案，许多人编写了自己的引导加载程序，试图兼容所有操作系统，而所有操作系统和独立的引导加载程序则像一团乱麻一样互相争斗。我的意思是，UEFI 之前的多启动简直是一团糟，根本不值得一提，从定义上来说，它就漏洞百出。

If UEFI - or a spec built on top of it - had just *mandated* that everybody follow the conventions UEFI carefully establishes, and *mandated* that firmwares provide a sensible user interface, the win would have been epic. But it doesn't, so it's entirely possible that in a UEFI world things will be *even worse than they were in the BIOS world*. If many more firmwares show up that don't present a good UI for the UEFI boot manager mechanism, what could happen is that OS vendors give up on the UEFI boot manager mechanism (or decide to support it *and* alternatives, because choice!) and just reinvent the entire goddamn

nightmare of BIOS multibooting *on top of UEFI* - and we'll all have to deal with all of that, *plus* the added complication of the UEFI boot manager layer. You'll have multiple bootloaders fighting to load multiple operating systems all on top of the whole UEFI boot manager mechanism which is just throwing a whole bunch of other variables into the equation.

如果 UEFI（或者基于 UEFI 构建的规范）能够强制所有厂商遵循其精心制定的约定，并强制固件提供合理的用户界面，那将是巨大的成功。但事实并非如此，因此在 UEFI 时代，情况很可能比 BIOS 时代更糟。如果出现更多无法为 UEFI 启动管理器机制提供良好用户界面的固件，那么操作系统厂商可能会放弃 UEFI 启动管理器机制（或者决定同时支持它和其他替代方案，毕竟选择多多！），转而在 UEFI 之上重新发明 BIOS 多启动这个令人头疼的噩梦——而我们所有人都将不得不面对这一切，再加上 UEFI 启动管理器层带来的额外复杂性。届时，多个引导加载程序将争相加载多个操作系统，而这一切都建立在 UEFI 启动管理器机制之上，这无疑会引入更多其他变量。

This is not a prospect filling the mind of anyone who's had to think about it with joy.

任何不得不考虑这件事的人，都不会为此感到高兴。

Still, it's important to recognize that the sins of UEFI in this area are sins of omission - they are not sins of commission, and they're not really the result of evil intent on anyone's part. The entity you should really be angry with if you have an idiotic system firmware that doesn't give you good access to the UEFI boot manager mechanism is not the UEFI forum, or Microsoft, and it *certainly* isn't Fedora and even more *certainly* isn't me ;). The entity you should be angry at is your system/motherboard manufacturer and the goddamn incompetents they hired to write the firmware, because the UEFI spec makes it really damn clear to anyone with two brain cells to rub together that it would be a very good idea to provide some kind of useful user interface to the UEFI boot manager, and any firmware which doesn't do so is crap code by definition. Yes, the UEFI forum should've realized that firmware engineers couldn't code their way out of a goddamned paper bag and just ordered them to do so, but still, it's ultimately the firmware engineers who should be lined up against the nearest wall.

不过，我们必须认识到，UEFI 在这方面的缺陷是疏忽之罪——并非故意为之，也并非任何人的恶意所致。如果你的系统固件愚蠢到无法让你很好地访问 UEFI 启动管理器机制，那么你真正应该感到愤怒的并非 UEFI 论坛、微软，当然也不是 Fedora，更不是我 ;)。你应该愤怒的对象是你的系统/主板制造商以及他们雇佣的那些该死的、不称职的固件编写者，因为 UEFI 规范已经非常明确地告诉任何一个稍有常识的人，为 UEFI 启动管理器提供某种有用的用户界面是非常明智的做法，任何不这样做的固件，从定义上来说，都是垃圾代码。是的，UEFI 论坛应该意识到固件工程师连个像样的代码都写不出来，然后命令他们去写，但归根结底，真正应该被罚站的还是固件工程师。

Wait, we can simplify that. "Any firmware is crap code". Usually pretty accurate.

等等，我们可以简化一下。“任何固件都是垃圾代码”。通常来说，这话相当准确。

Secure Boot 安全启动

So now we come, finally, to Secure Boot.

现在，我们终于要讲到安全启动了。

Secure Boot is not magic. It's not complicated. OK, that's a lie, it's incredibly complicated, but the *theory* isn't very complicated. And no, Secure Boot itself is not evil. I am entirely comfortable stating this as a fact, and you should be too, unless you think GPG is evil.

安全启动并非魔法，也不复杂。好吧，其实它非常复杂，但其理论并不难懂。而且，安全启动本身也并非邪恶。我可以很肯定地说这一点，你也应该如此，除非你认为 GPG 是邪恶的。

Secure Boot is defined in chapter 28 of the UEFI spec (2.4a, anyway). It's actually a pretty clever mechanism. But what it does can be described very, very simply. It says that the firmware can contain a set of signatures, and refuse to run any EFI executable which is not signed with one of those signatures.

安全启动在 UEFI 规范第 28 章（至少在 2.4a 版本中）有定义。它实际上是一种相当巧妙的机制。但它的工作原理可以非常简单地描述：固件可以包含一组签名，并拒绝运行任何未使用这些签名之一签名的 EFI 可执行文件。

That's it. Well, no, it really isn't, but that's a reasonably acceptable simplification. Security is hard, so there are all kinds of wibbly bits to implementing a really secure bootchain using Secure Boot, and mjb59 can tell you all about them, or you can pour another large shot of gin and read the whole of chapter 28. But that's the basic idea.

就是这样。嗯，其实远不止如此，但这算是一种比较合理的简化。安全问题很复杂，所以要用安全启动（Secure Boot）实现一个真正安全的启动链，其中有很多棘手的细节，mjb59 可以详细讲解，或者你也可以再倒一大杯杜松子酒，然后读完第 28 章。但这就是基本思路。

Using public key cryptography to verify the integrity of something is hardly a radical or evil concept. Pretty much all Linux distributions depend on it - we sign our packages and have our package managers go AWOOGA AWOOGA if you try to install a package which isn't signed with one of our keys. This isn't us being evil, and I don't think anyone's ever accused an OS of being evil for using public key cryptographic signing to establish trust in this way.

Secure Boot is literally this exact same perfectly widely accepted mechanism, applied to the boot chain. Yet because a bunch of journalists *wildly* grasped the wrong end of the stick, it's widely considered to be slightly more evil than Hitler.

使用公钥加密技术来验证文件的完整性，这算不上什么激进或邪恶的概念。几乎所有 Linux 发行版都依赖于此——我们对软件包进行签名，如果尝试安装未经我们密钥签名的软件包，我们的软件包管理器就会发出警告。这并非出于邪恶，而且我认为也从未有人指责操作系统使用公钥加密签名来建立信任是邪恶的。安全启动实际上就是将这种被广泛接受的机制应用于启动链。然而，仅仅因为一些记者曲解了其含义，它就被普遍认为比希特勒还要邪恶。

Secure Boot, as defined in the UEFI spec, says nothing at all about what the keys the firmware trusts should be, or where they should come from. I'm not going to go into all the goddamn details, because it gets stultifyingly boring and this post is too long already. But the executive summary is that the spec is utterly and entirely about defining a *mechanism* for doing cryptographic verification of a boot chain. It does not really even consider any kind of icky questions about the *policy* for doing so. It does nothing evil. It is as flexible as it could reasonably be, and takes care to allow for all the mechanisms involved to be configurable at multiple levels. The word 'Microsoft' is not mentioned. It is not in any way, shape, or form a secret agenda for Microsoft's domination of the world. If you doubt this, at the very bloody least, go and *read* it. I've given you all the necessary pointers. There is literally not a single legitimate reason I can think of for anyone to be *angry* with the idea "hey, it'd be neat if there was a mechanism for optional cryptographic verification of bootloader code in this firmware specification". None. Not one.

UEFI 规范中定义的安全启动完全没有提及固件信任的密钥应该是什么，或者它们应该来自哪里。我不想赘述所有细节，因为那实在太枯燥乏味了，而且这篇文章已经够长了。但简而言之，该规范完全是为了定义一种对启动链进行加密验证的 *机制*。它甚至没有考虑任何关于验证 *策略* 的棘手问题。它没有任何恶意。它尽可能地灵活，并且确保所有相关机制都可以在多个层面上进行配置。规范中没有提到“微软”这个词。它绝不是微软统治世界的秘密计划。如果你对此有所怀疑，至少去 *读读* 规范吧。我已经提供了所有必要的提示。我实在想不出任何正当理由，会有人对“如果固件规范中加入一个可选的引导加载程序代码加密验证机制就好了”这个想法感到 *愤怒*。一个也没有。

Secure Boot in the real world

现实世界中的安全启动

Most of the unhappiness about Secure Boot is not really about Secure Boot the mechanism - whether the people expressing that unhappiness *think* it is or not - but about specific implementations of Secure Boot in the real world.

大多数对安全启动的不满实际上并非针对安全启动机制本身——无论表达不满的人是否这么认为——而是针对安全启动在现实世界中的具体实现方式。

The only one we really care about is Secure Boot as it's implemented on PCs shipped with Microsoft Windows 8 or higher pre-installed.

我们真正关心的只有安全启动，因为它已在预装了 Microsoft Windows 8 或更高版本的 PC 上实现。

Microsoft has these things called the [Windows Hardware Certification Requirements](#). There they are. They are not Top Secret, Eyes Only, You Will Be Fed To Bill Gates' Sharks After Reading - they're right there on the Internet for anyone to read.

微软有一套叫做“[Windows 硬件认证要求](#)”的东西。就在那里。它们并非什么绝密文件，不是什么只有少数人才能看的“秘密武器”，也不是那种看完就会被喂给比尔·盖茨的投资人的“鲨鱼”——它们就公开在互联网上，任何人都可以查阅。

If you want to get cheap volume licenses of Windows from Microsoft to pre-install on your computers and have a nice "reassuring" 'Microsoft Approved!' sticker or whatever on the case, you have to comply with these requirements. That's all the force they have: they are not *actually* a part of the law of the United States or any other country, whatever some people seem to believe. Bill Gates cannot feed you to his sharks if you sell a PC that doesn't comply with these requirements, so long as you don't want a cheap copy of Windows to pre-install and a nice sticker. There is literally no requirement for a PC sold *outside* the Microsoft licensing program to configure Secure Boot in any particular way, or include Secure Boot at all. A PC that claims to have a UEFI 2.2 or later compliant firmware must implement Secure Boot, but can ship with it configured in literally absolutely any way it pleases (including turned off).

如果你想从微软获得廉价的 Windows 批量许可，预装在你的电脑上，并在机箱上贴上一个醒目的“微软认证！”标签或其他类似的标志，你就必须遵守这些要求。这就是他们所能施加的全部约束：这些要求实际上并非美国或其他任何国家的法律，尽管有些人似乎这么认为。只要你不想要预装廉价 Windows 和漂亮的标签，比尔·盖茨就无法因为你出售不符合这些要求的电脑而把你喂鲨鱼。对于在微软许可计划之外销售的电脑，没有任何规定要求必须以任何特定方式配置安全启动，或者必须包含安全启动。声称拥有符合 UEFI 2.2 或更高版本固件的电脑必须实现安全启动，但可以以任何方式配置（包括关闭）进行发货。

If you're going to have very loud opinions about Secure Boot, you have zero excuse for not going and reading the Microsoft certification requirements. Right now. I'll wait. You can search for "Secure Boot" to get to the relevant bit. It starts at "System.Fundamentals.Firmware.UEFISecureBoot".

如果你对安全启动有非常强烈的看法，那就没有任何理由不去阅读微软的认证要求。现在就去。我等你。你可以搜索“安全启动”找到相关内容。它从“System.Fundamentals.Firmware.UEFISecureBoot”开始。

You should read it. But here is a summary of what it says.

你应该读读这篇文章。不过，这里我先给你总结一下它的内容。

Computers complying with the requirements must:

符合要求的计算机必须：

- Ship with Secure Boot turned on (except for servers)
出厂时应启用安全启动（服务器除外）
- Have Microsoft's key in the list of keys they trust
将微软的密钥添加到他们信任的密钥列表中。
- Disable BIOS compatibility mode when Secure Boot is enabled (actually the UEFI spec requires this too, if I read it correctly)
启用安全启动时禁用 BIOS 兼容模式（实际上，如果我理解正确的话，UEFI 规范也要求这样做）。
- Support signature blacklisting
支持签名黑名单

x86 computers complying with the requirements must additionally:

符合要求的 **x86 计算机**还必须：

- **Allow** a physically present person to disable Secure Boot
允许现场人员禁用安全启动
- **Allow** a physically present person to enable Custom Mode, and modify the list of keys the firmware trusts
允许现场人员启用自定义模式，并修改固件信任的密钥列表。

ARM computers complying with the requirements must additionally:

符合要求的 **ARM 计算机**还必须：

- **NOT allow** a physically present person to disable Secure Boot
禁止允许在场人员禁用安全启动。
- **NOT allow** a physically present person to enable Custom Mode, and modify the list of keys the firmware trusts
禁止允许在场人员启用自定义模式并修改固件信任的密钥列表。

Yes. You read that correctly. The Microsoft certification requirements, for x86 machines, *explicitly require implementers to give a physically present user complete control over Secure Boot* - turn it off, or completely control the list of keys it trusts. Another important note here is that while the certification requirements state that the out-of-the-box list of trusted keys must *include* Microsoft's key, they don't say, for e.g., that it must *not include* any other keys. The requirements explicitly and intentionally allow for the system to ship with any number of other trusted keys, too.

是的，你没看错。微软针对 x86 架构机器的认证要求**明确规定，实现者必须让实际在场的用户完全控制安全启动**——可以将其关闭，也可以完全控制其信任的密钥列表。这里还需要特别注意的是，虽然认证要求规定默认的受信任密钥列表必须**包含**微软的密钥，但并没有规定**不能包含**其他密钥。这些要求明确且有意地允许系统附带任意数量的其他受信任密钥。

These requirements aren't present entirely out of the goodness of Microsoft's heart, or anything - they're present in large part because other people explained to Microsoft that if they weren't present, it'd have a hell of a lawsuit on its hands² - but they are present. Anyone who actually understands UEFI and Secure Boot cannot possibly read the requirements any other way, they are extremely clear and unambiguous. They both clearly *intend to* and *succeed in* ensuring the owner of a certified system has complete control over Secure Boot.

这些要求并非完全出于微软的善意——它们之所以存在，很大程度上是因为其他人向微软解释说，如果没有这些要求，微软将会面临一场棘手的诉讼²——但它们确实存在。任何真正了解 UEFI 和安全启动的人都不可能以其他方式解读这些要求，它们非常清晰明确。它们明确**旨在**确保认证系统的所有者对安全启动拥有完全的控制权，并且也**确实**做到了这一点。

If you have an x86 system that claims to be Windows certified but does not allow you to disable Secure Boot, it is in direct violation of the certification requirements, and you should certainly complain very loudly to someone. If a lot of these systems exist then we clearly have a problem and it might be time for that giant lawsuit, but so far I'm not aware of this being the case. All the x86-based, Windows-certified systems I've seen *have* had the 'disable

Secure Boot' option in their firmwares.

如果你的 x86 系统声称通过了 Windows 认证，但却不允许用户禁用安全启动，这直接违反了认证要求，你绝对应该强烈抗议。如果这类系统数量众多，那显然是个大问题，或许是时候提起巨额诉讼了，但就我目前所知，这种情况并不常见。我见过的所有基于 x86 架构且通过 Windows 认证的系统，其固件中 *都* 提供了“禁用安全启动”的选项。

Now, for ARM machines, the requirements are significantly more evil: they state exactly the opposite, that it must *not* be possible to disable Secure Boot and it must *not* be possible for the system owner to change the trusted keys. This is bad and wrong. It makes Microsoft-certified ARM systems into a closed shop. But it's worth noting it's no *more* bad or wrong than most other major ARM platforms. Apple locks down the bootloader on all iDevices, and most Android devices also ship with locked bootloaders.

现在，对于 ARM 架构的机器来说，要求则更加苛刻：它们恰恰相反，规定 *不能* 禁用安全启动，系统所有者也 *不能* 更改受信任的密钥。这很糟糕，也很错误。它将微软认证的 ARM 系统变成了一个封闭的圈子。但值得注意的是，这与其他大多数主流 ARM 平台相比，并没有什么特别之处。苹果公司锁定了所有 iDevice 设备的引导加载程序，大多数安卓设备也预装了锁定的引导加载程序。

If you're planning to buy a Microsoft-certified ARM device, be aware of this, and be aware that you will not be in control of what you can boot on it. If you don't like this, don't buy one. But also don't buy an iDevice, or an Android device with a locked bootloader (you can buy Android devices with unlocked or unlockable bootloaders, still, but you have to do your research).

如果您打算购买微软认证的 ARM 设备，请注意这一点：您将无法控制设备上可以启动的系统。如果您无法接受这一点，请不要购买。但也不要购买 iOS 设备或引导程序锁定的安卓设备（您仍然可以购买引导程序已解锁或可解锁的安卓设备，但您需要事先做好调查）。

As far as x86 devices go, though, right now, Microsoft's certification requirements actually *explicitly protect* your right to determine what can boot on your system. This is good.

就 x86 设备而言，目前微软的认证要求实际上 *明确保护* 了您决定系统启动项的权利。这很好。

Recommendations 建议

The following are AdamW's General Recommendations On Managing System Boot, offered with absolutely no guarantees of accuracy, purity or safety.

以下是 AdamW 关于管理系统启动的一般建议，但绝对不保证其准确性、完整性或安全性。

- **If you can possibly manage it, have one OS per computer.** If you need more than one OS, buy more computers, or use virtualization. If you can do this everything is very simple and it doesn't much matter if you have BIOS or UEFI firmware, or use UEFI-native or BIOS-compatible boot on a UEFI system. Everything will be nice and easy and work. You will whistle as you work, and be kind to children and small animals. All will be sweetness and light. Really, do this.

如果条件允许，每台电脑只安装一个操作系统。 如果需要多个操作系统，那就多买几台电脑，或者使用虚拟化技术。如果能做到这一点，一切都会变得非常简单，BIOS 固件还是 UEFI 固件，或者在 UEFI 系统上使用 UEFI 原生启动还是 BIOS 兼容启动，都无关紧要。一切都会轻松愉快，运行良好。你会一边工作一边哼着小曲，还会善待孩子和小动物。一切都会充满美好和光明。真的，试试吧。

- **If you absolutely must have more than one OS per computer, at least have one OS *per disk*.** If you're reasonably comfortable with how BIOS-style booting works and you don't think you need Secure Boot, it's pretty reasonable to use BIOS-compatible booting rather than UEFI-style booting in this situation on a UEFI-capable system. You'll probably have less pain to deal with and you won't really lose anything. With one OS per disk you can also mix UEFI-native and BIOS-compatible installations.

如果每台电脑必须安装多个操作系统，至少每个硬盘只安装一个操作系统。 如果您对 BIOS 启动方式比较熟悉，并且认为不需要安全启动，那么在支持 UEFI 的系统上，使用 BIOS 兼容的启动方式而不是 UEFI 启动方式是相当合理的。这样处理起来可能更省事，而且也不会损失什么。每个硬盘只安装一个操作系统，您还可以混合安装 UEFI 原生系统和 BIOS 兼容系统。

- **If you absolutely insist on having more than one OS per disk,** understand *everything* written on this page, understand that you are making your life much more painful than it needs to be, lay in good stocks of painkillers and gin, and don't go yelling at your OS vendor, whatever breaks. Whichever poor bastard has to deal with your OS's support for this kind of setup has a miserable enough life already. And for the love of *cookies*, don't mix UEFI-native and BIOS-compatible OS installations, you have enough pain to deal with already.

如果你非要坚持每块硬盘装多个操作系统， 请务必理解本页 *所有* 内容，明白你这样做只会让自己的生活变得比原本更痛苦，所以最好多备些止痛药和杜松子酒，而且无论系统出了什么问题，都不要去找操作系统厂商理论。负责处理这种配置的倒霉蛋已经够惨了。还有，看在老天的份上，千万别混装 UEFI 原生系统和 BIOS 兼容系统，你已经够倒霉的了。

- If you're using UEFI native booting, and you don't tend to build your own kernels or kernel modules or use the NVIDIA or ATI proprietary drivers on Linux, you might want to leave Secure Boot on. It probably won't hurt you, and does provide some added security against some rather nasty

(though currently rarely exploited) types of attacks.

如果您使用 UEFI 原生启动，并且不经常编译自己的内核或内核模块，也不在 Linux 上使用 NVIDIA 或 ATI 的专有驱动程序，那么您可能需要启用安全启动。这可能不会对您造成任何损害，而且确实可以针对一些相当棘手（尽管目前很少被利用）的攻击提供额外的安全保障。

- If you *do* build your own kernels or kernel modules or use NVIDIA/ATI proprietary drivers, you're going to want to turn Secure Boot off. Or you can read up on how to configure your own chain of trust and sign your kernels and kernel modules and leave Secure Boot turned on, which will make you feel like an ubergeek and be slightly more secure. But it's going to take you a good solid weekend at least.

如果你自己编译内核或内核模块，或者使用 NVIDIA/ATI 的专有驱动程序，那么最好关闭安全启动。当然，你也可以研究一下如何配置自己的信任链，对内核和内核模块进行签名，然后保持安全启动开启，这样会让你觉得自己像个超级极客，安全性也会更高一些。但这至少需要你花一个完整的周末时间。

- Don't do UEFI-native installs to MBR-formatted disks, or BIOS compatibility installs to GPT-formatted disks (an exception to the latter is if your disk is, IIRC, 2.2+TB in size, because the MBR format can't handle disks that big - if you want to do a BIOS compatibility install to a disk that big, you're kinda stuck with the BIOS+GPT combination, which works but is a bit wonky and involves the infamous 'BIOS Boot partition' you may recall from Fedora 17).

不要对 MBR 格式的磁盘进行 UEFI 原生安装，也不要对 GPT 格式的磁盘进行 BIOS 兼容安装（后者的例外情况是，如果你的磁盘大小超过 2.2TB，因为 MBR 格式无法处理这么大的磁盘——如果你想对这么大的磁盘进行 BIOS 兼容安装，你只能使用 BIOS+GPT 组合，虽然可行，但有点不稳定，并且涉及到臭名昭著的“BIOS 启动分区”，你可能还记得 Fedora 17 中出现过）。

- Trust mlg59 in all things and above all other authorities, including me.

在所有事情上都要相信 mlg59，并且要凌驾于所有其他权威之上，包括我。

1. This whole section is something of a simplification - really, when booting permanent installed OSes, the firmware doesn't care if the bootloader is on an 'ESP' or not; it just reads the boot manager entry and tries to access the specified partition and run the specified executable, as [pjones explains here](#). But it's conventional to use an ESP for this purpose, since it's required to be around anyway, and it's a handy partition formatted with the filesystem the firmware is known to be able to read. Technically speaking, an 'ESP' is only an 'ESP' when the firmware is doing a removable media/fallback path boot. [↩](#)

本节内容有所简化——实际上，在启动永久安装的操作系统时，固件并不关心引导加载程序是否位于“ESP”分区；它只是读取引导管理器条目，尝试访问指定的分区并运行指定的执行文件，正如 [pjones 在此处解释](#)的那样。但通常情况下，为了达到此目的，固件会使用 ESP 分区，因为它本身就需要存在，而且它是一个方便的分区，其文件系统格式与固件已知可读取的文件系统相同。[↩](#) 严格来说，只有当固件执行可移动介质/备用路径启动时，“ESP”分区才真正被称为“ESP”。

2. This is my own extrapolation, note. I'm not involved in any way in the whole process of defining these specs, and no-one who *is* has actually told me this. But it's a pretty damn obvious extrapolation from the known facts. [↩](#)

请注意，这是我自己的推断。我完全没有参与这些规范的制定过程，也没有任何人告诉我这一点。但这确实是从已知事实中显而易见的推断。[↩](#)

Mandriva Personal Red Hat Technical

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Comments 评论

kldixon wrote on 2014-01-26 10:54:

kldixon 于 2014-01-26 10:54 写道:

"You do not have a 'UEFI BIOS'. No-one has a 'UEFI BIOS'. Please don't ever say 'UEFI BIOS'."
<http://www.intel.com/content/www/us/en/motherboards/desktop-motherboards/desktop-boards-software-visual-bios.html>
<https://sites.google.com/site/visualbios/> nuff said.

“你没有‘UEFI BIOS’。没有人有‘UEFI BIOS’。请永远不要再说‘UEFI BIOS’。”
<http://www.intel.com/content/www/us/en/motherboards/desktop-motherboards/desktop-boards-software-visual-bios.html>
<https://sites.google.com/site/visualbios/> 无需多言。

[adamw](#) wrote on 2014-01-26 17:14:

[adamw](#) 于 2014-01-26 17:14 写道:

What, like Intel can't be wrong? Calling it a UEFI BIOS is incoherent by definition. That's just *not what it is*. I don't care who does it. :)

什么？难道英特尔就一定不会犯错吗？把它叫做 UEFI BIOS 本身就是自相矛盾的。它根本就不是 UEFI BIOS。我不在乎是谁做的。:)

[Rod Smith](#) wrote on 2014-02-16 03:27:

[Rod Smith](#) 于 2014 年 2 月 16 日 03:27 写道:

I'm with Adam on this one (and with almost all of what he wrote on this page, for that matter). If we're to do a battle of the links, try this one: <https://en.wikipedia.org/wiki/Uefi> To quote: "UEFI is meant to replace the Basic Input/Output System (BIOS) firmware interface" ...or, from the UEFI Forum (<http://www.uefi.org/faq>): "BIOS is typically used to refer to an Intel® Architecture firmware implementation rooted in the IBM PC design. Based on older standards and methods, BIOS was originally coded in 16-bit real mode x86 assembly code and remained substantially unchanged until its recent decline in use. By contrast, UEFI standards reflect the past 30 years of PC evolution by describing an abstract interface set for transferring control to an operating system or building modular firmware from one or more silicon and firmware suppliers. The abstractions of UEFI Forum specifications are designed to decouple development of producer and consumer code, allowing each to innovate more independently and with faster time-to-market for both. UEFI also overcame the hardware scaling limitations that the IBM PC design assumed, allowing its broad deployment across high-end enterprise servers to the embedded devices. UEFI is “processor architecture-agnostic,” supporting x86, x64, ARM and Itanium." I suspect that manufacturers use "BIOS" because the general public is familiar with the acronym and wouldn't know what's meant by "EFI" or "UEFI." Unfortunately, I believe that this choice is doing a lot of harm -- people think that their new computers' "BIOSes" are basically the same as they've always been, just with a new "UEFI feature" that might be something like AHCI support or USB keyboard support in previous BIOS iterations. In other words, the "BIOS" terminology masks the fact that EFI is fundamentally different from BIOS, as Adam has gone to great lengths to explain on this page.

在这一点上，我同意亚当的观点（事实上，我也同意他在这页上写的几乎所有内容）。如果要进行链接比拼，不妨试试这个：<https://en.wikipedia.org/wiki/Uefi> 引用原文：“UEFI 旨在取代基本输入/输出系统 (BIOS) 固件接口”……或者，摘自 UEFI 论坛 (<http://www.uefi.org/faq>): “BIOS 通常指的是基于 IBM PC 设计的 Intel® 架构固件实现。基于较早的标准和方法，BIOS 最初是用 16 位实模式 x86 汇编代码编写的，并且在近期使用率下降之前基本保持不变。相比之下，UEFI 标准反映了过去 30 年 PC 的发展历程，它描述了一个抽象接口集，用于将控制权移交给操作系统，或从一个或多个芯片和固件供应商构建模块化固件。UEFI 论坛规范的抽象旨在解耦生产者和消费者代码的开发，使双方都能更独立地进行创新，并加快产品上市速度。UEFI 还克服了 IBM PC 设计中存在的硬件扩展性限制，使其能够在高端企业服务器上广泛部署。”对于嵌入式设备而言，UEFI 是“处理器架构无关的”，支持 x86、x64、ARM 和 Itanium 架构。我怀疑制造商使用“BIOS”是因为大众熟悉这个缩写，而不知道“EFI”或“UEFI”是什么意思。不幸的是，我认为这种做法弊大于利——人们会认为新电脑的“BIOS”与以往的 BIOS 基本相同，只是增加了一些新的“UEFI 功能”，例如之前 BIOS 版本中提供的 AHCI 支持或 USB 键盘支持。换句话说，“BIOS”这个术语掩盖了 EFI 与 BIOS 本质上不同的事实，Adam 在本页中对此进行了详尽的解释。

[adamw](#) wrote on 2014-02-16 03:50:

[adamw](#) 于 2014-02-16 03:50 写道:

Wow, thanks a lot for nailing *exactly* the problem with calling it a BIOS, far more clearly than I did! That's precisely it: calling it a 'BIOS' is yet another 'confusion vector'. I mean, if you think of your UEFI firmware as 'the BIOS', how exactly is your brain going to process the concept that your BIOS has a BIOS compatibility mode? It's not going to end well. :)

哇，非常感谢你一针见血地指出了把它叫做 BIOS 的问题所在，比我解释得清楚多了！没错，就是这么回事：把它叫做“BIOS”又会造成另一种“混淆”。我的意思是，如果你把 UEFI 固件当作“BIOS”，你的大脑要怎么理解 BIOS 也有 BIOS 兼容模式这个概念呢？肯定不会有好结果。:)

[org](#) wrote on 2014-12-11 09:20:

[org](#) 于 2014-12-11 09:20 写道:

Какая-то жеванина. Когда читаешь, создается впечатление что автор хочет сказать "ты всё равно нихуя не поймешь".

这本书简直就是垃圾。读完之后，你会觉得作者想说的是：“你还是什么都看不懂。”

Neil Darlow wrote on 2014-01-26 11:15:

Neil Darlow 于 2014 年 1 月 26 日 11:15 写道:

Hi Adam, I first came across UEFI in November 2010 when I decided to build a new home server. I purchased an ASRock E350-M1 EPIA motherboard and was oblivious to the fact it had UEFI firmware. Back at that time there were few GNU/Linux distributions that could actually complete a native UEFI install. Both Fedora and OpenSUSE could boot DVD media in UEFI mode but the installations always failed. I actually ended-up using ArchLinux to manage the UEFI components and ran everything else in KVM virtual machines under that. Of course things are much different now and UEFI distribution installs are often successful. I was just pleased to note that, after reading this excellent tome, that I had somehow managed to implement UEFI boot corectly using my manual process. Actually there were a cartload of other technologies I implemented at the same time (e.g. unified login using OpenLDAP and winbind etc.) which have all become mainstream after I did them the hard way. The one thing I have learned from this experience, which you pointed-out, is to always use the definitive reference material for a particular technology. Guides and HOWTOs found on the Internet are always a source of either inaccurate information or information based on experimentation and deduction rather than hard fact. Keep up the good work my friend!

Regards, Neil Darlow

嗨，Adam，我第一次接触 UEFI 是在 2010 年 11 月，当时我决定搭建一台新的家用服务器。我买了一块华擎 E350-M1 EPIA 主板，却完全不知道它带有 UEFI 固件。那时候，能够真正完成原生 UEFI 安装的 GNU/Linux 发行版寥寥无几。Fedora 和 OpenSUSE 虽然都能以 UEFI 模式从 DVD 启动，但安装总是失败。最后，我干脆用 ArchLinux 来管理 UEFI 组件，其他所有程序都运行在 Arch Linux 下的 KVM 虚拟机里。当然，现在情况大不相同了，UEFI 发行版的安装通常都能成功。读完这本好书后，我很高兴地发现，我竟然成功地通过手动方式实现了 UEFI 启动。实际上，当时我还同时实现了一大堆其他技术（比如使用 OpenLDAP 和 winbind 实现统一登录等等），这些技术在我费尽周折完成之后，如今都已经成为主流了。正如你所指出的，我从这次经历中学到的最重要的一点是，对于特定技术，一定要使用权威的参考资料。互联网上的指南和操作教程要么信息不准确，要么是基于实验和推断而非确凿事实。我的朋友，继续努力！此致，尼尔·达洛

emmet.curran wrote on 2014-01-26 12:29:

emmet.curran 于 2014-01-26 12:29 写道：

Thanks Adam. I've had some headaches at work trying to fix broken machines running Win8 and had to wrestle with UEFI for a bit before I got the hang of it. This post makes it a bit clearer what exactly I'm wrestling with. Win8, UEFI and Secure Boot etc seem to work perfectly fine when everything is working correctly, but can become a bit of a nightmare (at the start) when you're trying to do something so simple as boot to a live cd or even safe mode when the stupid thing won't start. Most people hate Win8 for its Metro UI - I hate it because I can't get into safe mode without pleading with it first.

谢谢 Adam。我在工作中遇到了一些麻烦，因为要修复运行 Win8 的故障机器，而且在掌握 UEFI 之前也费了不少劲。这篇文章让我更清楚地了解了遇到的问题。Win8、UEFI 和安全启动等等，在一切正常的情况下似乎都运行良好，但当你尝试执行一些简单的操作，比如从 Live CD 启动，或者甚至在系统无法启动时进入安全模式时，它们就会变成一场噩梦（尤其是在启动初期）。大多数人讨厌 Win8 的 Metro UI——而我讨厌它，是因为我每次都得费尽心思才能进入安全模式。

Somebody wrote on 2014-01-26 15:05:

有人于 2014 年 1 月 26 日 15:05 写道：

So effectively what uefi is, is ramming two more levels of complicated bootloader code down into the board's soldered on flashrom, and requiring that those two complicated bootloaders follow a set of poorly described specifications. I really don't see how this makes anything easier than bios, particularly since a good SSD is a heck of a lot faster to read than a crappy flashrom. Now for my story; I have a laptop that supports some form of EFI/UEFI. Up to Fedora 17, I was able to pass the installer's kernel a parameter or two (might have been something like noefi or nogpt) to force the installer to work in legacy bios mode, and that would work. It would set up the standard partition format boot/root/home/swap, and the firmware would happily boot in bios compatibility mode. So couple of days ago, I found out that the F20 installer absolutely does not support this any more. So I tried wiping the disks and letting it do it the way it wanted to, and a very funny thing happened; it would only actually boot about 50% of the time. Completely unacceptable. So I pulled the disk and shoved it into another machine that has a most-definitely-not-[U]EFI firmware and did the install there before transferring back to the laptop. That works 100%. It would seem that Fedora is using [U]EFI to load some weird kind of shim that loads GRUB, effectively using EFI *as* a BIOS. Either use it correctly, or don't use it at all. If working as [U]EFI, grub really shouldn't exist at all.

所以，UEFI 的本质就是把两层复杂的引导加载程序代码硬塞进主板焊接的闪存芯片里，并要求这两层复杂的引导加载程序遵循一套描述不清的规范。我实在看不出它比 BIOS 有什么优势，尤其是在好的固态硬盘读取速度远超劣质闪存芯片的情况下。说说我的情况吧：我有一台支持某种 EFI/UEFI 的笔记本电脑。在 Fedora 17 之前，我可以通过给安装程序内核传递一两个参数（可能是 noefi 或 nogpt 之类的）来强制安装程序以传统 BIOS 模式运行，而且确实有效。它会设置标准的 boot/root/home/swap 分区格式，固件也能顺利地以 BIOS 兼容模式启动。但几天前，我发现 F20 安装程序完全不支持这种方式了。于是我尝试擦除磁盘，让它按照自己的方式进行操作，结果发生了一件非常奇怪的事情；它只有大约 50% 的概率能启动，完全无法接受。所以我把硬盘取出来，插到另一台绝对不是 [U]EFI 固件的机器上，在那里安装好系统后再传回笔记本电脑。这次完全成功。看来 Fedora 是利用 [U]EFI 加载某种奇怪的 shim 来加载 GRUB，实际上是把 EFI 当作 BIOS 来用。要么就正确使用，要么就干脆别用。如果真的像 [U]EFI 那样工作，GRUB 根本就不应该存在。

[adamw](#) wrote on 2014-01-26 19:13:

[adamw](#) 于 2014-01-26 19:13 写道：

"So effectively what uefi is, is ramming two more levels of complicated bootloader code down into the board's soldered on flashrom" Two more? No, it's only one level. The implementation isn't really soldered on anything - any given firmware's implementation can be adjusted / fixed with a firmware update, like any other element of the firmware. The *configuration* of the UEFI boot manager is done via NVRAM variables (that's what efibootmgr is actually twiddling). "I really don't see how this makes anything easier than bios, particularly since a good SSD is a heck of a lot faster to read than a crappy flashrom." You're not going to notice a speed difference in loading something like a couple of KiB of configuration. The reason it *potentially* makes it better than BIOS land is that the UEFI level is a much more sensible level for 'boot target selection' to take place than 'in the first few bytes of whichever disk you're booting from', a concept which comes with massively obvious layering violations. There are about fifty different ways to do boot target selection with MBR-based booting, none of which is compatible with any of the others, and all of which will happily fight with each other like cats in a sack if you don't know exactly what you're doing as you install all your OSes. "Now for my story; [...]" You're misunderstanding quite a few things there, but the headline is that 'noefi' isn't a Fedora installer parameter. It's a Linux kernel parameter. It's kind of a problematic thing to do, really, because you're essentially doing a UEFI native boot and then pretending you didn't. I'd recommend avoiding it. What you really want to do in your case is instruct your firmware to do a BIOS compatibility mode boot of your install media, not a UEFI native boot. The complications of this are discussed in the post. If your firmware absolutely does not allow you to do this, and you can't use efibootmgr or similar to do it, what you can do is write your install medium in such a way that it's not EFI bootable - doesn't contain an ESP - and then the firmware will usually 'automatically' boot it in BIOS compatibility mode when you ask the firmware to boot it. To do this with a Fedora image when writing to USB, for instance, use livecd-iso-to-disk and do *not* pass --efi. When doing a

UEFI native install, Fedora uses grub2-efi as its OS-specific bootloader. You *can* do 'direct' boot of a Linux kernel using UEFI - i.e. have your Linux kernel be an EFI executable and write a UEFI boot manager entry which points to that kernel, so you booted direct from the firmware layer to the OS kernel. (Really, the kernel contains a 'stub' EFI bootloader which does the job of loading the kernel - it's much like having grub2-efi, just that the bits are all baked into the kernel). This is the 'UEFI stubs' thing a later commenter mentions. But it's not very common to do this, and it's really not an Inarguably Better approach than having an EFI bootloader between the firmware and the OS kernel.

“所以，UEFI 的本质就是把两层复杂的引导加载程序代码硬塞进主板焊接的闪存 ROM 里。”多两层？不，只有一层。它的实现并没有真正焊接在任何东西上——任何固件的实现都可以像固件的其他元素一样，通过固件更新进行调整/修复。UEFI 引导管理器的*配置*是通过 NVRAM 变量完成的（这才是 efibootmgr 实际上在操作的）。“我真的不明白这比 BIOS 有什么优势，尤其是考虑到好的 SSD 读取速度比劣质闪存 ROM 快得多。”加载几 KiB 的配置数据，你根本不会注意到速度上的差别。它之所以*可能*比 BIOS 更好，是因为 UEFI 层对于“启动目标选择”来说比“在启动磁盘的前几个字节”要合理得多，后者显然存在严重的层级冲突。基于 MBR 的启动方式大约有五十种不同的启动目标选择方法，它们彼此互不兼容，如果你在安装操作系统时不清楚自己在做什么，它们就会像一团乱麻一样互相争斗。“现在来说说我的故事；[...]”你误解了很多东西，但重点是“noefi”不是 Fedora 安装程序的参数，而是 Linux 内核的参数。这样做其实挺麻烦的，因为你实际上是在执行 UEFI 原生启动，然后假装没做。我建议你避免这样做。你真正想做的是指示固件以 BIOS 兼容模式启动安装介质，而不是以 UEFI 原生模式启动。帖子中讨论了其中的复杂性。如果你的固件绝对不允许你这样做，而且你也无法使用 efibootmgr 或类似工具，那么你可以将安装介质写入非 EFI 可启动文件（不包含 ESP 分区），这样当你请求固件启动时，它通常会“自动”以 BIOS 兼容模式启动。例如，要将 Fedora 镜像写入 USB，可以使用 livecd-iso-to-disk 命令，并且不要传递 `--efi` 参数。在进行 UEFI 原生安装时，Fedora 使用 grub2-efi 作为其操作系统特定的引导加载程序。你*可以*使用 UEFI 直接启动 Linux 内核——也就是说，将你的 Linux 内核设置为 EFI 可执行文件，并写入一个指向该内核的 UEFI 启动管理器条目，这样就可以直接从固件层启动到操作系统内核。（实际上，内核包含一个“存根”EFI 引导加载程序，它负责加载内核——这很像使用 grub2-efi，只是所有组件都集成到了内核中。）这就是后来一位评论者提到的“UEFI 存根”。但这样做并不常见。而且，这实际上并不比在固件和操作系统内核之间添加 EFI 引导加载程序更好。

Somebody wrote on 2014-01-27 03:11:

有人于 2014-01-27 03:11 写道：

I'm not actually misunderstanding. Regardless of which level the parameter affected, the end result was something that *worked*. "that" of course, being past-tense. Yes, obviously I could build my own install disk, but for an end user to have to do that is really pushing too far, and in my case, far far more complicated than what I did to solve the issue. Like you've mentioned, the installer really has to be able to deal with all the corner cases of inappropriate behavior. Also, yes, I do mean two extra levels of bootloader. MBR and whatever actually does the multiboot. I am not disagreeing about there being some interesting aspects of UEFI, but it is quite a major problem that all of the implementations of it are proprietary. The less proprietary code I have to depend on, the better, even if the end result is marginally less efficient. My laptop is a perfect illustration of this.

我并没有误解。无论参数影响的是哪个层级，最终结果都是*有效*的。“有效”当然是过去式。没错，我当然可以自己制作安装盘，但最终用户这么做就太过分了，而且就我的情况而言，比我解决问题的方法要复杂得多。正如你所说，安装程序必须能够处理所有异常行为的特殊情况。另外，是的，我指的是两层额外的引导加载程序：MBR 和实际执行多重启动的程序。我并不否认 UEFI 有一些有趣的方面，但它的所有实现都是专有的，这确实是一个很大的问题。我需要依赖的专有代码越少越好，即使最终结果效率略低一些。我的笔记本电脑就是一个很好的例子。

[adamw](#) wrote on 2014-01-27 03:36:

[adamw](#) 于 2014-01-27 03:36 写道：

"Regardless of which level the parameter affected, the end result was something that *worked*. “that” of course, being past-tense." The problem with this is: <https://xkcd.com/1172/> You can find all kinds of ways of doing things that work, and are bad. Sometimes those ways stop working, and no-one wants to fix them, because they're bad. This may annoy you, but it's not wrong. If you want to do a BIOS compatibility mode install of your OS, boot its installation medium in BIOS compatibility mode. Really. That's the *right way* to do it. Your firmware should make this possible relatively easily, and if it doesn't, this post covers all kinds of things that should help you achieve it. Doing a UEFI native boot of your install media and then attempting to fool it into thinking you didn't boot in UEFI native mode is really a messy way to go about it. If noefi really isn't working for you any more that's some kind of bug somewhere, sure, but I don't develop an immediate urge to investigate that and fix it, because it's just not the mechanism you really want to be using for what you're trying to achieve.

“无论参数影响的是哪个层级，最终结果都是*有效*的。当然，这里的“有效”指的是过去式。”问题在于：<https://xkcd.com/1172/> 你可以找到各种各样有效的方法，以及各种糟糕的方法。有时这些方法会失效，但没人愿意修复它们，因为它们本身就不好。这可能会让你恼火，但并没有错。如果你想以 BIOS 兼容模式安装操作系统，那就用 BIOS 兼容模式启动安装介质。真的。这才是*正确的*方法。你的固件应该能比较轻松地实现这一点，如果不能，这篇文章涵盖了各种可以帮助你实现的方法。先用 UEFI 原生模式启动安装介质，然后试图欺骗系统让它以为你没有以 UEFI 原生模式启动，这真的是一种很糟糕的做法。如果 noefi 真的不再对你起作用了，那肯定是某个地方出现了 bug，但我不会立刻想去调查并修复它，因为它并不是你真正想要用来实现你想要的目标的机制。

[adamw](#) wrote on 2014-01-27 03:38:

[adamw](#) 于 2014-01-27 03:38 写道：

"Also, yes, I do mean two extra levels of bootloader. MBR and whatever actually does the multiboot." What? In the BIOS/MBR world you have some very simple logic in the firmware layer, and very complex bootloaders in MBRs. In the UEFI world you have more complex logic in the firmware layer - but really it's still just executing bootloader code it finds on the hard disk, there's just more potential to configure this - and, ideally, somewhat simpler bootloaders on EFI system partitions. Neither system viewed as a whole is inherently more complex. All the complexity that exists in the UEFI layers I describe in this post *also* exists in the BIOS/MBR world*, often duplicated many, many times between different implementations, and with no

standardization. Have you read the modifications to the page I've made over the last day or so? They may explain this more clearly.

“是的，我的确指的是两层额外的引导加载程序。MBR 和实际执行多重启动的那个。”什么？在 BIOS/MBR 系统中，固件层的逻辑非常简单，而 MBR 中的引导加载程序却非常复杂。在 UEFI 系统中，固件层的逻辑更复杂——但实际上它仍然只是执行硬盘上找到的引导加载程序代码，只是配置的可能性更大——理想情况下，EFI 系统分区上的引导加载程序也更简单一些。从整体上看，这两个系统本身并没有哪个更复杂。我在这篇文章中描述的 UEFI 各层中存在的所有复杂性，*在 BIOS/MBR 系统中也同样存在*，而且在不同的实现中经常重复出现很多次，没有任何标准化。你有没有看过我最近一两天对页面做的修改？它们或许能更清楚地解释这一点。

[adamw](#) wrote on 2014-01-28 04:27:

[adamw](#) 于 2014-01-28 04:27 写道：

As it happens, someone else complained about noefi not working, and I was poking about in that code today anyway, so I went and looked at it. Turns out the kernel's behaviour with 'noefi' changed in a way which broke anaconda's check for UEFI in this case (UEFI-native boot, but 'noefi' passed on the cmdline). This kind of thing is exactly why I'd suggest not relying on noefi. :) But if you try booting the installer with 'noefi inst.updates=http://www.happyassassin.net/updates/updates-1047904.img', it should work. See https://bugzilla.redhat.com/show_bug.cgi?id=1047904.

碰巧有人也抱怨过 noefi 不起作用，而我今天正好也在研究那部分代码，所以就去看了一下。结果发现内核对 'noefi' 的处理方式发生了变化，导致 Anaconda 在这种情况下无法正确检查 UEFI（虽然命令行参数为 'noefi'，但系统仍然会选择 UEFI 原生启动）。正是因为这种情况，我才建议不要依赖 noefi。:) 不过，如果你尝试使用 'noefi inst.updates=http://www.happyassassin.net/updates/updates-1047904.img' 来启动安装程序，应该就能正常工作了。参见 https://bugzilla.redhat.com/show_bug.cgi?id=1047904。

kevin wrote on 2017-06-20 14:16:

kevin 于 2017-06-20 14:16 写道：

Bit of a necro revival here, but I'm install RHEL7.1 and booting anaconda with tboot (1.8.2). It seems that tboot requires 'noefi' to be on the cmdline or the kernel panics. Unfortunately, I can't find much information on why that is. Of course adding 'noefi' also causes anaconda to break because as you said, it's a native UEFI boot and so anaconda is trying to use grub2-efi, which I guess fails when EFI runtime services are disabled. Does that all sound correct? Any suggestions here? What's the correct solution in this type of case? I'm thinking the code issue you called out in the linked bugreport is not the problem. I have not tried supplying the updates image you linked, however I have modified that exact code manually inside the squashfs image and based on the stacktrace I'm seeing, it's actually grub2 that's failing - specifically, when pyanaconda tries to execute "grub2-install --no-floppy /dev/sda", because I'm getting: "/usr/lib/grub/x86_64-efi/modinfo.sh doesn't exist." At least, that's the error I get when running the command manually that I think pyanaconda is executing

这里有点挖坟了，但我正在安装 RHEL 7.1，并使用 tboot (1.8.2) 启动 Anaconda。似乎 tboot 需要在命令行中指定 'noefi'，否则内核会崩溃。可惜的是，我找不到太多关于这方面原因的信息。当然，添加 'noefi' 也会导致 Anaconda 崩溃，因为正如你所说，它是原生 UEFI 启动，所以 Anaconda 会尝试使用 grub2-efi，而我猜想当 EFI 运行时服务被禁用时，grub2-efi 会失败。我的理解都对吗？有什么建议吗？在这种情况下，正确的解决方案是什么？我认为你在链接的错误报告中提到的代码问题并不是问题所在。我没有尝试使用您提供的更新镜像，但我已经手动修改了 squashfs 镜像中的相应代码。根据我看到的堆栈跟踪信息，实际上是 grub2 出了问题——具体来说，是 pyanaconda 尝试执行“grub2-install --no-floppy /dev/sda”命令时出错，因为我收到的错误信息是：“/usr/lib/grub/x86_64-efi/modinfo.sh 不存在”。至少，这是我手动运行该命令时（我认为 pyanaconda 正在执行该命令）得到的错误。

kevin wrote on 2017-06-20 14:31:

kevin 于 2017-06-20 14:31 写道：

Also, per your question from that bug report: "What would be the actual use case for the behaviour you describe? (Genuine question, not snark - I wish to know). Who would want to boot in UEFI native mode, pass 'noefi', and have the installer do what you suggest?" The case that I have is a necessity to boot in UEFI mode because of 4k sector drives, and needing to use tboot (apparently requires 'noefi'), and yet anaconda fails. Maybe this is a use case you were wondering about.

另外，关于您在那个错误报告中提出的问题：“您描述的行为的实际应用场景是什么？（我是真心发问，没有讽刺的意思——我真的很想知道）。谁会想要以 UEFI 原生模式启动，传递 'noefi' 参数，然后让安装程序执行您建议的操作？”我的情况是，由于我的硬盘是 4K 扇区，所以必须以 UEFI 模式启动，并且需要使用 tboot（显然需要 'noefi' 参数），但 Anaconda 却失败了。也许这正是您想了解的应用场景。

[adamw](#) wrote on 2017-06-29 21:03:

[adamw](#) 于 2017-06-29 21:03 写道：

"The case that I have is a necessity to boot in UEFI mode because of 4k sector drives"

“由于我的硬盘是 4K 扇区硬盘，所以我的情况必须以 UEFI 模式启动。”

I'm not quite sure what you mean here; to my knowledge there's nothing about 4K sector sizes that requires the use of UEFI. Could you clarify?

我不太明白你的意思；据我所知，4K 扇区大小并不需要使用 UEFI。你能解释一下吗？

q2dg wrote on 2014-01-26 15:30:

q2dg 于 2014-01-26 15:30 写道：

Hello. Using UEFISTubs is a good idea?

您好。使用 UEFISTubs 是个好主意吗？

[adamw](#) wrote on 2014-01-27 02:13:

[adamw](#) 于 2014-01-27 02:13 写道：

It's an *interesting* idea. I haven't played with it enough myself to tell you whether it's a good one or not. I guess I'd say that in my experience all the really tricky problems and misunderstandings people have with UEFI happen at the firmware layer; I haven't really seen people have much trouble at the EFI bootloader layer. So I don't think using an EFI stub kernel boot approach is something that would make the lives of most people struggling with UEFI any easier. But it's sure a technically interesting approach, and if you feel like playing with it, have fun.

这确实是个*有趣的*想法。我自己还没怎么尝试过，所以无法判断它到底好不好。就我的经验而言，人们在使用 UEFI 时遇到的所有棘手问题和误解都发生在固件层；我很少看到有人在 EFI 引导加载程序层遇到麻烦。因此，我认为使用 EFI stub 内核启动方式并不能让大多数在 UEFI 上苦苦挣扎的人轻松多少。但这确实是一个很有意思的技术方案，如果你想尝试一下，那就好好享受吧。

Tom wrote on 2014-02-10 15:12:

Tom 于 2014 年 2 月 10 日 15:12 写道：

The gummiboot and refind boot managers use the efi stub

gummiboot 和 refind 启动管理器使用 efi 存根。

[Rod Smith](#) wrote on 2014-02-16 03:42:

[Rod Smith](#) 于 2014 年 2 月 16 日 03:42 写道：

There are system-to-system differences in how specific boot loaders work. A couple of years ago, GRUB 2 was hideously unreliable, in my experience; it would often fail to boot kernels, would hang, and would otherwise misbehave. It's settled down a lot recently and tends to be much more reliable these days, but it's still the most ungainly and difficult-to-configure boot loader I've ever seen. It has the benefit of being worked on by many smart people so that it works reasonably well "out of the box" on MOST peoples' systems; but in those cases when it doesn't work, GRUB becomes a nightmare. In such cases, using almost anything else is likely to be easier than struggling with GRUB. My Web page on the options can be helpful:

<http://www.rodsbooks.com/efi-bootloaders/> The EFI stub loader is one of these options, and in my experience it's quite reliable; however, some people on the Arch Linux forum have reported sporadic difficulties with it with 3.7 and later kernels. I have yet to see a single report of such problems on other distributions, though, so I suspect that there's something odd about the way the Arch kernels are being built that's contributing to those problems.

不同系统使用的引导加载程序存在差异。几年前，GRUB 2 的可靠性极差，就我个人经验而言，它经常无法启动内核，会卡死，还会出现其他各种问题。虽然最近情况有所改善，可靠性也提高了不少，但它仍然是我见过的最笨拙、最难配置的引导加载程序。它的优势在于由许多聪明人共同开发，因此在大多数人的系统上都能“开箱即用”地运行良好；但一旦出现问题，GRUB 就会变成一场噩梦。在这种情况下，使用几乎任何其他引导加载程序都比费力地使用 GRUB 要容易得多。我的网页上列出了各种选项，或许对您有所帮助：

<http://www.rodsbooks.com/efi-bootloaders/> EFI stub loader 就是其中一种选择，就我的经验而言，它相当可靠；然而，Arch Linux 论坛上的一些用户反映，在使用 3.7 及更高版本的内核时，会出现一些零星的问题。不过，我还没有在其他发行版上看到过类似的问题报告，所以我怀疑是 Arch 内核的构建方式存在一些特殊之处，导致了这些问题。

[adamw](#) wrote on 2014-02-16 03:52:

[adamw](#) 于 2014-02-16 03:52 写道：

I'm tempted to channel Churchill in describing grub2 - it's the worst bootloader we have, except for all the others...

我真想借用丘吉尔的话来形容 grub2——它是我们拥有的最糟糕的引导加载程序，除了其他所有程序之外.....

[Rod Smith](#) wrote on 2014-02-16 04:34:

[Rod Smith](#) 于 2014 年 2 月 16 日 04:34 写道：

Personally, I'd leave out the last five words. I'm not a fan of GRUB 2. Under EFI, I prefer using ANYTHING else to boot Linux.

就我个人而言，我会去掉最后五个字。我不喜欢 GRUB 2。在 EFI 下，我宁愿用任何其他方式启动 Linux。

eggplant wrote on 2014-01-26 17:32:

eggplant 于 2014-01-26 17:32 写道：

Congratulations! you finished the boring spec giving ?? tries! :) A huge "thanks.rpm" for filtering & accumulating just what end users need to know for a good workable knowledge. The main things to remember for a prospective computing device buyer nowadays: 1. "... it's entirely possible that in a UEFI world things will be even worse than they were in the BIOS world ... you should be angry at is your system/motherboard manufacturer ...". 2. The big evil declaration is "... Disable BIOS compatibility mode when Secure Boot is enabled ..." to be Microsoft Certified specially for ARM computers. 3. "Apple locks down the bootloader ... most Android devices also ship with locked bootloaders" 4. Microsoft compliant Secure Boot is not the only evil, there is

FOTA looking for you. Probably it already caught you. :) I think just "painkillers and gin" are not sufficient enough in this modern age <(-)

恭喜！你终于完成了枯燥的规格说明！:) 非常感谢你筛选并整理出最终用户真正需要了解的实用知识。如今，对于潜在的电脑设备买家来说，以下几点至关重要：1. “.....在 UEFI 时代，情况很可能比 BIOS 时代更糟.....你应该责怪的是你的系统/主板制造商.....”。2. 最邪恶的声明是“.....启用安全启动时禁用 BIOS 兼容模式.....”，这是微软专门针对 ARM 架构电脑的认证要求。3. “苹果锁定了引导加载程序.....大多数安卓设备也出厂时就锁定了引导加载程序。”4. 符合微软标准的“安全启动”并非唯一的“邪恶”，还有 FOTA（固件更新）在等着你。或许你已经中招了。:) 我觉得在这个时代，“止痛药和杜松子酒”已经不足以解决问题了 <(-)

[adamw](#) wrote on 2014-01-26 19:01:

[adamw](#) 于 2014-01-26 19:01 写道:

BIOS compatibility mode really isn't relevant to the ARM world. No-one's ever shipped an ARM system with a BIOS. I don't think any ARM UEFI firmwares implement BIOS compatibility at all, it would be a ridiculous thing to do. The evil thing about the Windows ARM case is that the Windows certification requirements *for ARM* specifically state that the user must not be able to disable or re-configure Secure Boot - precisely the opposite of the requirements *for Windows*, which state that the user *must* be able to do those things. Understood from a 'mobile world' perspective, where people are comfortable with the concept of a locked bootloader, what that effectively does is enshrine bootloader locking on *ARM* Windows devices as a part of the platform definition. To be a Microsoft-certified ARM Windows device, the bootloader must be locked. As I said, that is indeed bad, though only exactly *as* bad as Apple.

BIOS 兼容模式对 ARM 世界来说其实无关紧要。从来没有人推出过带有 BIOS 的 ARM 系统。我认为没有任何 ARM UEFI 固件实现了 BIOS 兼容性，这样做简直荒谬。Windows ARM 问题的关键在于，Windows 对 ARM 的认证要求明确规定用户不得禁用或重新配置安全启动——这与 Windows 本身的认证要求截然相反，后者规定用户必须能够执行这些操作。从“移动领域”的角度来看，人们已经习惯了锁定引导加载程序的概念，而这实际上相当于将引导加载程序锁定作为平台定义的一部分，并写入了 ARM Windows 设备。要成为微软认证的 ARM Windows 设备，引导加载程序必须锁定。正如我所说，这确实很糟糕，但与苹果的情况相比，糟糕程度也仅次于苹果。

chip wrote on 2014-01-26 18:13:

chip 于 2014-01-26 18:13 写道:

You base several arguments that microsoft (and others) have nothing to do with conspiring to produce and support poor technology.. I would like to introduce you to a new concept: <http://en.wikipedia.org/wiki/Cartel> Historically: <http://en.wikipedia.org/wiki/Wintel> ... Nothing should be written about UEFI other than it is constraining and complicated solution to projects like coreboot (Just give us the specs and docs please, thank you)... This does nothing to help motherboard manufacturers that will continue to produce crap bioses and boards.

你提出了几个论点，声称微软（以及其他公司）与串谋生产和支持劣质技术无关.....我想向你介绍一个新概念：<http://en.wikipedia.org/wiki/Cartel>.....历史背景：<http://en.wikipedia.org/wiki/Wintel>.....关于 UEFI，除了它对像 coreboot 这样的项目来说是一个限制性很强且复杂的解决方案之外，不应该再写任何其他内容（请直接提供规格和文档，谢谢）.....这对主板制造商没有任何帮助，他们将继续生产劣质的 BIOS 和主板。

[adamw](#) wrote on 2014-01-26 19:03:

[adamw](#) 于 2014-01-26 19:03 写道:

Thanks for the condescension. I have a degree in history including several economic history modules. I am comfortable with the concept of a cartel. Intel designed EFI as a technical solution to a range of technical problems. BIOS is not a good standard. It isn't even a standard, in the first place, just a convention. It has its own huge set of problems and limitations which I didn't go into in this article because it was out of scope. There is a genuine need on a technical level for a replacement for the BIOS, and UEFI is what we wound up with. Its imperfections are cock-ups, not conspiracies. All the bits in the post where I complain about how the UEFI spec isn't sufficiently prescriptive? Well, that goes in spades for BIOS because *there is no BIOS spec*. All you have to go on in writing a BIOS is 'make it behave like other BIOSes'. This is ridiculous, of course. There are certainly BIOS implementations as idiotic as any UEFI implementation.

谢谢你的居高临下。我拥有历史学学位，其中还包括几门经济史课程。我对卡特尔的概念并不陌生。英特尔设计 EFI 是为了解决一系列技术问题。BIOS 并非一个好的标准。它根本就不是标准，而只是一种约定俗成。它自身存在着诸多问题和局限性，由于超出本文范围，我并未详述。从技术层面来说，确实需要一个替代 BIOS 的方案，而 UEFI 正是我们最终的选择。它的缺陷是设计上的失误，而非阴谋。我在文章中抱怨 UEFI 规范不够明确？其实，BIOS 的情况更糟，因为*根本没有 BIOS 规范*。编写 BIOS 时，你唯一能做的就是“让它像其他 BIOS 一样运行”。这当然很荒谬。当然，也存在一些和 UEFI 一样愚蠢的 BIOS 实现。

chip wrote on 2014-01-26 19:21:

chip 于 2014-01-26 19:21 写道:

Bios doesn't need a spec. The hardware does. The manufacturers must clear the communication lines to people who develop with this hardware (from small to large organizations). UEFI doesn't help this. It increases reliance on black box technology.

BIOS 不需要规范，硬件才需要。制造商必须确保使用该硬件进行开发的人员（从小型到大型组织）都能获得清晰的沟通渠道。UEFI 对此并无帮助，反而增加了对黑盒技术的依赖。

[adamw](#) wrote on 2014-01-27 02:14:

[adamw](#) 于 2014-01-27 02:14 写道:

Your comment is entirely incoherent and contains no information or argument of any value.

你的评论完全语无伦次，没有任何有价值的信息或论点。

chip wrote on 2014-01-29 23:06:

chip 于 2014-01-29 23:06 写道：

What i was referring to was your tone on coreboot and that BIOS is not a good standard, or that it needs a standard. As i said, BIOS does not need a standard or specification. The hardware which it initializes should be clearly specified and documented. Standardizing a BIOS does not help anyone.

我指的是你对 Coreboot 的语气，以及你认为 BIOS 不是一个好的标准，或者说它需要一个标准。正如我所说，BIOS 不需要标准或规范。它初始化的硬件应该有明确的规范和文档。BIOS 标准化对任何人都没有好处。

[Rod Smith](#) wrote on 2014-02-16 03:50:

[Rod Smith](#) 于 2014 年 2 月 16 日 03:50 写道：

By the logic that "Bios doesn't need a spec," it should be fine to write a BIOS that loads the boot loader code from the LAST sector on the disk, or the SECOND sector on the disk, rather than from the first sector on the disk. This would work, of course, if boot loader code were found there, but it would be completely incompatible with every disk that holds a BIOS boot loader. That's just one example of the need for standards in the BIOS arena; the BIOS includes dozens of system calls that are documented in an ad-hoc manner.

按照“BIOS 不需要规范”的逻辑，编写一个从磁盘最后一个扇区或第二个扇区加载引导加载程序代码的 BIOS，而不是从磁盘第一个扇区加载，应该也没问题。当然，如果引导加载程序代码确实存在于第一个扇区，这种方法是可行的，但它将与所有包含 BIOS 引导加载程序的磁盘完全不兼容。这只是 BIOS 领域需要标准的一个例子；BIOS 包含数十个系统调用，而这些调用的文档都是临时性的。

Mantas wrote on 2014-01-27 06:39:

Mantas 于 2014-01-27 06:39 写道：

"Space for the MBR" is a bit wrong. The MBR is always exactly 512 bytes, the bootstrap code 440 bytes (rest is the partition table). The space before first partition is not part of the MBR. (Which just makes it even worse to assume that it'll be available...) On the other hand, not all bootloaders use that space. For example, syslinux does not – its MBR code jumps directly to a file in the boot partition.

“MBR 空间”的说法有点不准确。MBR 始终为 512 字节，引导代码为 440 字节（其余部分是分区表）。第一个分区之前的空间不属于 MBR。（这使得假设它可用变得更加错误.....）另一方面，并非所有引导加载程序都会使用这部分空间。例如，syslinux 就不会使用——它的 MBR 代码会直接跳转到引导分区中的一个文件。

[adamw](#) wrote on 2014-01-27 07:41:

[adamw](#) 于 2014-01-27 07:41 写道：

Oh, you're right, I do believe - we usally call it the 'bootloader embedding space' or something like that, right? I'll have to refresh my memory on that and come back to it tomorrow, I'm just going off my memory of all the 'fun' we had with that crap several releases back. Thanks for the note.

哦，你说得对，我确实记得——我们通常把它叫做“引导加载程序嵌入空间”之类的，对吧？我得回忆一下，明天再来处理，我现在只是凭着几个版本前我们处理那玩意儿时的“乐趣”记忆来推测的。谢谢你的提醒。

Mantas wrote on 2014-01-27 16:39:

Mantas 于 2014-01-27 16:39 写道：

Personally I call it the space where my GPT is located... I think "embedding" is a common term with GRUB, yes.

我个人称之为我的 GPT 所在的空间.....我认为“嵌入”是 GRUB 中常用的术语，是的。

Tom wrote on 2014-02-10 15:16:

Tom 于 2014 年 2 月 10 日 15:16 写道：

'something like that' = post-mbr gap = "gap" between the end of the mbr and the beginning of the first partition

类似这样的内容 = 主引导记录 (MBR) 后间隙 = MBR 末尾和第一个分区开头之间的“间隙”

[Rod Smith](#) wrote on 2014-02-16 03:53:

[Rod Smith](#) 于 2014 年 2 月 16 日 03:53 写道：

Today, though, not all disks use 512-byte sectors. I don't know offhand how a BIOS treats such disks from a boot perspective, although I'd wager that a lot of true BIOSes can't boot from such disks at all.

不过，如今并非所有磁盘都使用 512 字节扇区。我不太清楚 BIOS 在启动过程中如何处理这类磁盘，但我估计很多真正的 BIOS 根本无法从这类磁盘启动。

[adamw](#) wrote on 2014-02-16 03:57:

[adamw](#) 于 2014-02-16 03:57 写道：

Back when we tried doing GPT by default in Fedora 16 (IIRC), I think we found that only disks larger than 2TB were using 4K logical sectors. I *think* this still holds true today - smaller disks use 512B logical sectors, i.e. they present a 512B sector size to the system even if they really use 4K sectors internally (4K physical sectors).

我记得当初我们在 Fedora 16 中尝试默认使用 GPT 分区表时，发现只有大于 2TB 的磁盘才会使用 4K 逻辑扇区。我*认为*这一点至今仍然成立——较小的磁盘使用 512B 逻辑扇区，也就是说，即使它们内部实际使用了 4K 扇区（4K 物理扇区），系统看到的也是 512B 的扇区大小。

[Rod Smith](#) wrote on 2014-02-16 04:31:

[Rod Smith](#) 于 2014 年 2 月 16 日 04:31 写道：

Lots of USB enclosures these days seem to be using 4KiB logical sectors, no matter what the disk's size. At least, I've seen problem reports in online forums related to this. For instance, some enclosures translate to 4KiB logical sectors when using USB interfaces, but present the drive's native (usually 512-byte) logical sector size when connected via eSATA. Of course, that's a recipe for disaster!

现在很多 USB 硬盘盒似乎都使用 4KiB 的逻辑扇区，不管硬盘实际容量有多大。至少我在网上论坛上看到过相关的问题报告。例如，有些硬盘盒在使用 USB 接口时会显示 4KiB 的逻辑扇区，但通过 eSATA 连接时却显示硬盘本身的逻辑扇区大小（通常为 512 字节）。这当然会造成很多问题！

PoMo wrote on 2014-02-01 14:44:

PoMo 于 2014-02-01 14:44 写道：

Thank your for this great essay, I enjoyed reading it a lot. * You strongly discourage multibooting from the same disk. Could you shed some light on this quote in particular: "understand that you are making your life much more painful than it needs to be, [...], and don't go yelling at your OS vendor, whatever breaks." From what I read in your essay, your main beef with implementations of uefi is that motherboards may _display_ the boot-choice badly or not at all. Apart from this UI issue, is there anything else that's painful? And what else is likely/known to break? Your text gave me the (possibly wrong) impression that you could do everything with efibootmgr, and with that, bypassing whatever downsides the motherboard UI has. * This leads me to a dreamy crazy question: is it imaginable that we'll see a grub (or something that looks like grub but isn't), that massively uses what efibootmgr uses. I could imagine this having its own entry in the efi list, and being the default boot entry. The displayed list would just show the efi entries, selecting an entry would use the nextboot efi feature, a timeout would nextboot a custom default (not changing the efi default, which would keep pointing to this grub-like thing). Letting my mind roam freely, this could bypass the problem of motherboards not displaying the list in the same fashion, and OSes being able to rely on it being there and knowing how it presents choice to the user, making it not being every individual OSes headache. (Also gone be the days of OSes overwriting each other's bootmanagers). Any installed OS would just need to add itself to the efi boot list. Done. (And add such grub-like thing if there isn't one yet, and make it the default). With a few naming conventions, it could even add temporary boot entries (like for one-time modifying kernel params) which it would delete again on next occasion. The one obvious downside would be that you're always rebooting at least once. first boot going into this grub-like thing, and second into the os you pick in it. But I'd accept that for all the advantages I can think of ;)

感谢您这篇精彩的文章，我读得很开心。* 您强烈反对从同一块硬盘进行多系统启动。能否请您特别解释一下这句话：“要知道，这样做会让你的生活变得比原本更痛苦，[...]，而且无论出现什么问题，都不要责怪你的操作系统供应商。”从我读到的文章来看，您对 UEFI 实现的主要不满在于主板可能显示启动选项的方式很糟糕，甚至根本不显示。除了这个用户界面问题之外，还有其他令人头疼的地方吗？还有哪些方面容易/已知会出现问题？您的文章让我（可能错误地）觉得您可以使用 efibootmgr 完成所有操作，从而绕过主板用户界面的所有缺点。* 这让我想到一个异想天开的问题：我们是否有可能看到一个 GRUB（或者看起来像 GRUB 但实际上不是的东西），它大量使用 efibootmgr 的功能？我可以想象它在 EFI 启动列表中拥有自己的条目，并成为默认启动项。显示的列表只会显示 EFI 启动项，选择一个启动项会使用 NextBoot EFI 功能，超时后会启动一个自定义的默认启动项（不会更改 EFI 默认启动项，因为这样会指向类似 GRUB 的程序）。我的想法是，这可以绕过主板显示启动列表方式不一致的问题，操作系统可以依赖这个列表的存在，并了解它如何向用户提供选择，从而避免每个操作系统都为此头疼。（操作系统之间互相覆盖启动管理器的情况也将不复存在）。任何已安装的操作系统只需将自身添加到 EFI 启动列表中即可。搞定。（如果还没有类似 GRUB 的程序，则添加一个并将其设置为默认启动项）。只要遵循一些命名规则，它甚至可以添加临时启动项（例如用于一次性修改内核参数），并在下次启动时将其删除。一个显而易见的缺点是，你总是至少要重启一次。第一次启动会进入类似 GRUB 的引导程序，第二次启动则会进入你在其中选择的操作系统。但考虑到我能想到的所有优点，我愿意接受这一点；)

[adamw](#) wrote on 2014-02-01 17:05:

[adamw](#) 于 2014-02-01 17:05 写道：

"Apart from this UI issue, is there anything else that's painful? And what else is likely/known to break?" If you manage to get a given multiboot setup working, and don't go poking it with any sticks, it *ought* to keep working. It's mainly the deployment time where people have trouble, and then understanding what actually constitutes their boot config (and hence should not be poked with sticks) after that. I mean, it can certainly work; I just see so many people struggling with it and misunderstanding stuff that I get concerned. That section was slightly tongue-in-cheek, but with a serious point: it really is easier if you can just stick to an OS per machine or per disk if you can. It may just be part of my general inclination after years of fiddling with PCs and trying to help other people fiddle with them: I really, really believe in the 'choice is an excellent way to shoot yourself in the foot' argument, and try to keep my setups as simple as possible. There's enough damn complexity in dealing with computers without you going out and voluntarily adding more on top, IMO :) "Your text gave me the (possibly wrong) impression that you could do everything with efibootmgr, and with that, bypassing whatever downsides the motherboard UI has." Yes, this is basically true - *as long as you make sure you can get into a UEFI OS to poke efibootmgr*, of course :) If you take the time to learn efibootmgr, and you make sure you always have a handy way of running it, yes, it's a very handy get-out-of-jail-free card to have around. "This leads me to a dreamy crazy question:..." That's actually a rather interesting design. I don't know of anyone who's done it yet, but I don't know everything. Rod (if he's reading? hi, Rod) may do. People have certainly written things that are meant to sit at the UEFI bootloader level and

intermediate between you and all this craziness. I have to admit that I tend to view them as yet another layer of craziness ;), but some people prefer to take the approach of picking one and making it their primary interface to the whole shebang. Rod's Refind - <http://www.rodsbooks.com/efi-bootloaders/refind.html> - is one of these, and he has a general page on UEFI bootloaders at <http://www.rodsbooks.com/efi-bootloaders/> .

除了这个用户界面问题，还有其他令人头疼的地方吗？还有哪些地方容易出问题？如果你成功配置好多系统启动，并且不去随意改动，它*应该*能一直正常工作。人们遇到的问题主要集中在部署阶段，以及部署之后理解启动配置的构成（因此不应该随意改动）。我的意思是，多系统启动当然可以运行；我只是看到很多人为此苦苦挣扎，并且误解了很多东西，这让我很担心。刚才那段话有点开玩笑的成分，但其中也包含一个严肃的观点：如果可以的话，每台机器或每个磁盘只安装一个操作系统，确实会更简单。这可能只是我多年来摆弄电脑并帮助别人摆弄电脑后养成的一种习惯：我真的非常相信“选择越多越容易出错”的说法，并尽量让我的配置保持简单。处理电脑已经够复杂了，我觉得你没必要再给自己找麻烦：“你的文字让我（可能错了）以为可以用 efibootmgr 做所有事，从而绕过主板 UI 的所有缺点。”是的，这基本上没错——*前提是你能进入 UEFI 系统来操作 efibootmgr*，当然:) 如果你花时间学习 efibootmgr，并确保始终有方便的运行方式，那么它确实是一张非常方便的“免死金牌”。“这让我想到一个有点疯狂的问题：.....”这确实是一个相当有趣的设计。我不知道有人已经这么做了，但我也不是什么都懂。Rod（如果他看到了吗？嗨，Rod）或许可以。确实有人编写了一些旨在置于 UEFI 引导加载程序层级、作为你与所有这些复杂机制之间的桥梁的工具。我必须承认，我倾向于将它们视为又一层复杂的机制;)，但有些人更喜欢选择一个工具，并将其作为与整个系统交互的主要界面。Rod 的 Refind (<http://www.rodsbooks.com/efi-bootloaders/refind.html>) 就是其中之一，他还在 <http://www.rodsbooks.com/efi-bootloaders/> 上提供了一个关于 UEFI 引导加载程序的通用页面。

PoMo wrote on 2014-02-15 18:40:

PoMo 于 2014-02-15 18:40 写道：

Thanks for the Rodbooks.com links. Unfortunately those bootmanagers/loaders (refind, gummyboot, ..) are, as you say, adding "yet another layer layer of craziness". In short, they seem to mostly leave available features aside and instead focus on what EFI requires an OS to conform to and make use of that. And they're adding some more requirements on top of that (like naming schemes, certain min kernel version, or even limitations to certain OSes). Why don't those bootmanagers "simply" manage efi boot entries? (and present them in a nice way) I can only assume that that wouldn't always work, but i'd love to hear/read from someone who knows. (like Rod) I fear that as long as there's no bootmanager that works with the motherboard's efi boot entries (reading from it, possibly also with the ability to add/modify/delete/backup/restore entries), and that can boot all entries that can than be booted from the motherboard's efi bootmanager, OSes cannot rely on available features and by that will always bring along their own bootmanager/loader that will confuse/add complexity by only working with their OS and perhaps 2-3 others, and that will possibly also mess with what other OSes put in place. If there was such a bootmanager (and it was free and opensource ;)), each OS would only have to care about adding a working EFI-entry for itself. Of course the OS-installer could (and should) also offer to optionally install that bootmanager during installation. (and not in a grub-way where they roll their own version with adjustments for their OS) I know it's a lot of wishful thinking ;) I'd love to read about that topic from someone who knows this stuff though ;)

感谢您提供的 Rodbooks.com 链接。可惜的是，正如您所说，那些引导管理器/加载器（例如 refind、gummyboot 等）又增加了一层“混乱”。简而言之，它们似乎大多忽略了现有功能，而是专注于 EFI 对操作系统的要求，并利用这些要求。而且它们还在此基础上增加了一些额外的要求（例如命名规则、特定的最低内核版本，甚至对某些操作系统的限制）。为什么这些引导管理器不“简单地”管理 EFI 启动项呢？（并以友好的方式呈现它们）我只能猜测这样做并非总是有效，但我很想听听/读读这方面的专家的意见。（就像 Rod 一样）我担心，只要没有一个能够与主板 EFI 启动项兼容（读取启动项，可能还具备添加/修改/删除/备份/恢复启动项的功能）的启动管理器，并且能够启动所有可以从主板 EFI 启动管理器启动的启动项，操作系统就无法依赖现有功能，因此总是会自带启动管理器/加载器，而这些启动管理器/加载器只能与自身操作系统以及可能另外两三个操作系统兼容，这不仅会造成混乱和增加复杂性，还可能干扰其他操作系统已安装的启动项。如果存在这样一个启动管理器（而且它是免费开源的），那么每个操作系统只需关心添加一个适合自身的 EFI 启动项即可。当然，操作系统安装程序也可以（而且应该）在安装过程中提供安装该启动管理器的选项。（而且不是那种自己开发版本并针对操作系统进行调整的粗糙做法）我知道这只是美好的愿望;) 不过我很想听听懂这方面知识的人怎么说;)

[Rod Smith](#) wrote on 2014-02-16 03:17:

[Rod Smith](#) 于 2014 年 2 月 16 日 03:17 写道：

A boot manager that presents a menu based on what's in the firmware's NVRAM is certainly do-able. I've toyed with the idea of adding such a mode of operation to rEFInd, but I haven't done so because I don't believe it would add anything to what rEFInd already does, and in many ways it would in fact be limiting. For instance, if you use rEFInd to boot Linux kernels directly, the approach you suggest would require using efibootmgr every time you add a kernel. Since no distribution does this, it would be an extra manual step for the user -- in other words, it would DEGRADE functionality. Another problem with this approach -- and a flaw with EFI generally -- is that placing boot loaders on the hard disk and information about them in NVRAM creates two points of failure in the process of launching boot loaders. This is a practical real-world problem -- I've both heard of and personally encountered systems that "forget" their boot loader entries on a regular basis. On such systems, the only practical solution is to use the fallback boot loader (EFI/BOOT/bootx64.efi), and if it were to rely on the NVRAM entries, it would fail miserably. Even systems that don't USUALLY forget their NVRAM entries can do so occasionally. One of my computers does so if a hard disk is temporarily unplugged, for instance. There can also be OS-driven bugs. Some versions of shim can create an ever-expanding boot list, for instance, and this would be a nightmare for a boot loader following your suggested design. That's not to say that such an approach doesn't have its merits, and might not appeal to some people -- it does have merits, like being configurable from any OS using the standard EFI mechanisms. IMHO, though, it's not preferable to the auto-scanning that rEFInd does, at least not in general. I'll consider adding it as an option to rEFInd; in fact, in writing this reply, I'm contemplating ways that the NVRAM-based data might be integrated with the auto-scanned data in ways that might be useful. Some additional comments on your post, and on Adam's response to it: * My own "main beef" with EFI is with the number of bugs found in the EFI world. The poor boot manager menus that some implementations present are an issue, and they represent a good argument for using rEFInd, gummiboot, GRUB, or some other boot manager. (Note that rEFInd and gummiboot are both add-on boot managers only, whereas GRUB does double duty as both a boot manager and a boot loader.) Other bugs, like EFIs that forget NVRAM entries, are less common but often much more painful than poor user interfaces. * The type of boot manager you describe would NOT require rebooting to boot another OS; it could,

like rEFInd, gummiboot, or even GRUB, boot the next boot loader directly. (Yes, GRUB can chainload to another EFI program.) * IMO, relying on efibootmgr and the firmware's built-in boot manager is not a good approach to managing a dual-boot computer, except possibly in those rare cases when the built-in boot manager is really good. The built-in boot managers are usually extremely limited in what they can do. Most notably, most of them require hitting a key (which varies from one computer to another, no less!) at JUST the right time in the boot process to boot anything but the default OS. This sort of design has always struck me as brain-dead, and because of system-to-system differences, documenting it is a nightmare. Fedora's philosophy at one point was to rely on the built-in EFI boot manager for multi-booting, but if I'm not mistaken, the Fedora developers have come to their senses on this one. * I understand the view of rEFInd or gummiboot as being "another layer of craziness," but I believe it's misplaced. Recall the point I made that GRUB is both a boot loader and a boot manager. Thus, when dual-booting Linux and Windows, the boot path for Linux is likely to be EFI->GRUB->Linux kernel; and when booting Windows, it will be either EFI->Windows boot loader->Windows kernel or EFI->GRUB->Windows boot loader->Windows kernel, depending on whether the installation relies on the EFI boot manager (bad choice) or GRUB (better choice) to select which OS to boot. The boot path with rEFInd or gummiboot will look EXACTLY THE SAME, except that you replace GRUB with rEFInd or gummiboot, and the details of how the Linux kernel launches are different. rEFInd CAN launch Linux via GRUB, but GRUB can also launch Linux via rEFInd. Thus, when analyzed fully and configured optimally, neither rEFInd nor gummiboot is "another layer of craziness"; they're both alternatives to GRUB, at least functionally. Yes, rEFInd and gummiboot are both boot managers but not boot loaders; but with a boot loader built into 3.3.0 and later kernels, you don't really need a separate boot loader program to launch Linux. * You mentioned the minimum kernel version as a limitation of rEFInd and gummiboot. It's true that both rely on the EFI stub loader, which was added with the 3.3.0 kernel. Not many distributions ship with kernels older than that at this point, though. Thus, this is no longer really a practical concern. * Compared to GRUB, file naming is NOT a limitation of rEFInd; both boot programs can be configured with manual boot stanzas, which can launch kernels and boot loaders with any filename. The difference is that rEFInd can auto-detect boot loaders (including Linux kernels) in certain locations and with certain filenames. This is an ADDITIONAL FEATURE of rEFInd compared to GRUB. Thus, when comparing the two, it's GRUB that has the deficit, not rEFInd! Really, my intent in designing rEFInd (or expanding rEFIt's OS-scanning features; it's only fair to give credit to Christoph Pfisterer, rEFIt's creator, for designing the basic framework) is the same as what you want, at a broad level: To launch any OS's boot loader with little or no extra configuration. rEFInd does this without relying on the NVRAM entries because they're fallible, but rEFInd scans the locations where boot loaders are supposed to go, plus some other locations where they (especially Linux kernels) are commonly found in practice. People who use rEFInd tell me that it finds boot loaders and Linux kernels quite reliably; most of the problem reports I see relate to EFI bugs or difficulties installing rEFInd in exotic setups, not to the boot loader scanning features.

根据固件 NVRAM 中的内容生成菜单的启动管理器当然是可行的。我曾考虑过为 rEFInd 添加这种操作模式，但最终没有付诸行动，因为我认为它不会给 rEFInd 增加任何新功能，而且在很多方面实际上还会限制其功能。例如，如果您使用 rEFInd 直接启动 Linux 内核，您建议的方法需要在每次添加内核时都使用 efibootmgr。由于没有任何发行版会这样做，因此这将为用户增加一个额外的手动步骤——换句话说，它会降低功能。这种方法的另一个问题——也是 EFI 的一个普遍缺陷——是将启动加载程序放在硬盘上，并将相关信息放在 NVRAM 中，这在启动启动加载程序的过程中会造成两个故障点。这是一个实际存在的问题——我听说过也亲身遇到过系统经常“忘记”其启动加载程序条目的情况。在这样的系统上，唯一可行的解决方案是使用备用引导加载程序（EFI/BOOT/bootx64.efi），如果它依赖 NVRAM 条目，则会彻底失败。即使是通常不会忘记 NVRAM 条目的系统，偶尔也会发生这种情况。例如，我的一台电脑在硬盘暂时拔出时就会出现这种情况。此外，操作系统本身也可能存在 bug。例如，某些版本的 shim 会创建一个不断扩展的引导列表，这对按照您建议的设计运行的引导加载程序来说将是一场噩梦。但这并不意味着这种方法毫无优点，它或许对某些用户没有吸引力——它确实有优点，例如可以使用标准 EFI 机制从任何操作系统进行配置。不过，恕我直言，至少在大多数情况下，它不如 rEFInd 的自动扫描功能。我会考虑将其作为 rEFInd 的一个选项；事实上，在撰写此回复时，我正在思考如何将基于 NVRAM 的数据与自动扫描的数据进行整合，以期找到一些有用的方法。关于您的帖子以及 Adam 的回复，我还有一些补充说明：* 我个人对 EFI 的主要不满在于 EFI 世界中存在的大量 bug。某些实现中糟糕的启动管理器菜单就是一个问题，这也充分说明了使用 rEFInd、gummiboot、GRUB 或其他启动管理器的必要性。（请注意，rEFInd 和 gummiboot 都只是附加的启动管理器，而 GRUB 则兼具启动管理器和引导加载程序的功能。）其他 bug，例如 EFI 忘记 NVRAM 条目，虽然不太常见，但通常比糟糕的用户界面更令人头疼。* 您描述的那种启动管理器不需要重启即可启动其他操作系统；它可以像 rEFInd、gummiboot 甚至 GRUB 一样，直接启动下一个引导加载程序。（是的，GRUB 可以链式加载到另一个 EFI 程序。）* 我认为，除非在极少数情况下内置引导管理器确实非常出色，否则依赖 efibootmgr 和固件内置的引导管理器并不是管理双启动计算机的好方法。内置引导管理器的功能通常非常有限。最值得注意的是，它们中的大多数都需要在启动过程中按下某个键（而且这个键在不同的计算机上都不一样！）才能启动除默认操作系统之外的任何其他程序。这种设计在我看来一直很愚蠢，而且由于系统之间的差异，编写相关文档简直是噩梦。Fedora 曾经的理念是依赖内置的 EFI 引导管理器来实现多启动，但如果我没记错的话，Fedora 的开发者们已经意识到了这一点。我理解有人认为 rEFInd 或 gummiboot 是“又一层混乱”，但我认为这种看法并不准确。请记住我之前提到的，GRUB 既是引导加载程序又是引导管理器。因此，在 Linux 和 Windows 双启动时，Linux 的启动路径很可能是 EFI->GRUB->Linux 内核；而启动 Windows 时，启动路径可能是 EFI->Windows 引导加载程序->Windows 内核，也可能是 EFI->GRUB->Windows 引导加载程序->Windows 内核，具体取决于安装程序是依赖 EFI 引导管理器（糟糕的选择）还是 GRUB（更好的选择）来选择启动哪个操作系统。使用 rEFInd 或 gummiboot 的启动路径看起来完全相同，只是将 GRUB 替换成了 rEFInd 或 gummiboot，Linux 内核的启动方式有所不同。rEFInd 可以通过 GRUB 启动 Linux，但 GRUB 也可以通过 rEFInd 启动 Linux。因此，经过全面分析和优化配置后，rEFInd 和 gummiboot 都不是“又一层复杂的东西”；它们都是 GRUB 的替代方案，至少在功能上是如此。没错，rEFInd 和 gummiboot 都是引导管理器，但不是引导加载程序；不过，由于 3.3.0 及更高版本的内核都内置了引导加载程序，所以你实际上不需要单独的引导加载程序来启动 Linux。* 你提到最低内核版本是 rEFInd 和 gummiboot 的一个限制。的确，它们都依赖于 EFI 存根加载程序，该加载程序是在 3.3.0 内核中添加的。不过，现在很少有发行版会使用比这更旧的内核。因此，这不再是一个实际问题。* 与 GRUB 相比，文件名并不是 rEFInd 的限制；这两个引导程序都可以通过手动配置引导节来启动任何文件名的内核和引导加载程序。区别在于，rEFInd 可以自动检测特定位置和特定文件名的引导加载程序（包括 Linux 内核）。这是 rEFInd 相对于 GRUB 的一个附加功能。因此，比较两者，不足之处在于 GRUB，而不是 rEFInd！实际上，我设计 rEFInd（或者说扩展 rEFIt 的操作系统扫描功能；当然，我们应该感谢 rEFIt 的创建者 Christoph Pfisterer 设计了基础框架）的初衷与你的需求在总体上是一致的：以极少的额外配置启动任何操作系统的引导加载程序。rEFInd 不依赖 NVRAM 条目来实现这一点，因为 NVRAM 条目可能存在误差，但 rEFInd 会扫描引导加载程序应该所在的位置，以及一些引导加载程序（尤其是 Linux 内核）在实践中常见的其他位置。使用 rEFInd 的用户告诉我，它能非常可靠地找到引导加载程序和 Linux 内核；我看到的大多数问题报告都与 EFI 错误或在特殊设置中安装 rEFInd 的困难有关，而不是与引导加载程序扫描功能有关。

[Steve Riley](#) wrote on 2014-03-28 00:48:

[Steve Riley](#) 于 2014 年 3 月 28 日 00:48 写道:

At Kubuntu Forums (where I'm an admin), I have documented my UEFI experiences over time. I have come to appreciate its superiority over BIOS. Initially, I had preferred booting the kernel directly via NVRAM variables until I discovered rEFInd. I concur with everything Rod writes about how rEFInd simplifies managing multi-boot. On one system I have set up rEFInd to boot Kubuntu, openSUSE, Arch, and Windows 8. Actually, "set up" implies too much work -- the only default I changed was to use text mode boot rather than graphical. Otherwise, rEFInd is a cinch because its autodetection is so good. I no longer need to mess with NVRAM variables because rEFInd presents a "just right" abstraction layer. For these reasons, I've been encouraging our members to ditch GRUB completely and go with rEFInd instead. (Yes, I am gushing. Oh well!)

我在 Kubuntu 论坛（我是管理员）上记录了我使用 UEFI 的种种体验。我逐渐体会到它比 BIOS 更胜一筹。最初，我更喜欢直接通过 NVRAM 变量启动内核，直到我发现了 rEFInd。我完全赞同 Rod 关于 rEFInd 如何简化多系统启动管理的观点。在一个系统上，我设置了 rEFInd 来启动 Kubuntu、openSUSE、Arch 和 Windows 8。实际上，“设置”这个词听起来有点复杂——我唯一更改的默认设置是使用文本模式启动而不是图形模式启动。除此之外，rEFInd 非常便捷，因为它的自动检测功能非常出色。我不再需要修改 NVRAM 变量，因为 rEFInd 提供了一个“恰到好处”的抽象层。正因如此，我一直鼓励我们的成员彻底放弃 GRUB，转而使用 rEFInd。（是的，我有点激动。好吧！）

Doug wrote on 2014-02-03 21:22:

Doug 于 2014-02-03 21:22 写道:

One thing I _don't_ understand: these FAT partitions: are they little bitty ones that just hold UEFI code? They can't be system partitions, because FAT doesn't support permissions. So what and where are they?

有一件事我_不太明白_: 这些 FAT 分区：它们是用来存放 UEFI 代码的小分区吗？它们不可能是系统分区，因为 FAT 不支持权限控制。那么它们是什么？它们在哪里？

[adamw](#) wrote on 2014-02-03 23:11:

[adamw](#) 于 2014-02-03 23:11 写道:

"EFI system partition" is the name the spec uses to refer to them. They're not "system partitions" in the sense of Unix file permissions or something like that. The sense is more "the stuff here is just uninteresting bits necessary to make the system boot, not anything you're likely to be interested in". The article explains what they're for: put very generically, it's a place where the OS layer can write stuff for the firmware layer to read. The obvious thing that falls into that category is bootloader code, and indeed that is mostly what is put in ESPs. There are other potential uses for it, but the main thing is bootloaders. As the article (I hope) explains, the role of the ESP is essentially to do the job of: * the MBR * the empty space between the MBR/partition table and the first partition in a BIOS/MBR boot. In a BIOS/MBR boot, the bootloader is located in those spaces. In a UEFI boot, it's located on the EFI system partition. The spec explicitly states that you can have as many EFI system partitions as you like on whatever disks you like in whatever locations you like on those disks. There is a specific GPT partition type that identifies them as ESPs. BTW, as a Fun Educational Side Note, this isn't specific to UEFI - lots of other things do this, in fact it seems like every boot method invented since the BIOS/MBR style has done something like this. If you were to dig a PowerPC system out of your local junk heap and install Fedora on it, you'd find you needed either a 'PReP Boot Partition' or an 'Apple Bootstrap Partition'. ARM systems that use uboot need a 'U-Boot Partition'.

规范中用“EFI 系统分区”来指代它们。它们并非像 Unix 文件权限之类的“系统分区”。更确切地说，这里存放的是一些无关紧要的、系统启动所必需的数据，并非用户可能感兴趣的内容。文章解释了它们的用途：简单来说，它是操作系统层写入固件层可读取数据的地方。最明显的例子就是引导加载程序代码，事实上，ESP 分区中主要存放的就是引导加载程序代码。ESP 分区还有其他潜在用途，但主要用途是存储引导加载程序。正如文章（我希望如此）所解释的，ESP 分区的作用本质上是：* 充当主引导记录 (MBR) * 在 BIOS/MBR 启动模式下，MBR/分区表与第一个分区之间的空白区域。在 BIOS/MBR 启动模式下，引导加载程序位于这些空白区域。在 UEFI 启动模式下，引导加载程序位于 EFI 系统分区上。规范明确指出，你可以在任何磁盘上的任何位置创建任意数量的 EFI 系统分区。有一种特定的 GPT 分区类型可以将其标识为 ESP 分区。顺便提一下，这并非 UEFI 特有的——很多其他系统也采用这种做法，事实上，自 BIOS/MBR 启动方式出现以来，几乎所有其他启动方式都采用了类似的机制。如果你从本地的废旧电脑堆里翻出一台 PowerPC 系统，并在上面安装 Fedora 系统，你会发现你需要一个“PReP 启动分区”或“Apple Bootstrap 分区”。使用 u-boot 的 ARM 系统则需要一个“U-Boot 分区”。

[Rod Smith](#) wrote on 2014-02-16 04:05:

[Rod Smith](#) 于 2014 年 2 月 16 日 04:05 写道:

At least some EFIs give the option to read firmware updates from the ESP, so that's another example of what can go there. (Apple's EFI does this, as does the firmware in my ASUS motherboard.) It's possible to place EFI drivers on the ESP so that the firmware can read them. At the moment, this is probably most common in conjunction with a rEFInd installation; rEFInd comes with several EFI filesystem drivers so that rEFInd can read Linux kernels from the Linux /boot directory. In principle, though, you could put a driver for a plug-in Ethernet card, a video card, or whatever on the ESP so as to give the firmware the ability to use that hardware, even if the hardware lacks EFI-compatible firmware itself. A few such drivers do exist, but they're pretty rare. Some Linux boot loaders, such as ELILO, require that the Linux kernel reside on an EFI-readable partition, which in practice makes the ESP the easiest location. Along those lines, in some cases it makes sense to mount the ESP at /boot in Linux, so that the kernel will go in the ESP's root. Some distributions don't like this because they rely on symbolic links or other Unix filesystem features in /boot, but this approach is quite popular with Arch Linux users. The freedesktop.org boot loader proposal (<http://www.freedesktop.org/wiki/Specifications/BootLoaderSpec/>) would encourage, but not

require, mounting the ESP at /boot.

至少有些 EFI 允许从 ESP 分区读取固件更新，这便是 ESP 分区可以容纳的内容的又一个例子。（苹果的 EFI 就是这样做的，我的华硕主板上的固件也是如此。）可以将 EFI 驱动程序放在 ESP 分区上，以便固件可以读取它们。目前，这在配合 rEFInd 安装时最为常见；rEFInd 自带多个 EFI 文件系统驱动程序，以便 rEFInd 可以从 Linux 的 /boot 目录读取 Linux 内核。原则上，您可以将以太网卡、显卡或其他任何硬件的驱动程序放在 ESP 分区上，使固件能够使用该硬件，即使该硬件本身缺少 EFI 兼容的固件。虽然确实存在一些这样的驱动程序，但它们非常少见。一些 Linux 引导加载程序，例如 ELILO，要求 Linux 内核位于 EFI 可读分区上，而实际上，ESP 分区是最便捷的位置。基于此，在某些情况下，将 ESP 挂载到 Linux 的 /boot 目录是合理的，这样内核就会位于 ESP 的根目录。一些发行版不支持这种做法，因为它们依赖于 /boot 目录下的符号链接或其他 Unix 文件系统特性，但这种方法在 Arch Linux 用户中非常流行。freedesktop.org 的引导加载程序提案（<http://www.freedesktop.org/wiki/Specifications/BootLoaderSpec/>）鼓励（但并非强制要求）将 ESP 挂载到 /boot 目录。

[SteveSi](#) wrote on 2014-02-07 13:30:

[SteveSi](#) 于 2014-02-07 13:30 写道：

Hi Very nice article. I am unclear about 'Removable devices' and 'Removable Media'. Most USB hard disks are of the 'Fixed disk' type - the SCSI Removable Media Bit (RMB) is 0. Most USB Flash drives are of the 'Removable' type - the SCSI RMB is 1. Some USB Flash drives (e.g, WinToGo Certified drives) have RMB=0. What does the UEFI BIOS consider to be a 'Removable' drive - does it look at what the drive reports in the RMB - or does it merely assume any mass storage device on an external interface (eSATA, USB, etc.) is a 'Removable' drive?

您好，文章写得很好。我对“可移动设备”和“可移动介质”的概念不太明白。大多数 USB 硬盘属于“固定磁盘”类型——SCSI 可移动介质位（RMB）为 0。大多数 U 盘属于“可移动”类型——SCSI RMB 为 1。有些 U 盘（例如 WinToGo 认证的 U 盘）的 RMB 为 0。UEFI BIOS 是如何判断“可移动”设备的呢？它会查看 RMB 的值吗？还是仅仅假定所有通过外部接口（eSATA、USB 等）连接的大容量存储设备都是“可移动”设备？

[adamw](#) wrote on 2014-02-07 17:29:

[adamw](#) 于 2014-02-07 17:29 写道：

That's an interesting question, and I don't have the answer on hand. Looking through the spec, I don't believe it ever defines what should be considered 'removable' or 'non-removable'. The sensible thing for a firmware to do would of course be to respect the RMB. My opinions on how commonly firmwares do sensible things, are, I believe, on the public record ;) It does explicitly list "Hard Drive" in section 12.3.4 - "This section describes how booting from different types of removable media is handled." - so it clearly considers the case of a 'removable hard drive', but it does not quantify precisely what it means by that. Of course, the distinction isn't actually incredibly important; as I read it, what it basically boils down to is that when everything in the boot list is invalid and the firmware's doing fallback path processing it is required to check removable media before fixed media, and removable media are required in the spec to have only a single EFI system partition.

这是一个很有意思的问题，但我手头没有答案。查阅规范后，我认为它并没有明确定义什么是“可移动”或“不可移动”。固件理应遵循 RMB（可移动资源管理）原则。至于我对固件通常如何行事的看法，我想应该已经公开了；) 规范在 12.3.4 节中明确列出了“硬盘驱动器”——“本节描述了如何处理从不同类型的可移动介质启动的情况。”——所以它显然考虑到了“可移动硬盘驱动器”的情况，但并没有具体说明它的含义。当然，这种区别实际上并不十分重要；我的理解是，当启动列表中的所有选项都无效，固件执行回退路径处理时，它必须先检查可移动介质，然后再检查固定介质，而规范要求可移动介质只能有一个 EFI 系统分区。

[SteveSi](#) wrote on 2014-02-07 17:37:

[SteveSi](#) 于 2014-02-07 17:37 写道：

I suspect the BIOS interrogates devices on external interfaces first. Where the spec. says 'removable devices' it should really say 'external devices'??? It should be simple enough to test, but of course, it will depend on how different BIOS vendors interpret the spec.!

我怀疑 BIOS 会优先检测外部接口上的设备。规范中写的是“可移动设备”，但实际上应该写成“外部设备”才对？测试起来应该不难，当然，这取决于不同 BIOS 厂商对规范的解读！

[adamw](#) wrote on 2014-02-07 17:38:

[adamw](#) 于 2014-02-07 17:38 写道：

Yeah, as the spec doesn't clearly state it, you can't assume all firmwares will do the same thing. And you don't mean "the BIOS". It is not a BIOS. :)
是的，由于规范中没有明确说明，你不能假设所有固件都会做同样的事情。而且你说的不是“BIOS”。它不是 BIOS。：)

Tom wrote on 2014-02-10 15:11:

Tom 于 2014 年 2 月 10 日 15:11 写道：

The gummiboot and refind boot managers use the efi stub
gummiboot 和 refind 启动管理器使用 efi 存根。

John King wrote on 2014-02-14 19:34:

John King 于 2014 年 2 月 14 日 19:34 写道：

Dear Adam, Your description of UEFI is quite the thing. There is so much garbage on the web on UEFI. Your article is both articulate, accurate, and a joy to read. I'm going to use this in my lectures if you don't (and if you do, _' mind). It would be nice to go into a bit more detail on how GNU-Linux and/or Windows 8 boot loads but that would probably overstretch your aim. I love you

亲爱的亚当，你对 UEFI 的描述真是太棒了。网上关于 UEFI 的文章简直是垃圾。你的文章条理清晰、准确无误，读起来也赏心悦目。如果你不打算用这篇文章，我会把它用在我的讲课中（如果你用了，我当然不介意）。如果能更详细地讲解一下 GNU/Linux 和/或 Windows 8 的启动过程就更好了，不过那样可能就有点超出你的写作范围了。爱你！

mossywell wrote on 2014-03-13 12:58:

mossywell 于 2014-03-13 12:58 写道：

You say that "In the BIOS world, absolutely all forms of multi-booting are handled above the firmware layer". Mikhail Ranish actually wrote a multi-boot loader that fit entirely within the MBR code section (Cylinder 0, Head 0, Sector 1) in the late 1990s. I'm not referring to the XOSL implementation and other similar ones that booted code in the partition boot sector, this work of art fit entirely into the MBR. The "GUI" was limited of course, but even to this day, it is a work of poetry, not IT. :)

你说“在 BIOS 世界里，所有形式的多系统启动都是在固件层之上处理的”。但实际上，Mikhail Ranish 在 20 世纪 90 年代末编写了一个完全位于 MBR 代码段（0 柱面，0 磁头，1 扇区）内的多系统启动加载器。我指的不是 XOSL 实现和其他类似的、将启动代码放在分区引导扇区的实现，这个作品完全位于 MBR 内。当然，它的“图形用户界面”功能有限，但即使到了今天，它仍然是一部杰作，而非 IT 作品。：)

Paul wrote on 2014-04-04 13:14:

Paul 于 2014 年 4 月 4 日 13:14 写道：

1. Currently reading the huge web page. Very impressive and delightfully educational. 2. Pale GREY coloured http links should be in a more easily readable colour. 3. Paragraph numbers should be added - to assist referencing and to assist readers returning to the web page to resume reading.

1. 正在阅读这个篇幅很长的网页。内容非常精彩，而且很有教育意义。2. 浅灰色的 http 链接应该换成更易辨认的颜色。3. 应该添加段落编号，以便于引用，也方便读者返回网页继续阅读。

demon wrote on 2014-04-14 05:22:

demon 于 2014-04-14 05:22 写道：

One thing that's worth mentioning re: UEFI vs. BIOS booting is the fact that, with a BIOS boot, you're heavily constrained by the fact that the CPU is starting in 16-bit real mode. The same way 8088- and 8086-based IBM PCs booted over 30 years ago, and it hasn't changed. That's a real pain in the ass for loading things like the Linux kernel and initrd images into extended RAM (or as it's called on a real OS, "RAM"). You have to dick around with poking things through the A20 address gate using the keyboard controller, effectively using it to "ship-in-a-bottle" load the kernel and initrd bit by bit into extended RAM, then either jump into the kernel or flip into protected mode (or long mode on an x86_64 processor) and then jump into the kernel. UEFI, however, is running in protected mode (or long mode), so you're not working with your hands tied behind your back. Otherwise, good and thorough information, thanks!

关于 UEFI 和 BIOS 启动，值得一提的是，BIOS 启动时，CPU 以 16 位实模式启动，这限制了你的操作。就像 30 多年前基于 8088 和 8086 的 IBM PC 一样，至今依然如此。这使得将 Linux 内核和 initrd 镜像等文件加载到扩展 RAM（或像真正的操作系统中那样称为“RAM”）变得非常麻烦。你必须通过键盘控制器，小心翼翼地操作 A20 地址门，相当于用“瓶中船”的方式将内核和 initrd 逐位加载到扩展 RAM 中，然后跳转到内核或切换到保护模式（或 x86_64 处理器上的长模式），然后再跳转到内核。而 UEFI 则直接运行在保护模式（或长模式）下，因此你无需受到这些限制。其他方面信息都很好很全面，谢谢！

[woodelf](#) wrote on 2014-04-17 01:49:

[woodelf](#) 于 2014-04-17 01:49 写道：

Hello Adam Williamson. This is really a great essay! May I re-distribute this essay in my mother language? I like it very much, and it is a good essay for us to learn. If it is OK, I may combine your essay with some of the comments.

你好，亚当·威廉姆森。这篇文章写得真好！我可以把它翻译成我的母语吗？我很喜欢这篇文章，它对我们来说是一篇很好的学习范例。如果可以的话，我可能会把你的文章和一些评论结合起来。

[adamw](#) wrote on 2014-04-17 01:53:

[adamw](#) 于 2014-04-17 01:53 写道：

Yes, absolutely, I give you permission, thanks! Please just drop a comment here with a link to the translation when you're done :)

当然可以，我同意，谢谢！翻译完成后请在这里留言，附上翻译链接即可：)

[woodelf](#) wrote on 2014-05-28 02:24:

[woodelf](#) 于 2014-05-28 02:24 写道：

Here it is: <http://blog.woodelf.org/2014/05/28/how-uefi-works-translated/>. I referred to your source link at the beginning of my post. Anyway, the translation work takes much more time than I expected lol.

链接在此：<http://blog.woodelf.org/2014/05/28/how-uefi-works-translated/>。我在文章开头就提到了你的链接。总之，翻译工作比我想象的要花更多时间，哈哈。

[jtmorris](#) wrote on 2014-06-27 22:21:

[jtmorris](#) 于 2014-06-27 22:21 写道：

I think I might be in love with you. This was totally awesome. I think I might print it out and frame it. Seriously, after years of struggling and researching, I had a pretty good grasp on BIOS and a reasonable grasp on UEFI. But that came at the cost of blood, sweat, tears, and massive headaches. This condensed it into a short, well-written, easy to understand post. I learned a lot, and what I already knew was greatly solidified. Thank you, thank you, thank you!

我觉得我可能爱上你了。这篇文章简直太棒了！我甚至想把它打印出来裱起来。说真的，经过多年的摸索和研究，我对 BIOS 有了相当不错的了解，对 UEFI 也有了一定的认识。但这可是付出了无数心血、汗水、泪水和无数次的头疼。这篇文章把所有内容浓缩成了一篇简短、易懂、条理清晰的文章。我学到了很多，而且我原有的知识也得到了极大的巩固。谢谢，谢谢，谢谢！

[adamw](#) wrote on 2014-06-27 22:28:

[adamw](#) 于 2014-06-27 22:28 写道：

I am deeply offended that you would suggest this post is "short" ;) Seriously, thanks very much, and I'm glad to help!

你居然觉得这篇文章“短”，我感到非常冒犯 ;) 说真的，非常感谢，很高兴能帮到你！

[jtmorris](#) wrote on 2014-06-28 04:53:

[jtmorris](#) 于 2014-06-28 04:53 写道：

Ha ha! Okay, it's not short. It's extremely long. But I read it all in one sitting without alcohol or caffeine, and I finished with a much better understanding of the subject matter. Which is more than I can say for any other document on the boot process I've ever looked at. They're either lacking in information, plain wrong, or stunningly dull, long, and difficult to understand. So... not short. Concise.

哈哈！好吧，它确实不短，非常长。但我一口气读完了，期间没喝酒也没喝咖啡，读完后我对这个主题有了更深入的理解。这比我之前看过的任何其他关于启动流程的文档都要好得多。那些文档要么信息不足，要么完全错误，要么枯燥乏味、冗长晦涩。所以.....它不是短，而是简洁。

Sergio Belkin wrote on 2014-07-06 17:25:

Sergio Belkin 于 2014 年 7 月 6 日 17:25 写道：

Hi Adam, I love your job since Mandriva days. Thanks for explaining this complex topic in such a friendly way. However I have a doubt. We here the vboxdrv kernel module, it's not signed so only works with Secure Boot disabled. What I cannot understand if I should recompile and sign my own kernel, or if can I sign only the module vboxdrv... Thanks in advance!

嗨，Adam，我从 Mandriva 时代就一直很喜欢你的工作。感谢你用如此友好的方式解释这个复杂的问题。不过我有个疑问。我们这里有个 vboxdrv 内核模块，它没有签名，所以只能在禁用安全启动的情况下工作。我不明白的是，我是应该重新编译并签名我自己的内核，还是可以只对 vboxdrv 模块签名.....提前感谢！

David wrote on 2014-07-11 20:41:

David 于 2014 年 7 月 11 日 20:41 写道：

I'd really love to get my hands on (a series of) books about computer hardware history from as far as the first PC onward to cover as much as possible as clear and deep as your article. Thank you very much.

我非常想找到一套关于计算机硬件发展史的书籍，从第一台个人电脑开始，尽可能全面、清晰、深入地介绍，就像您的文章一样。非常感谢。

M. Green wrote on 2014-07-19 23:29:

M. Green 于 2014 年 7 月 19 日 23:29 写道：

Thanks... best article I've found on UEFI and secure boot. I'm an IT professional with over 35 years of hardware and software experience. I've been working with hard drives and disk booting since we had to low level format each drive and scan for sector errors before shipping a PC. Back then you could only boot from hard drives if you had a BIOS extension ROM plugged in to a PC slot. Ah... BIOS extension ROMs... those were the days. LOL Anyway, I've been trying to figure out for months if I can enable secure boot and still swap drives in my laptop. There was so much conflicting info on UEFI out there that I was concerned that turning SB on would tie the laptop to just a single bootable drive and that I wouldn't be able to boot from the old drive once the OS was installed on the new one. Now after reading this I understand that as long as both drives have signed operating systems on them I can swap them at will. Now I can finally format a new hybrid drive with Win8.1 Pro and still boot from the Win8.1 Home drive that came with the

system if I realize I left something important on it. Again, thank you!

谢谢.....这是我找到的关于 UEFI 和安全启动最好的文章。我是一名拥有超过 35 年硬件和软件经验的 IT 专业人士。我从电脑出厂前必须对每个硬盘进行低级格式化并扫描扇区错误的时候就开始接触硬盘和磁盘启动了。那时候，只有在电脑插槽中插入 BIOS 扩展 ROM 才能从硬盘启动。啊.....BIOS 扩展 ROM.....真是怀念啊。哈哈。总之，几个月来我一直在研究能否在笔记本电脑中启用安全启动并同时更换硬盘。关于 UEFI 的信息太多太杂，我担心启用安全启动后，笔记本电脑就只能从单个启动盘启动，一旦操作系统安装到新硬盘上，我就无法从旧硬盘启动了。现在读了这篇文章，我明白了，只要两个硬盘上都安装了签名的操作系统，我就可以随意更换它们。现在我终于可以格式化一块装有 Win8.1 专业版的混合硬盘了，而且如果我发现系统自带的 Win8.1 家庭版硬盘里留了重要文件，仍然可以从上面启动。再次感谢！

Mariano S. wrote on 2014-07-22 10:26:

Mariano S. 于 2014-07-22 10:26 写道：

A tear rolled in my eye when I find that "someone else" has gotten to appreciate the magic of this program. I'm using it to this date, safely booting 7 OS from a single drive. Actually I use the beta-beta-beta and I've been using it for so many years that I want to find the coder to tell him he can remove that "beta" tag from it :)

看到“其他人”也领略到这款程序的魅力，我感动得热泪盈眶。直到今天，我还在用它，安全地从单个硬盘启动了 7 个操作系统。实际上，我用的是 beta-beta-beta 版本，而且已经用了这么多年，我都想找到开发者，告诉他可以把“beta”标签去掉了:)

Sergio Belkin wrote on 2014-08-03 03:25:

Sergio Belkin 于 2014 年 8 月 3 日 03:25 写道：

One more question, if I understood well, UEFI can access to kernel, so Do we still need GRUB to boot Linux? Thanks in advance!

还有一个问题，如果我理解正确的话，UEFI 可以访问内核，那么我们还需要 GRUB 来启动 Linux 吗？提前感谢！

Kalashnikov wrote on 2014-08-12 09:51:

Kalashnikov 于 2014-08-12 09:51 写道：

You do have some explanation here, but it appears that you have failed to exclude the garbage on the other sites too. Here is how your article appears to readers: Garbage about other sites something about UEFI garbage about other misconceptions something about UEFI garbage and (something about uefi here)funny talk something about UEFI here garbage Try keeping the nonsense entirely out. People are actually spending their precious time reading stuff on the internet. Make it count.

你确实有一些解释，但似乎你也没能把其他网站上的垃圾信息也剔除掉。读者看到你的文章是这样的：关于其他网站的垃圾信息，一些关于 UEFI 的内容，一些关于其他误解的垃圾信息，一些关于 UEFI 的内容，一些垃圾信息，以及（一些关于 UEFI 的内容）一些有趣的闲聊，一些关于 UEFI 的内容，垃圾信息。尽量把这些无意义的内容完全剔除掉。人们花在互联网上阅读的时间是宝贵的，要让时间更有价值。

[adamw](#) wrote on 2014-08-12 11:20:

[adamw](#) 于 2014-08-12 11:20 写道：

This is my personal blog. I'll include as much garbage as I want. You are free, at any point, not to read it.

这是我的个人博客。我会在上面写任何我想写的东西，你想写多少乱七八糟的东西都行。你随时都可以选择不看。

Ralph wrote on 2014-12-13 22:27:

Ralph 于 2014 年 12 月 13 日 22:27 写道：

Hi Adam, GREAT WORK! MANY THANKS! I NEVER learned so much about ANY subject in such a SHORT article! And I LOVE the garbage passages – they really hit the point. I will build a Hackintosh within the next days (for the first time – after beeing pissed off by Apple more and more), and your elucidation will surely be helpful. I will dual boot with Linux – but as I have learned now, I shall use separate disks, what I will observe, though using rEFInd (many thanks to you, Rod, too!) for 2 years without any troble on my Mac mini on ONE disk. Dear greetings from Germany Ralph

嗨，亚当，太棒了！非常感谢！我从未在如此短小的文章中学到过这么多东西！而且我特别喜欢那些“垃圾”段落——它们真的切中要害。我会在接下来的几天内组装一台黑苹果（这是我第一次组装——自从被苹果越来越惹恼之后），你的讲解肯定会很有帮助。我会安装 Linux 双系统——但正如我现在所了解到的，我会使用单独的硬盘，我会观察一下，尽管我在我的 Mac mini 上使用 rEFInd（也非常感谢你，罗德！）已经两年了，而且只用了一块硬盘，一直都没出过问题。来自德国的问候，拉尔夫

henal wrote on 2014-08-24 14:37:

henal 于 2014-08-24 14:37 写道：

Thanks for this information on UEFI. However I have a question concerning the handling of more than one EFI-System partition. Which one will be used when booting ? Will it be selected as we know from BIOS according to the boot sequence list defined in the NVRAM of the motherboard ?

感谢您提供的关于 UEFI 的信息。但我有一个关于处理多个 EFI 系统分区的问题。启动时会使用哪个分区？它会像我们所知的 BIOS 那样，根据主板 NVRAM 中定义的启动顺序列表来选择吗？

Andy wrote on 2014-09-16 10:38:

Andy 于 2014 年 9 月 16 日 10:38 写道:

Thanks for the article. I struggled with UEFI on an ASUS desktop - it came with Windows 8.1 preinstalled. I then rashly installed the latest version of PCLinuxOS (this is not EFI aware) without understanding the implications. This installed itself in BIOS mode with GRUB as the bootloader and gave me options to boot PCLinuxOS and Windows. The option to boot windows failed (seems obvious now) but I then lost the ability to view the Windows UEFI boot manager. In the firmware I tried forcing it just to boot via UEFI but it just refused to start. I tried booting via Ubuntu's boot-repair but this gave an error because the grub-efi package wasn't installed on PCLinuxOS (PCLinuxOS has the package grub2-efi). Finally I installed UEFI aware Ubuntu and it detected the PCLinuxOS and fixed everything up - I can now boot Ubuntu, Windows and PCLinuxOS. What I don't understand is how I got stuck in BIOS mode and couldn't see the Windows EFI boot manager and how Ubuntu fixed that up?

感谢这篇文章。我在一台预装了 Windows 8.1 的华硕台式机上遇到了 UEFI 启动的问题。之后，我草率地安装了最新版本的 PCLinuxOS（它不支持 EFI），当时并没有意识到这样做的后果。PCLinuxOS 默认以 BIOS 模式安装，并使用 GRUB 作为引导程序，同时提供了启动 PCLinuxOS 和 Windows 的选项。启动 Windows 的选项失败了（现在看来很明显），之后我再也无法查看 Windows 的 UEFI 启动管理器了。我尝试在固件中强制系统仅通过 UEFI 启动，但它就是无法启动。我又尝试使用 Ubuntu 的启动修复工具，但出现了错误，因为 PCLinuxOS 上没有安装 grub-efi 软件包（PCLinuxOS 使用的是 grub2-efi 软件包）。最后，我安装了支持 UEFI 的 Ubuntu 系统，它检测到了 PCLinuxOS 并修复了所有问题——现在我可以正常启动 Ubuntu、Windows 和 PCLinuxOS 了。我不明白的是，为什么我会卡在 BIOS 模式，看不到 Windows EFI 启动管理器，以及 Ubuntu 是如何解决这个问题的？

Lu wrote on 2014-09-18 15:32:

Lu 于 2014-09-18 15:32 写道:

Thankyou so so very much for this info.

非常感谢您提供这些信息。

fru franklin wrote on 2014-10-03 09:45:

fru franklin 于 2014-10-03 09:45 写道:

i cant access the BIOS or boot menu of my laptop acer aspire V5-571. I'll be happy if you to get you're help.

我的宏碁 Aspire V5-571 笔记本电脑无法进入 BIOS 或启动菜单。如果您能提供帮助，我将不胜感激。

Mertsch wrote on 2014-10-05 13:21:

Mertsch 于 2014 年 10 月 5 日 13:21 写道:

Thank you very much for this article, I feel very informed about UEFI now. Also thanks for clearing up the "Secure Boot hate" which I only heard on the side and never really knew what it was about.

非常感谢您撰写这篇文章，我现在对 UEFI 有了更深入的了解。另外，也感谢您澄清了“安全启动”的种种误解，我之前只是略有耳闻，从未真正了解过它究竟是什么。

[Dean](#) wrote on 2014-10-11 01:34:

[Dean](#) 于 2014 年 10 月 11 日 01:34 写道:

There is a UEFI BIOS practically. For compatibility reasons(some OSes don't support UEFI), recent PC firmware always support both legacy BIOS and UEFI. A UEFI BIOS normally refers to a UEFI-compatible BIOS, or a mixture of UEFI and legacy BIOS.

实际上存在 UEFI BIOS。出于兼容性考虑（某些操作系统不支持 UEFI），最新的 PC 固件通常同时支持传统 BIOS 和 UEFI。UEFI BIOS 通常指的是兼容 UEFI 的 BIOS，或者 UEFI 和传统 BIOS 的混合体。

Steve Riley wrote on 2014-10-11 02:15:

Steve Riley 于 2014 年 10 月 11 日 02:15 写道:

No. You have UEFI _or_ BIOS. If you have UEFI, then you also have a BIOS compatibility mode, in which the UEFI emulates BIOS. "UEFI-compatible BIOS" means nothing. The only "mixture" I'm aware of is that goofy "Hybrid EFI" firmware that Gigabyte sold for a while. It was a nightmare, as Rod explains: <http://www.rodsbooks.com/gb-hybrid-efi/>

不，你只有 UEFI 和 BIOS 两种选择。如果你有 UEFI，那么你也有一个 BIOS 兼容模式，在这种模式下，UEFI 会模拟 BIOS。“UEFI 兼容 BIOS”毫无意义。我所知道的唯一“混合”方案是技嘉曾经销售过的那个奇葩的“混合 EFI”固件。正如 Rod 解释的那样，那简直是一场噩梦：<http://www.rodsbooks.com/gb-hybrid-efi/>

[adamw](#) wrote on 2014-10-11 09:09:

[adamw](#) 于 2014-10-11 09:09 写道:

Dean: Steve is correct, except that some recent systems are now being sold with UEFI firmwares that have no CSM (BIOS compatibility mode).

迪恩：史蒂夫说得对，只是现在一些新推出的系统配备了没有 CSM（BIOS 兼容模式）的 UEFI 固件。

fisher wrote on 2014-10-19 06:21:

fisher 于 2014-10-19 06:21 写道:

Hi Adam, I have a questuion . If a USB flash drive or an optical disc are not GPT partitioned , but it does have the file EFIBOOTBOOT{machine type short-name}.EFI on it , Can it boot in UEFI-native mode? can uefi detect the bootloader automatically? Thank you.

嗨，Adam，我有个问题。如果 U 盘或光盘没有 GPT 分区，但里面有 EFIBOOTBOOT{机器类型简称}.EFI 文件，它能以 UEFI 原生模式启动吗？UEFI 能自动检测到引导加载程序吗？谢谢。

[adamw](#) wrote on 2014-10-19 15:40:

[adamw](#) 于 2014-10-19 15:40 写道:

The answer to that is 'possibly' => There's an MBR partition type for EFI system partitions, and strictly according to the spec, firmwares are supposed to respect it. But it's not something we think has been very widely used in The Real World and I haven't tested it for real myself. But yeah, in theory if a disk has a partition with the MBR ESP partition type and it's correctly laid out, the firmware should be capable of doing a UEFI native boot from it, AIUI.

答案是“可能”=> EFI 系统分区确实有 MBR 分区类型，严格按照规范，固件应该支持这种分区类型。但我们认为这种分区类型在实际应用中并不常见，我自己也没有进行过实际测试。不过，理论上来说，如果磁盘上有一个 MBR ESP 分区类型且布局正确，那么固件应该能够从该分区进行 UEFI 原生启动，至少我是这么认为的。

Honiix wrote on 2015-08-06 15:46:

Honiix 于 2015-08-06 15:46 写道:

Some tools that create a installation-media for native UEFI system, like Rufus, don't initialize the USB-stick as GPT but MBR !?! O_o I've tried to create a GPT USB stick with an ESP partition but I failed. On Windows, Diskpart crashes if you try to do so. I haven't tested with Linux yet. Because the Rufus's way works on every laptop and desktop I've inserted the MBR-USB-stick into. Also, I spend alot of time googling for a guide to create a USB in GPT with a ESP partition and found nothing. Why is that? My guess is, ESP partition is optional on removable media !?

有些工具，比如 Rufus，用来创建 UEFI 系统安装介质，但它们不会将 U 盘初始化为 GPT 分区表，而是 MBR 分区表！我尝试创建一个带有 ESP 分区的 GPT U 盘，但失败了。在 Windows 系统下，如果尝试这样做，Diskpart 会崩溃。我还没有在 Linux 系统下测试过。因为 Rufus 的方法在我插入过的所有笔记本电脑和台式机上都能正常工作，即使插入的是 MBR 分区的 U 盘。此外，我花了很多时间在網上搜索如何创建带有 ESP 分区的 GPT U 盘，但一无所获。这是为什么呢？我猜测，ESP 分区在可移动介质上是可选的！？

[SteveSi](#) wrote on 2014-10-19 15:51:

[SteveSi](#) 于 2014-10-19 15:51 写道:

Any UEFI system should be able to boot from a plain FAT16 or FAT32 MBR Primary partition. That is in the spec. For instance, make a FAT32 USB Drive and extract the files from a Win8 x64 ISO to it and it will UEFI boot. Easy2Boot uses this feature to boot multiple UEFI payloads from one multiboot USB drive. The partition should not need to be marked as Active (though some UEFI systems seem to require this!) and it should be the first Primary partition on the drive (though it may still boot on some systems if it is not the 1st partition).

任何 UEFI 系统都应该能够从普通的 FAT16 或 FAT32 MBR 主分区启动。这是规范中明确规定的。例如，制作一个 FAT32 格式的 U 盘，并将 Win8 x64 ISO 镜像文件提取到其中，即可实现 UEFI 启动。Easy2Boot 正是利用这一特性，从一个多启动 U 盘启动多个 UEFI 程序。该分区无需标记为活动分区（尽管某些 UEFI 系统似乎需要这样做！），并且应该是 U 盘上的第一个主分区（尽管在某些系统上，即使它不是第一个分区，也可能仍然可以启动）。

Max wrote on 2014-10-27 02:17:

Max 于 2014-10-27 02:17 写道:

"If you have a UEFI-based system whose firmware has the BIOS compatibility feature, and you decide to use it, and you apply this decision consistently, then as far as booting is concerned, you can pretend your system is BIOS-based, and just do everything the way you did with BIOS-style booting. If you're going to do this, though, just make sure you do apply it consistently. I really can't recommend strongly enough that you do not attempt to mix UEFI-native and BIOS-compatible booting of permanently-installed operating systems on the same computer, and especially not on the same disk. It is a terrible terrible idea and will cause you heartache and pain. If you decide to do it, don't come crying to me." I'm metaphorically crying ... Hello Adam, I've done a few things of the previous lines, and my computer is behaving kind of weird. I installed Debian as LVM and on BIOS mode with a GPT formatted disk, and It got really slow. After that, I deleted the entire disk and tried to install it again but in UEFI mode. However, when I turned on the UEFI mode, the UEFI boot manager didn't recognize my disk, only the CDDVD (It does recognize the Debian installation cd but does not initialize it) and USB's units. I thought it was because I didn't have an ESP, so I installed it and didn't work. I gave up and changed it to the BIOS mode , and magically, my CDDVD unit started working, but since I have read the UEFI mode advantages, I would like to know how I can fix the apparent problem. For the record, I have an USB with a UEFI Shell which I tried to use through STARTUP.nsh but the script doesn't start and makes me think I don't have the latest version. Also, I have

the gdisk tool in a Hiren's CD. Thanks in advanced for reading and your possible help.

“如果你有一个基于 UEFI 的系统，其固件具有 BIOS 兼容功能，并且你决定使用它，并且始终如一地执行此操作，那么就启动而言，你可以假装你的系统是基于 BIOS 的，并像使用 BIOS 启动一样进行所有操作。但是，如果你要这样做，请务必确保始终如一地执行此操作。我强烈建议你不要尝试在同一台计算机上，尤其是在同一块硬盘上，混合使用 UEFI 原生启动和 BIOS 兼容启动来启动永久安装的操作系统。这是一个极其糟糕的主意，会让你痛苦不堪。如果你执意要这样做，别来找我哭诉。”我简直要崩溃了……你好 Adam，我按照前面几行做了一些事情，但我的电脑现在有点奇怪。我使用 GPT 格式的硬盘，以 LVM 模式和 BIOS 模式安装了 Debian，结果电脑运行速度变得非常慢。之后，我删除了整个磁盘，并尝试以 UEFI 模式重新安装。然而，启用 UEFI 模式后，UEFI 启动管理器无法识别我的磁盘，只能识别 CD/DVD（它能识别 Debian 安装光盘，但无法初始化）和 U 盘。我以为是因为我没有 ESP 分区，所以安装了 ESP 分区，但仍然无效。我放弃了，切换到 BIOS 模式，神奇的是，我的 CD/DVD 驱动器开始工作了。但我读过一些关于 UEFI 模式优势的文章，所以想知道如何解决这个明显的问题。顺便一提，我有一个装有 UEFI Shell 的 U 盘，我尝试通过 STARTUP.nsh 脚本启动它，但脚本无法运行，这让我怀疑我的脚本版本可能不是最新的。另外，我还有一张 Hiren's CD，里面有 gdisk 工具。感谢您的阅读和可能的帮助。

Stinger wrote on 2014-10-27 11:41:

Stinger 于 2014 年 10 月 27 日 11:41 写道：

Hi Adam Thanks for a great 'essay' :) It's hard to come by relevant info regarding (U)EFI booting , so I'm glad I found this page. Most of what I have seen on fora's and websites are, as you so nicely put it, half baked truths, propaganda or downright lies. Many (I don't know how many) potential new Linux users hit a brick wall when they first try a Linux distro because their pc either can't boot the install/live media of the distro or, even worse, if they get it to install they cant see or boot the installed system. As I see it, this can be caused by two things ? 1. The distro has poor support for UEFI firmware. 2. The pc has a poorly designed UEFI firmware. Hypothetical, to get as many as possible to enjoy their first encounter with a distro, you would have to have good support for UEFI in general and beyond that, you would have to do your best to deal with poorly designed UEFI firmware. All this should, in my opinion be part of the live/installer media and incorporated in the installation routine, preferably dealt with automatically. With my limited knowledge, the first obstacle is booting the live/installer media on a UEFI system but many distros already have dealt with that so I guess that's not the biggest problem. As I see it, the biggest problem is that the installation routine of many distros doesn't deal gracefully with installing on a UEFI pc, leaving the user to try and deal with it themselves afterwards (even worse if its a multiboot scenario) . Now to my question: What would in your opinion be the best way to get a Linux distro installed on a UEFI system ? Is it as Rod Smith suggests to use 'the kernel's EFI stub loader (in conjunction with rEFInd or gummiboot, if necessary)' ? Your opinion would be highly appreciated :)

嗨，Adam，感谢你精彩的“文章”:) 关于 (UEFI) 启动的相关信息很难找到，所以我很高兴找到了这个页面。正如你所说，我在论坛和网站上看到的大部分内容都是不成熟的信息、宣传，甚至是彻头彻尾的谎言。很多（我不知道具体有多少）潜在的 Linux 新用户在第一次尝试 Linux 发行版时都会遇到瓶颈，因为他们的电脑要么无法启动发行版的安装/Live 介质，要么更糟糕的是，即使安装成功，也无法看到或启动已安装的系统。在我看来，这可能是由以下两个原因造成的：1. 该发行版对 UEFI 固件的支持不佳。2. 电脑的 UEFI 固件设计不佳。假设要让尽可能多的人享受他们第一次接触 Linux 发行版的体验，首先需要对 UEFI 提供良好的支持，其次，还要尽力解决设计不佳的 UEFI 固件问题。我认为所有这些都应该包含在 Live/安装程序介质中，并集成到安装流程中，最好是自动处理。以我有限的知识来看，第一个障碍是在 UEFI 系统上启动 Live/安装程序介质，但许多发行版已经解决了这个问题，所以我认为这并非最大的难题。在我看来，最大的问题是许多发行版的安装程序无法优雅地处理在 UEFI 电脑上的安装，导致用户之后不得不自行处理（如果是多系统启动的情况则更糟）。现在我的问题是：您认为在 UEFI 系统上安装 Linux 发行版的最佳方法是什么？是否如 Rod Smith 建议的那样，使用“内核的 EFI stub 加载器（必要时可配合 rEFInd 或 gummiboot）”？非常感谢您的意见:)

SomeName wrote on 2014-10-28 16:16:

SomeName 于 2014-10-28 16:16 写道：

Really nice article! I like your sarcastic style of writig ;)

文章写得真好！我喜欢你这种略带讽刺的写作风格 ;)

Rajan K Srivastava wrote on 2014-12-02 00:33:

Rajan K Srivastava 于 2014-12-02 00:33 写道：

This is an excellent blog, it addresses several doubts -- a big THANKS. One question: when an x86 is powered on, which component (software/hardware) of x86 reads the GPT and then transfers the control to first instruction of UEFI executable?

这是一个很棒的博客，解答了我的几个疑问——非常感谢！还有一个问题：当 x86 系统上电时，x86 的哪个组件（软件/硬件）读取 GPT 分区表，然后将控制权移交给 UEFI 可执行文件的第一条指令？

Yash Pal wrote on 2014-12-24 06:47:

Yash Pal 于 2014-12-24 06:47 写道：

Thanks for the article. It does give a good overall introduction to UEFI.

感谢这篇文章。它对 UEFI 进行了很好的总体介绍。

Kan wrote on 2014-12-30 04:42:

Kan 于 2014-12-30 04:42 写道：

This is really a bad article: * It's not organized well, the concepts do not follow natural logical flow of understanding. I'd like to know, at the high level, how BIOS works vs how UEFI works, and highlight the differences. * The author added a lot of useless words/sentences which do not actually help understanding, but only to distract the readers. * The author did not use concrete examples to illustrate important points. 5 Minutes into the reading, I still don't get any clue of how UEFI works even at the high level. Waste of my time!

这篇文章真是太糟糕了：* 文章结构混乱，概念之间的逻辑混乱，让人难以理解。我只想了解 BIOS 和 UEFI 的工作原理，并重点说明它们的区别。* 作者添加了很多无用的词句，这些词句不仅没有帮助理解，反而分散了读者的注意力。* 作者没有使用具体的例子来说明重点。读了 5 分钟，我仍然完全不明白 UEFI 的工作原理，哪怕是从宏观层面来说。真是浪费时间！

Nate wrote on 2015-01-29 17:16:

Nate 于 2015-01-29 17:16 写道：

For what it's worth, in the industry, on dell's newest servers you must enter system bios and within you can select bios or uefi as your boot mood, so they are using the same term for two different things (the term for the basic input/output system and also the firmware).

值得一提的是，在业内，戴尔最新的服务器上，你必须进入系统 BIOS，然后在其中选择 BIOS 或 UEFI 作为启动模式，所以他们用同一个术语来指代两件不同的事情（基本输入/输出系统和固件）。

oolu wrote on 2015-01-31 17:29:

olu 于 2015-01-31 17:29 写道:

It is possible to transform GPT into MBR fluently without data loss, see gdisk (maybe use "m" or "x" command then choose Transofrm MBR to GPT and vice versa)

可以将 GPT 格式流畅地转换为 MBR 格式而不会丢失数据，请参阅 gdisk（可以使用“m”或“x”命令，然后选择“从 MBR 转换为 GPT”或“从 MBR 转换为 GPT”）。

Simon Trangmar wrote on 2015-02-08 23:39:

Simon Trangmar 于 2015-02-08 23:39 写道：

Dear Adam Came across your article googling 'GPT and MBR - differences' - while trying to work out why in hell I wasn't able to restore a GHOST image to the same machine. Found it very helpful (although I had to read it copious times and with copious amounts of gin - as recommended!) Thanks for all the time and effort you have put into this article - I really appreciate it. Simon / Adelaide | Oz

亲爱的亚当，我在谷歌上搜索“GPT 和 MBR - 区别”时偶然发现了你的文章——当时我正在努力弄明白为什么我无法将 GHOST 镜像恢复到同一台机器上。这篇文章对我帮助很大（虽然我不得不反复阅读，而且还喝了不少杜松子酒——正如文章建议的那样！）。感谢你为这篇文章付出的时间和精力——我非常感激。西蒙/阿德莱德 | 澳大利亚

Joe Pesco wrote on 2015-02-14 23:07:

Joe Pesco 于 2015 年 2 月 14 日 23:07 写道：

Humphrey Bogart was regular everyday folk and George Clooney is regular everyday folk at least in their movie roles. adamW gets the Peter Venkman award for saving us time with these updates!

亨弗莱·鲍嘉和乔治·克鲁尼在电影角色中都扮演着普通人的角色。adamW 因这些更新节省了我们的时间，荣获彼得·文克曼奖！

Anon wrote on 2015-03-08 12:24:

匿名用户于 2015 年 3 月 8 日 12:24 写道：

If you absolutely insist on having more than one OS per disk, understand everything written on this page, understand that you are making your life much more painful than it needs to be, lay in good stocks of painkillers and gin, and don't go yelling at your OS vendor, whatever breaks.

如果你坚持要在每个磁盘上安装多个操作系统，请理解本页所写的所有内容，明白你正在让自己的生活变得比原本应该的痛苦得多，准备好充足的止痛药和杜松子酒，无论系统出现什么问题，都不要对你的操作系统供应商大喊大叫。

This seems to imply that UEFI is doing a pretty damn bad job, if something as simple as having a dual boot has to be a massive pain and recommended-against.

这似乎暗示 UEFI 的表现非常糟糕，因为像双启动这样简单的操作都会变得极其麻烦，甚至不建议使用。

[adamw](#) wrote on 2015-03-08 21:44:

[adamw](#) 于 2015-03-08 21:44 写道：

Oh, that's intended to apply equally to BIOS. Dual boot isn't really simple and never has been, it's an excellent source of unnecessary annoyances in my book.

哦，这同样适用于 BIOS。双启动其实并不简单，从来就不简单，在我看来，它只会带来很多不必要的麻烦。

Chris Chua wrote on 2015-03-08 18:34:

Chris Chua 于 2015-03-08 18:34 写道:

+1 for using the word "janky"

+1, 因为用了“janky”这个词。

Dave Thompson wrote on 2015-03-11 08:27:

Dave Thompson 于 2015 年 3 月 11 日 08:27 写道:

About the statement that all x86 MUST be able to turn off UEFI/secure boot For at least AO725 there was no entry in the bios, after much search you MUST set supervisor password in bios before these entries appear PLEASE, please choose a simple password or you will have to do an emergency bios restore (Fn-Esc, right name for bios, in this case in / of UFD, not/EFI/BOOT) My friend wanted a complicated password, I couldn't be bothered to remember, he forgot!!!!

关于“所有 x86 架构的电脑都必须能够关闭 UEFI/安全启动”的说法，至少在 AO725 上，BIOS 里没有相关选项。经过一番搜索，我发现必须先要在 BIOS 里设置管理员密码，这些选项才会出现。请务必选择一个简单的密码，否则你将不得不进行紧急 BIOS 恢复（Fn-Esc，BIOS 的正确名称，在本例中位于 U 盘根目录下，而不是 EFI/BOOT）。我的朋友想要一个复杂的密码，我懒得记，结果他自己也忘了！

[Richard](#) wrote on 2015-04-02 11:55:

[Richard](#) 于 2015 年 4 月 2 日 11:55 写道:

Very well Written. And Clear. And after reading, I can safely say.... What the hell is INTEL doing! I can understand the desperate need to replace the aging BIOS with a more robust and functional TIER system, but this was FAR from what i was expecting to happen. Ill share my point, 15 years ago i stubled across "Terabyte Unlimiteds BootIT Next Generation" software. used for boot management. It fits on a 1440kb floppy disk. It installs using the MBR a hugely improved MBR loader, affectionatly named EMBR. Back then it was was EMBR 1.0. its since been improved. EMBR allowed the Boot Manager (Part of BOOTIT) Boot any partition from any drive, anywhere. (including removable devices) Have near UNLIMITED partitions of any kind on near UNLIMIED attached drives. TRUE hide drive(s) and partition(s) from the Operating System you choose to boot. And best of all it was Completely Configurable with SECURITY with a SIMPLE and CLEAN LOOKING Bootloader Menu, intergrated PARTITION AND DRIVE MANAGER able to size, resize, Move, copy, create, delete partitions of ANY kind with SPEED and 100% reliablity. And all this fit on 1 floppy disk which could have been made in to a firmware BIOS. I still use it today on BLADE SERVERS !! as well as home PC's Intel should have looked around FIRST before trying to build a better BIOS, because others out there had the STANDARD in place ALREADY. Im sticking with EMBR, it works without complication, and its a hell of a lot easier to understand.

写得很好，也很清晰。读完之后，我可以很肯定地说.....英特尔到底在干什么！我能理解他们迫切需要用更强大、功能更完善的 TIER 系统来替换老旧的 BIOS，但这远远超出了我的预期。我来分享一下我的观点：15 年前，我偶然发现了“Terabyte Unlimited BootIT Next Generation”这款用于启动管理的软件。它的大小可以装在一张 1440KB 的软盘上。它使用 MBR 安装一个经过大幅改进的 MBR 加载器，我们亲切地称之为 EMBR。当时是 EMBR 1.0 版本，后来又进行了改进。EMBR 允许启动管理器（BootIT 的一部分）从任何驱动器的任何分区启动（包括可移动设备）。它允许在几乎无限数量的连接驱动器上创建几乎无限数量的任何类型的分区。它还可以真正地对操作系统隐藏驱动器和分区，使其不被操作系统识别。最棒的是，它完全可配置，安全性极高，拥有简洁明了的引导加载程序菜单，集成了分区和驱动器管理器，能够快速可靠地调整、移动、复制、创建和删除任何类型的分区。所有这些功能都集成在一张软盘上，这张软盘完全可以做成固件 BIOS。我至今仍在刀片服务器和家用电脑上使用它！英特尔在尝试开发更好的 BIOS 之前，应该先看看其他厂商已经提供了标准。我还是坚持使用 EMBR，它运行稳定，而且更容易理解。

[Richard](#) wrote on 2015-04-02 12:03:

[Richard](#) 于 2015-04-02 12:03 写道:

I neglected to add, EMBR also blows away the drive space limits i.e 2TB, It was past that issue 9 years ago

我忘了补充一点，EMBR 也突破了硬盘空间限制，比如 2TB。这个问题 9 年前就已经解决了。

Joe wrote on 2015-04-06 17:58:

Joe 于 2015-04-06 17:58 写道:

There is nothing here that sells me on using UEFI on my computer. I am not changing the way I describe computers, to recognize UEFI as some kind of important development, at least until I know why it is a good idea. A BIOS is a firmware that starts a computer. Therefore, a UEFI is a BIOS as far as I'm concerned. When I change my BIOS so that I can start my system on the basis of dependable 1980s technology, I disable UEFI. Why is it better for the UEFI bootloader code to be located on the motherboard instead of on the hard drive anyway? It just scatters the data in the computer. I don't feel a better sense of security over my data by having Microsoft offload its operating system onto my computer's motherboard instead of keeping it contained on the hard drive. (I know that Microsoft did not invent the part of the Windows which has been offloaded onto the motherboard. But Microsoft insists that this non-Microsoft code be incorporated into the design, so it is part of the Windows operating system. Otherwise, Windows could work without it. UEFI gets pushed in the way of other operating systems which are forced to work around the intrusion.) I am annoyed with Microsoft about my lack of good access to the firmware boot manager. My hardware vendor may have been able to mitigate my inconvenience by designing better support, but I dwell more on the fact that Microsoft pressured my vendor to implement UEFI in the first place, and I do not want UEFI. Also, Microsoft has been refusing to license the sale of Windows on ARM devices that allow for legacy boot or Insecureboot. If it's not Secureboot, it must be Insecureboot, right? The blog post here observes that it is not a good idea to mix legacy boot with UEFI native boot. I need legacy boot, and I don't have the choice to keep it unmixed with native

UEFI boot. My BIOS is designed to scan the computer for internal and external drives that are natively bootable by UEFI. Then, it preferentially boots one of those drives, even if I have legacy boot turned on. It only boots legacy-style if a drive that is natively bootable by UEFI cannot be found.

这里没有任何理由让我选择在我的电脑上使用 UEFI。我不会改变我对电脑的描述方式，也不会把 UEFI 视为某种重要的发展，至少在我弄明白它到底好在哪里之前不会。BIOS 是启动电脑的固件。因此，在我看来，UEFI 也是 BIOS。当我修改 BIOS，让系统能够基于可靠的 80 年代技术启动时，我就会禁用 UEFI。无论如何，UEFI 引导加载程序代码放在主板上而不是硬盘上有什么好处呢？这只会让电脑里的数据更加分散。微软把操作系统放到我的电脑主板上，而不是放在硬盘上，并不能让我感到数据更安全。（我知道微软并没有发明 Windows 系统中被卸载到主板上的那部分代码。但微软坚持要求将这段非微软代码整合到设计中，使其成为 Windows 操作系统的一部分。否则，Windows 完全可以脱离它运行。UEFI 的强推阻碍了其他操作系统的发展，迫使它们不得不绕过这种“入侵”。）我对微软让我无法有效访问固件启动管理器感到恼火。我的硬件供应商或许可以通过设计更好的支持来减轻我的不便，但我更在意的是，微软当初施压我的供应商来实现 UEFI，而我并不想要 UEFI。此外，微软一直拒绝授权销售支持传统启动或不安全启动的 ARM 设备上的 Windows 系统。如果不是安全启动，那肯定就是不安全启动，对吧？这篇博文指出，将传统启动与 UEFI 原生启动混合使用并非明智之举。我需要传统启动，而且没有办法将其与原生 UEFI 启动完全分开。我的 BIOS 设计为扫描计算机中支持 UEFI 原生启动的内置和外置硬盘。然后，即使我启用了传统启动，它也会优先从这些硬盘启动。只有当找不到支持 UEFI 原生启动的硬盘时，才会采用传统启动方式。

[adamw](#) wrote on 2015-04-06 18:47:

[adamw](#) 于 2015-04-06 18:47 写道：

None of that reads like you actually read or understood my blog post, so my response is to tell you to actually read it and understand it.

你的回复完全不像是在认真阅读或理解我的博客文章，所以我建议你认真阅读并理解它。

[SteveSi](#) wrote on 2015-04-06 18:30:

[SteveSi](#) 于 2015-04-06 18:30 写道：

UEFI is needed because the old BIOS+MBR spec only copes with disks of <2TB. Also it caters for different machine/CPU architectures and UEFI can be expanded in the future. If your BIOS will not MBR\CSM boot, then that is the fault of the BIOS/system manufacturer. Many non-UEFI (MBR) systems are still being produced today that cannot boot from a USB drive that has the active partition past 137GB - this is due to poor coding of the BIOS USB driver and the 'feature' has been around for >10 years. Some MBR BIOSes even completely ignore the MBR boot code and just execute the PBR code instead - totally against the original IBM PC spec! You could now blame all this on the UEFI-spec being too vague - i.e. it would have been an *ideal opportunity* to clearly specify both UEFI and MBR booting requirements once and for all and then UEFI\CSM BIOS writers would have a clear spec. to follow. Instead they just copped out by saying 'we will not define the CSM compatibility functions!' There is no official MBR-booting spec. (it just sort of grew with all sorts of geometry translation schemes and bugs for 8GB/137GB limits, etc.!) and AFAIK, there was no developers guide produced on how to code a UEFI-BIOS with examples in pseudo-code, function descriptions, etc. plus what mandatory interfaces to provide to the user. Big mistake!

之所以需要 UEFI，是因为旧的 BIOS+MBR 规范只能处理 2TB 左右的磁盘。此外，UEFI 还能兼容不同的机器/CPU 架构，并且未来可以扩展。如果您的 BIOS 无法启动 MBR/CSM 系统，那是 BIOS/系统制造商的问题。如今，许多非 UEFI（MBR）系统仍然无法从活动分区超过 137GB 的 USB 驱动器启动——这是由于 BIOS USB 驱动程序的编码不佳，而且这个“特性”已经存在了十年之久。一些 MBR BIOS 甚至完全忽略 MBR 启动代码，直接执行 PBR 代码——这完全违背了最初的 IBM PC 规范！现在，您可以将所有这些问题归咎于 UEFI 规范过于模糊——也就是说，这本是一个绝佳的机会，可以一次性明确规定 UEFI 和 MBR 的启动要求，这样 UEFI/CSM BIOS 编写者就能有一个清晰的规范可以遵循。他们却敷衍了事地说：“我们不会定义 CSM 兼容性函数！”目前还没有官方的 MBR 启动规范（它只是随着各种几何转换方案和针对 8GB/137GB 限制等问题的 bug 而逐渐形成的！），而且据我所知，也没有开发者指南指导如何编写 UEFI BIOS，包括伪代码示例、函数描述等等，以及必须向用户提供的接口。这真是个大错误！

Joe wrote on 2015-04-06 19:52:

Joe 于 2015-04-06 19:52 写道：

Whatever nice stuff UEFI does that BIOS does not do could be coded and stored on the hard drive of every Windows system. That way, if I didn't want it, I could simply delete it and still be using native BIOS on my machine. I would have preferred having this UEFI in some "arbitrary" pre-partition area instead of jammed into my motherboard. Then old things could automatically still work on my new computer. My computer came with Windows 8. I went through all the trouble to determine how to make the system bootable in legacy mode. (It is lots and lots of trouble, and I eventually got it working.) My computer now seems to boot every bit as efficiently as it did before the switch. How do I know that I succeeded? I got Diskcryptor full disk encryption to work properly on my computer. It needs the disk to be BIOS bootable. That company that Richard mentioned 2 comments behind my first one, Terabyte, has excellent instructions on how to reconfigure Windows 8 from native UEFI boot to legacy boot. And I too remember Terabyte's EMBR to handle large partitions (>2TB), and bypass the limit of 4 primary partitions. The solution to that problem is to create a standard where partition information does not have to be stuck within 64 bytes in sector 0. Just use the next sector for partitions. BIOS was not causing the 64 byte limit on partition tables. The lack of operating system support for partitions outside of sector 0 was enforcing the limits. All Microsoft had to do to keep BIOS was to invent a partition scheme that uses the next sector, and make their new release of Windows aware of the partitions there. All of the facilities for "understanding" the partitions could have been written in the bootloader. That bootloader could have been in the partition containing Windows. BIOS bootloaders do not all need to be saved in any unpartitioned space except for sector 0. Grub4DOS used sector 0, the PBR of the active partition, and some files in the partition,

if my memory serves me right.

UEFI 的一些 BIOS 无法实现的强大功能，完全可以编写代码并存储在每个 Windows 系统的硬盘上。这样，如果我不需要这些功能，就可以直接删除，继续使用电脑自带的 BIOS。我更希望 UEFI 能放在某个预先分区的“任意”区域，而不是直接集成到主板上。这样，旧程序就能在新电脑上自动运行。我的电脑预装的是 Windows 8。我费了九牛二虎之力才搞清楚如何让系统以传统模式启动。（这真是费了九牛二虎之力，不过最终我还是成功了。）现在我的电脑启动效率和切换之前一样高。我怎么知道我成功了呢？因为我的电脑上 Diskcryptor 全盘加密功能运行正常。它需要磁盘能够从 BIOS 启动。理查德在我第一条评论后面两条提到的那家公司 Terabyte，提供了非常详细的教程，教你如何将 Windows 8 从原生 UEFI 启动重新配置为传统启动。我也记得 Terabyte 的 EMBR 技术，它能够处理大于 2TB 的大分区，并绕过 4 个主分区的限制。解决这个问题的方法是创建一个标准，让分区信息不必局限于 0 号扇区的 64 字节，而是使用下一个扇区。BIOS 并非造成分区表 64 字节限制的原因，而是因为操作系统不支持 0 号扇区之外的分区才导致了这一限制。微软为了保留 BIOS，只需发明一种使用下一个扇区的分区方案，并让新版 Windows 识别这些分区即可。所有“理解”分区的功能都可以写入引导加载程序中，而这个引导加载程序可以放在包含 Windows 的分区里。除了扇区 0 之外，BIOS 引导加载程序并非都需要保存在任何未分区的空间中。如果我没记错的话，Grub4DOS 使用了扇区 0、活动分区的 PBR 以及分区中的一些文件。

[adamw](#) wrote on 2015-04-06 21:10:

[adamw](#) 于 2015-04-06 21:10 写道：

You keep talking about UEFI like it's somehow attached to Windows. It isn't. That's about 50% of the point of the post, which is why I keep saying it sounds like you haven't read it.

你一直把 UEFI 说得好像它跟 Windows 有什么关系似的。其实不然。这大概占了这篇文章要点的一半，所以我才一直说你好像没读过这篇文章。

Joe wrote on 2015-04-09 22:43:

Joe 于 2015-04-09 22:43 写道：

If Microsoft will not allow PC vendors to certify that PCs without UEFI have Windows 8 preinstalled, I take this to mean that UEFI has been absorbed into "Windows", even if it is an outsourced component from Microsoft's frame of reference, and even if UEFI is reusable by incorporating it into other systems. When I follow a guide to make Windows BIOS bootable, I am removing a Windows component to improve the performance of my PC. The latter is a familiar experience to a lot of people. UEFI is not evil, insofar as it is a standard for the behavior of a bootloader. It is bad, at least currently, for it to be on motherboards instead of on hard drives. Being in the firmware, it can abet plans of manufacturers to bind computers to specific operating systems, excluding others. Having UEFI in my PC's firmware does not give me a better sense of security. It gives me a worse one. Bad actors who wrote rootkits used to target the code on hard drives. It was easy to prepare for a recovery from the changes that a rootkit could make by copying the hard drive. The binary backup of the hard drive could be copied back to restore the computer. Now, are the same bad actors going to be induced into finding ways to reflash a victim's firmware? Rootkits that change firmware would most probably get computers bricked. Maybe this risk could be mitigated if it were easy for users to back up and reflash firmware.

如果微软不允许电脑厂商认证那些没有 UEFI 的电脑预装了 Windows 8，我认为这意味着 UEFI 已经被“Windows”吸收了，即便它并非微软自主开发的组件，即便 UEFI 可以通过集成到其他系统中而复用。当我按照指南让 Windows BIOS 可启动时，我实际上是在移除一个 Windows 组件来提升电脑性能。很多人都有过类似的经历。UEFI 本身并非邪恶，因为它是一种引导加载程序行为的标准。但至少目前来看，它位于主板而非硬盘上是不妥的。由于它存在于固件中，可能会助长厂商将电脑绑定到特定操作系统、排除其他操作系统的计划。电脑固件中包含 UEFI 并不会让我感到更安全，反而会更不安全。过去，编写 rootkit 的恶意程序通常会攻击硬盘上的代码。通过复制硬盘，可以轻松应对 rootkit 造成的更改。硬盘的二进制备份可以复制到电脑上，从而恢复系统。现在，那些恶意攻击者是否会想方设法重新刷写受害者的固件呢？修改固件的 rootkit 很可能导致电脑彻底损坏。如果用户能够轻松备份和重新刷写固件，或许可以降低这种风险。

[adamw](#) wrote on 2015-04-10 04:05:

[adamw](#) 于 2015-04-10 04:05 写道：

"If Microsoft will not allow PC vendors to certify that PCs without UEFI have Windows 8 preinstalled, I take this to mean that UEFI has been absorbed into “Windows”

“如果微软不允许 PC 厂商认证没有 UEFI 的 PC 预装了 Windows 8，我认为这意味着 UEFI 已经被‘Windows’吸收了。”

I can't stop you taking whatever you like, but that's clearly not what it means. Microsoft also doesn't certify any PCs without an x86 or ARM CPU; that doesn't mean the x86 and ARM CPU specifications have been 'absorbed into Windows'. UEFI is an independent industry standard; Microsoft chooses to require that Windows 8+ systems use UEFI firmwares, just like it chooses to require that they use x86 or ARM CPUs.

我无法阻止你随意取用，但这显然不是它的意思。微软也不会认证任何没有 x86 或 ARM CPU 的 PC；但这并不意味着 x86 和 ARM CPU 规范已经被“融入 Windows”。UEFI 是一个独立的行业标准；微软选择要求 Windows 8 及更高版本的系统使用 UEFI 固件，就像它选择要求它们使用 x86 或 ARM CPU 一样。

"UEFI is not evil, insofar as it is a standard for the behavior of a bootloader."

“UEFI 本身并不邪恶，因为它是一种引导加载程序行为的标准。”

It is not that. It is a standard for the behaviour of *firmware*. System firmware involves far more than the boot process.

并非如此。这是**固件**行为的标准。系统固件涉及的内容远不止启动过程。

"Having UEFI in my PC's firmware does not give me a better sense of security."

“在我的电脑固件中使用 UEFI 并不能给我带来更好的安全感。”

I don't know why you would bring that up. UEFI does not exist in order to be 'more secure' than BIOS, it exists to be more capable.

我不明白你为什么会提起这个。UEFI 的存在并非为了比 BIOS “更安全”，而是为了功能更强大。

"Now, are the same bad actors going to be induced into finding ways to reflash a victim's firmware?"

“那么，这些不法分子是否会被诱导去寻找方法来重新刷新受害者的固件呢？”

That's exactly as possible with UEFI-without-Secure-Boot as it is with BIOS. No more and no less. On a system with Secure Boot, it is *substantially* less possible than on a UEFI-non-SB or BIOS system.

无论是在不启用安全启动的 UEFI 系统中，还是在 BIOS 系统中，这种情况都完全可能发生。不多也不少。在启用安全启动的系统中，这种情况发生的可能性远低于在不启用安全启动的 UEFI 系统或 BIOS 系统中。

Neill Rutherford wrote on 2015-07-31 12:57:

Neill Rutherford 于 2015 年 7 月 31 日 12:57 写道：

I do not have a UEFI firmware computer. I have an HP G61 64b machine all the rest are desktops and 32b. So my question is regarding using a USB bootable drive in various machines. As I don't have the new hardware I can not examine the how do i do this myself. **For the old legacy BIOS boot firmware** machines there is little problem. providing one can set boot order and it accepts booting from a USB port otherwise booting from an optical disk via chain-loading might be possible (yeah i have machines that old) **For UEFI firmware machines** 1) if the machine in question does not have an internal hard drive. 2) if the machine does have an internal hard drive but would prefer that the internal drive is not accessed in any way. Ideally I would want a bootable USB that would work in any machine. I think this should be possible by booting via MBR on BIOS machines and by booting via the UEFI partition on UEFI machines. What are you thoughts Sorry if my terminology is not correct. Neill

我没有 UEFI 固件的电脑。我有一台 HP G61 64 位机器，其他的都是 32 位台式机。我的问题是关于如何在不同的机器上使用 U 盘启动盘。由于我没有新的硬件，所以无法自己研究具体操作方法。 **对于老式的传统 BIOS 启动固件** 机器来说，问题不大。只要可以设置启动顺序，并且支持从 USB 端口启动，或者也可以通过光盘链式加载启动（是的，我有一些这么老的机器）。 **适用于 UEFI 固件机器** 1) 如果所讨论的机器没有内置硬盘驱动器。 2) 如果机器确实有内置硬盘，但希望不要以任何方式访问内置硬盘。理想情况下，我想要一个能在任何机器上运行的启动 U 盘。我认为可以通过在 BIOS 机器上通过 MBR 启动，在 UEFI 机器上通过 UEFI 分区启动来实现。你有什么想法？如果我的用词不当，请见谅。尼尔

[adamw](#) wrote on 2015-07-31 16:46:

[adamw](#) 于 2015-07-31 16:46 写道：

Well, sure, distributions have already had to deal with this. mjb59 wrote up how we keep the Fedora images bootable on multiple platforms a few years back: <https://mjb59.dreamwidth.org/11285.html>

当然，发行版早就需要处理这个问题了。几年前，mjb59 写过一篇关于我们如何保证 Fedora 镜像在多个平台上都能启动的文章：<https://mjb59.dreamwidth.org/11285.html>

There's a lot of detail, but it boils down to giving the media both MBR and GPT partition tables and including a bootloader in the EFI fallback path location for whatever arch it should boot on.

细节很多，但归根结底就是给介质提供 MBR 和 GPT 分区表，并在 EFI 回退路径位置中包含一个引导加载程序，以适应它应该启动的任何架构。

[SteveSi](#) wrote on 2015-07-31 13:07:

[SteveSi](#) 于 2015-07-31 13:07 写道：

First, if you use VirtualBox+VMM you can boot from a USB drive using UEFI in the Virtual Machine. So you can easily test UEFI booting. See www.rmprepusb.com - Tutorial #4. Next, if the USB drive is FAT32, then you can make it MBR-bootable (normal BIOS) and by placing the .efi boot files on it, UEFI-bootable. e.g. format a USB drive as FAT32, add a bootmgr boot sector (e.g. using RMPrepUSB or bootsect or just format it using Windows - ensure the partition is marked as Active) and then simply copy the contents of a Windows 8.1 64-bit DVD to it. To boot in any machine, you would also need to add 32-bit UEFI boot files. However, there is a fly in the ointment! Many UEFI systems have a 'bug' - if they 'see' a USB drive which has both MBR and EFI files on it, they will NOT allow you to MBR-boot from it (only the UEFI-boot option is offered). What this means is that, on these systems,

you can never MBR-boot from the USB drive! To MBR-boot, you would have to delete the EFI boot files first - and then it would not UEFI-boot!

首先，如果您使用 VirtualBox + VMUB，则可以在虚拟机中使用 UEFI 从 USB 驱动器启动。这样您就可以轻松测试 UEFI 启动。请参阅 www.rmprepusb.com - 教程 #4。其次，如果 USB 驱动器是 FAT32 格式，则可以将其设置为 MBR 启动（普通 BIOS），并通过将 .efi 启动文件放入其中，使其成为 UEFI 启动。例如，将 USB 驱动器格式化为 FAT32，添加一个引导扇区（例如使用 RMPrepUSB 或 bootsect，或者直接在 Windows 中格式化 - 确保分区标记为活动分区），然后将 Windows 8.1 64 位 DVD 的内容复制到其中。要在任何计算机上启动，您还需要添加 32 位 UEFI 启动文件。但是，美中不足的是.....许多 UEFI 系统存在一个“漏洞”——如果它们检测到 U 盘同时包含 MBR 和 EFI 文件，则不允许您从该 U 盘进行 MBR 启动（仅提供 UEFI 启动选项）。这意味着在这些系统上，您永远无法从该 U 盘进行 MBR 启动！要进行 MBR 启动，您必须先删除 EFI 启动文件——但即使删除了 EFI 启动文件，也无法进行 UEFI 启动！

Neill Rutherford wrote on 2015-07-31 20:45:

Neill Rutherford 于 2015 年 7 月 31 日 20:45 写道：

Hello Thank you both for your replies. Just so that you understand where I am coming from I want to create a USB disk or USB disks (One for root / and one for /home) Then in an unknown computer (x86, X86-64 and maybe Mac) which has no internal hard disk I want the USB disk to boot. To cope with low spec machines I would choose Ubuntu 14.04 LTS 32b. The USB will contain an installed working operating system not a liveUSB. Do you see any problems with what I am trying to attempt?

您好，感谢两位的回复。为了让您更清楚地了解我的想法，我想创建一个或多个 U 盘（一个用于根目录/，一个用于/home）。然后，我想在一台没有内置硬盘的未知电脑（x86、x86-64，甚至可能是 Mac）上启动这些 U 盘。为了兼容低配置电脑，我选择 Ubuntu 14.04 LTS 32 位系统。U 盘里会安装一个可正常运行的操作系统，而不是一个 Live USB。您觉得我的做法有什么问题吗？

Jenel wrote on 2015-08-08 14:11:

Jenel 于 2015-08-08 14:11 写道：

I can't get around this: "With a BIOS firmware, your firmware level 'boot menu' is, necessarily, the disks connected to the system at boot time". What is the boot menu, then? Is it "press F10 to choose boot device", or is it lilo / grub / "press F8 for safe boot"?

我一直搞不明白：“使用 BIOS 固件时，固件级别的‘启动菜单’必然是指启动时连接到系统的磁盘”。那么，启动菜单到底是什么？是“按 F10 选择启动设备”，还是 lilo/grub/“按 F8 进入安全启动”？

[adamw](#) wrote on 2015-08-08 15:36:

[adamw](#) 于 2015-08-08 15:36 写道：

"Press F10 to choose boot device". I'm just saying that since all BIOS knows how to do is 'run the boot sector on a disk', the only boot options you have at the BIOS level are 'this disk or that disk'. With UEFI that's not the case, since it's more capable.

“按 F10 选择启动设备”。我的意思是，由于 BIOS 只能“运行磁盘上的引导扇区”，因此在 BIOS 层面上，你唯一的启动选项就是“这个磁盘或那个磁盘”。而 UEFI 则不同，它的功能更强大。

joe jviden wrote on 2015-08-21 06:35:

joe jviden 于 2015-08-21 06:35 写道：

This blog post is aimed at regular everyday folks. I seem to be the only one you hit. Everyone seems to be an IT tech or something. I have had a few problems with a hard drive upgrade on a newer windows 8.1 computer with UEFI. I bought it for the i-5 4570 processor in an otherwise cheaply furnished package. I bought a 2TB SSHD to upgrade the hard drive. I, being a regular everyday folk, used a hard disk cloning device to migrate my operating system. It seemed to work well for a week or so but then I started to get a warning from HP that my hard drive was failing and I needed to run a bunch of diagnostic tests. This would happen at startup after hibernation or an overnight shutdown. I ordered another of the same SSHD and shipped the first one back to Amazon. The second one acted the same as the first one. I tried everything I could think of to fix it but to no avail. I then installed it in an older windows 7 computer and it worked fine and is still working fine many months latter. I then ordered an SSD that came with a disc to do the software migration. It is working fine after many months. I noticed that the disk management utility says that I have a healthy EFI partition. It used to say there was a UEFI partition. HP support assistant has messaged me several times about not having UEFI installed and wants to install it. I have refused to install it as I think I could cause the error messages to appear at boot again. I am still confused as to why an EFI partition would not be considered a UEFI partition and why HP is bugging me about it. I installed windows 10 on my old windows 7 machine yesterday and it got me thinking about how much trouble I could get into installing it on this newer computer with the SSD upgrade. Do you think It will cause trouble? Your article helped me understand why there was a problem with cloning a UEFI hard drive. Since UEFI-based computers have bootable information written in their motherboards, you may need to set related hard drive information into your target computer after migrating to a new hard disk. In the contrary, traditional BIOS-based computers don't need such step: you only need plug a cloned hard drive into a target computer and use it. Something is still wopperjawed somewhere or HP would not be bothering me. Any ideas on this subject? Where do you set related hard drive information into your computer after migrating to a new hard disk? Thanks

for shooting at me.

这篇博文是写给普通人的。结果好像只有我一个人遇到这个问题。大家好像都是 IT 技术人员之类的。我最近在一台较新的 Windows 8.1 电脑上升级硬盘时遇到了一些问题，这台电脑是 UEFI 的。我买这台电脑主要是看中了它的 i5 4570 处理器，其他配置都比较简陋。为了升级硬盘，我买了一块 2TB 的混合硬盘（SSHD）。作为一个普通人，我用硬盘克隆设备迁移了操作系统。一开始似乎运行良好，大概持续了一周左右，但之后我开始收到惠普的警告，说我的硬盘出现故障，需要运行一系列诊断测试。这种情况会在休眠后或夜间关机后启动时出现。我又订购了一块同样的混合硬盘，把第一块退回给了亚马逊。第二块硬盘的情况和第一块一样。我尝试了我能想到的所有方法来修复它，但都无济于事。后来我把它装到一台旧的 Windows 7 电脑上，运行正常，几个月过去了仍然运行良好。我随后订购了一块固态硬盘，附带一张光盘用于软件迁移。几个月过去了，它一直运行良好。我注意到磁盘管理工具显示我有一个健康的 EFI 分区。之前它显示的是 UEFI 分区。惠普支持助手多次提醒我没有安装 UEFI，并建议我安装。我拒绝了，因为我认为安装后可能会导致启动时再次出现错误信息。我仍然不明白为什么 EFI 分区不被视为 UEFI 分区，以及惠普为什么一直纠缠这个问题。昨天我在我的旧 Windows 7 电脑上安装了 Windows 10，这让我开始思考，如果在这台升级了固态硬盘的新电脑上安装 Windows 10，会不会出现什么问题。您认为会出问题吗？ 你的文章帮我理解了为什么克隆 UEFI 硬盘会出问题。由于基于 UEFI 的计算机将启动信息写入主板，因此在迁移到新硬盘后，可能需要在目标计算机中设置相关的硬盘信息。相反，传统的基于 BIOS 的计算机则不需要这一步骤：只需将克隆的硬盘插入目标计算机即可使用。肯定还有什么地方没搞清楚，否则惠普也不会来找我麻烦。关于这个问题，你有什么想法吗？在迁移到新硬盘后，应该在哪里设置相关的硬盘信息呢？谢谢你的指点。

maxevan wrote on 2016-02-23 15:35:

maxevan 于 2016-02-23 15:35 写道：

i have an image of os,full backup im notready to deploy(restore) it,yet to the new laptop windows 8.1 uefi,preinstalled the image was 7 64 bit,u said u got it working,how??? n what software u use

我有一个操作系统镜像，是完整的备份，但我还没准备好把它部署（恢复）到新笔记本电脑上，这台笔记本电脑预装的是 Windows 8.1 UEFI 系统，镜像里是 Windows 7 64 位系统。你说你已经搞定了，你是怎么做到的？你用的是什么软件？

Seppe Sol wrote on 2015-08-25 00:04:

Seppe Sol 于 2015-08-25 00:04 写道：

The firmware (InsydeH2O) on my Packard Bell easynote TE laptop has no support for Bios compatibility, thus: no VESA, VB2, OpenCL etc. The UEFI protocols only reveal one graphic and one text resolution for one screen, a primitive blt protocol without clipping feature, and a framebuffer in PCI space. Yet, windows 8 perfectly knows all resolutions of all the connected graphic devices. It even knows about the external screen and it's possibilities. I assume that MS (directx?) uses window device drivers to handle graphic devices without Bios support. How can UEFI developers obtain similar functionality without Bios support? Note: Microsoft's bootx64.efi takes 1.5Mb. Thank you Seppe

我的 Packard Bell easynote TE 笔记本电脑上的固件 (InsydeH2O) 不支持 BIOS，因此不支持 VESA、VB2、OpenCL 等。UEFI 协议仅允许一个屏幕使用一种图形分辨率和一个文本分辨率，采用不具备裁剪功能的原始 BLT 协议，并且帧缓冲区位于 PCI 空间。然而，Windows 8 却能完美识别所有已连接图形设备的全分辨率。它甚至能识别外接显示器及其功能。我猜测微软（可能是 DirectX? ）使用了 Windows 设备驱动程序来处理不支持 BIOS 的图形设备。UEFI 开发人员如何在没有 BIOS 支持的情况下实现类似的功能？注：微软的 bootx64.efi 文件大小为 1.5MB。谢谢 Seppe

Victor wrote on 2015-08-28 01:36:

Victor 于 2015-08-28 01:36 写道：

Good night! I just want to say "THANK YOU" !!! This article helped me a lot to get some lights concerning all this mess and misunderstandings boiling around UEFI. And yes, i have crapy firmware on my PC. I can't even imagine the time and effort it took you to write this. Thanks again and greetings from Portugal :) . P.S. Sorry for my English.

晚安！我只想道声“谢谢”！！！这篇文章帮了我大忙，让我对围绕 UEFI 的种种混乱和误解有了更清晰的认识。是的，我的电脑固件确实很糟糕。我无法想象您花了多少时间和精力来写这篇文章。再次感谢，来自葡萄牙的问候 :)。PS：我的英语不太好，请见谅。

Alan wrote on 2015-09-03 03:51:

Alan 于 2015-09-03 03:51 写道：

Thanks for the article, it was very informative. I have "crappy" firmware as well. I didn't fully understand UEFI until I read this article. It was long, but good long :)

感谢这篇文章，内容非常丰富。我的固件也“很糟糕”。在读这篇文章之前，我对 UEFI 并不完全了解。文章虽然很长，但很有价值 :)

bm wrote on 2015-09-12 11:29:

bm 于 2015-09-12 11:29 写道：

i have older pc. it has bios, not graphical uefi. it also has an option to boot from network. is this fact contradicting "In the BIOS world, absolutely all forms of multi-booting are handled above the firmware layer." or there is second bottom to it? i don't see any other layer involved in network booting other than

selecting this as an option. do i miss anything?

我有一台老电脑，它用的是 BIOS，不是图形化的 UEFI。它也有网络启动选项。这是否与“在 BIOS 世界里，所有形式的多启动都是在固件层之上处理的”这一说法相矛盾？或者说，它背后还有固件层之上的机制？除了选择网络启动选项之外，我没看到网络启动涉及其他任何层。我是不是漏掉了什么？

Janusz wrote on 2015-09-18 18:35:

Janusz 于 2015-09-18 18:35 写道：

Well, thank you for sharing your thoughts. I found your recommendation extremely into the point. I would appreciate some more technical information regarding: UEFI boot manager exact memory location, format and accessibility (by updater or installer) the way of interference with GPT or its backup, in the next version. Thank you

非常感谢您的分享。您的建议非常中肯。我希望您能提供更多关于以下方面的技术信息：UEFI 启动管理器的确切内存位置、格式和访问权限（通过更新程序或安装程序），以及下一版本中如何干扰 GPT 分区表或其备份。谢谢。

Sanjit Keskar wrote on 2015-09-20 06:23:

桑吉特·凯斯卡于 2015-09-20 06:23 写道：

Thanks AdamW. This blog article was TREMENDOUSLY useful and enlightening for me But may I ask you for a bottom line statement on a query of mine I have an acer switch 10 e laptop which has ONLY a UEFI boot (legacy boot not supported and no CSM). Yes it has a Secure boot option BUT THE UEFI BBOT ARCHITECTURE IS 32 BIT ONLY I cannot boot a multitude of boot disks that claim to be UEFI bootable (for imaging programs, disk management programs for formatting and disk resizing and data recovery. They of course do boot on my other UEFI boot laptop. This is probably because my uefi is 32 bit only. SO WHATS THE WORKAROUND (IF ANY)? Does the responsibility of booting a bootable disk on 32 bit uefi architecture lie with the software company ? i.e they have to write code for the same? say for example acronis Is it possible to modify the iso image (if it is not illegal) by introducing some ready to use files to get such cds to boot on a 32 bit uefi device? What is the bootia32.efi? and how is it relevant to my problem? Thanks

谢谢 AdamW。这篇博客文章对我帮助很大，也很有启发性。但我能否就我的一个疑问请教您一个最终答案？我有一台 Acer Switch 10 E 笔记本电脑，它只有 UEFI 启动（不支持传统启动，也没有 CSM）。是的，它有安全启动选项，但是 UEFI 启动架构仅支持 32 位。我无法启动许多声称支持 UEFI 启动的启动盘（用于映像程序、磁盘管理程序（用于格式化和调整磁盘大小）以及数据恢复）。当然，它们在我的另一台支持 UEFI 启动的笔记本电脑上可以启动。这可能是因为我的 UEFI 仅支持 32 位。那么有什么解决方法吗？在 32 位 UEFI 架构上启动启动盘的责任是否在于软件公司？也就是说，他们是否需要为此编写代码？例如，Acronis。是否可以修改 ISO 镜像（如果合法的话），通过添加一些现成的文件来使这些光盘能够在 32 位 UEFI 设备上启动？bootia32.efi 是什么？它与我的问题有什么关系？谢谢。

[adamw](#) wrote on 2015-09-20 06:41:

[adamw](#) 于 2015-09-20 06:41 写道：

Hi, Sanjit, thanks for the kind words.

嗨，Sanjit，谢谢你的赞美。

32-bit UEFI firmwares are oddballs. Up until the last couple of years, the only ones that existed were very early Mac firmwares, which pretty much no-one cared about supporting.

32 位 UEFI 固件很特殊。直到最近几年，市面上只有非常早期的 Mac 固件，而这些固件几乎无人问津。

The ones that have shown up recently are mainly tablet/convertible systems based on the Intel 'Baytrail' platform, like yours. These have 64-bit CPUs but shipped with 32-bit firmwares due to some kind of Microsoft fail, the details of which I've forgotten.

最近出现的主要是基于英特尔“Baytrail”平台的平板电脑/二合一电脑，就像你的那款一样。这些电脑配备了 64 位 CPU，但由于微软的某种失误（具体细节我已经忘记了），出厂时搭载的是 32 位固件。

It is, indeed, up to OS vendors to make images compatible with such systems - if they actually care to. There are two ways to do this:

的确，操作系统厂商有责任确保镜像与此类系统兼容——如果他们真的想这么做的话。有两种方法可以做到这一点：

- Build 32-bit UEFI bootable images
 - 构建 32 位 UEFI 可启动映像
- Build 64-bit images capable of booting on 32-bit UEFI firmwares (using a bootloader capable of booting a 64-bit OS from a 32-bit firmware)
 - 构建能够在 32 位 UEFI 固件上启动的 64 位映像（使用能够从 32 位固件启动 64 位操作系统的引导加载程序）

I actually built a 32-bit UEFI remix of Fedora called [Fedlet](#), because I also have one of these systems and wanted to play with getting Fedora running on it. I have not had much time to maintain it lately, though.

实际上，我构建了一个基于 Fedora 的 32 位 UEFI 重制版，名为 [Fedlet](#)，因为我也有一台这样的系统，想试试在上面运行 Fedora。不过，最近我一直没太多时间维护它。

Distributions are generally not very interested in building 32-bit UEFI bootable images. We usually keep 32-bit images around for *old* hardware, and some old hardware has trouble booting images that are set up for UEFI as well as BIOS, and no-one really wants to go to the trouble of building *two sets* of 32-bit images just to cater to some oddball systems that Microsoft screwed up.

发行版通常不太热衷于构建 32 位 UEFI 可启动镜像。我们通常会保留 32 位镜像以备 *旧* 硬件使用，而有些旧硬件在启动同时支持 UEFI 和 BIOS 的镜像时会遇到问题，而且也没人愿意为了迎合微软搞砸的某些特殊系统而费劲地构建 *两套* 32 位镜像。

Distros are more open in principle to the idea of making their 64-bit images bootable on these oddball 32-bit UEFI firmwares, but someone has to take the time to do the work, and for Fedora none of the few of us who are at all interested in the topic have been interested *enough* yet to do it. Matthew Garrett and Peter Jones did great work to make grub capable of the 64-on-32 trick, but all the distro infra around that still needs to be set up and tested, and it's just a lot of work for comparatively little return.

发行版原则上更乐于接受让其 64 位镜像能够在这些特殊的 32 位 UEFI 固件上启动的想法，但总得有人花时间去做，而对于 Fedora 来说，我们这些对此感兴趣的人中，目前还没有人愿意投入 *足够的* 精力去做这件事。Matthew Garrett 和 Peter Jones 为让 grub 能够实现 64 位启动 32 位系统做出了卓越的贡献，但围绕这方面的所有发行版基础设施仍然需要搭建和测试，而且这项工作投入巨大，回报却相对较小。

If you're really invested in making this work you can try running the latest Fedlet release on your system, but I make no guarantees about it, and don't have much time to update it. There are also folks maintaining hacks for other distros, I haven't kept up with them lately though. For stuff like Acronis, you're basically at the mercy of the company, and I doubt they care enough about the fairly small number of those systems out there to take the trouble.

如果你真的想让它正常工作，可以尝试在你的系统上运行最新的 Fedlet 版本，但我无法保证它一定有效，而且我也没有太多时间去更新它。也有人维护其他发行版的破解版，但我最近没怎么关注。至于像 Acronis 这样的软件，你基本上只能听天由命，而且我怀疑他们是否足够重视那些用户数量相对较少的系统，所以不会费心去维护。

bootia32.efi is the 'fallback path' bootloader name for 32-bit Intel. As described in the article, the 'fallback path' basically works by defining standard locations where the firmware can look for a bootloader; the spec defines filenames for various arches, because obviously if you want to make a disk bootable via the 'fallback path' on multiple arches, you have to be able to provide an appropriate executable for each arch, and the firmware has to be able to find the right one. So the name for 32-bit Intel is bootia32.efi. The name for 64-bit Intel is bootx64.efi, IIRC.

bootia32.efi 是 32 位 Intel 的“备用路径”引导加载程序名称。正如文章所述，“备用路径”的基本工作原理是定义固件可以查找引导加载程序的标准位置；规范为各种架构定义了文件名，因为显然，如果要使磁盘能够通过“备用路径”在多个架构上启动，则必须能够为每个架构提供相应的可执行文件，并且固件必须能够找到正确的文件。因此，32 位 Intel 的名称是 bootia32.efi。如果我没记错的话，64 位 Intel 的名称是 bootx64.efi。

(Amusingly, I think we've found both firmwares that require the name to be upper-case - BOOTX64.EFI or BOOTIA32.EFI - and firmwares that require it to be lower-case - bootx64.efi or bootia32.efi . Sigh.)

（有趣的是，我们似乎既找到了要求固件名称大写的固件——BOOTX64.EFI 或 BOOTIA32.EFI——也找到了要求固件名称小写的固件——bootx64.efi 或 bootia32.efi。唉。）

Sanjit Kesar wrote on 2015-09-20 09:49:

桑吉特·凯斯卡于 2015-09-20 09:49 写道：

OK Got that (and more than I bargained for)! I was always under the impression that if I saw the bootia32.efi file in the boot folder, that such bootable media will always boot on a 32 bit uefi firmware. Please do address the following doubts I have So the ball is in he software owner's court to add compatibility for the bootable media to boot on a 32 bit UEFI firmware How would FEDLET help me in my particular predicament? i.e. the need to run bootable media on a 32bit uefi firmware? Is there anyway to inspect the iso of an uefi bootable media and know that it will o will not boot on m 32bit uefi baytrail laptop Can you confirm that this limitation in booting will not be applicable in the following case. I windows install a disk management software on my baytrail laptop. The software allows me to change the size of the system partition. The laptop has to reboot to do the same and perform this task from outside windows so to speak. May I assume that the (re)booting will not be a problem and task will be performed. Thanks again adamW

好的，明白了（而且比我预想的还要明白）！我一直以为，如果我在启动文件夹中看到 bootia32.efi 文件，那么这种启动介质就一定能在 32 位 UEFI 固件上启动。请解答我以下几个疑问：现在是否应该让启动介质兼容 32 位 UEFI 固件，取决于软件开发者。FEDLET 能如何帮助我解决我目前遇到的问题？也就是如何在 32 位 UEFI 固件上运行启动介质？有没有办法检查 UEFI 启动介质的 ISO 文件，从而判断它是否能在我的 32 位 UEFI Baytrail 笔记本电脑上启动？能否确认以下情况不受此启动限制的影响？我在 Windows 系统下在我的 Baytrail 笔记本电脑上安装了一个磁盘管理软件。该软件允许我更改系统分区的大小。笔记本电脑必须重启才能执行此操作，也就是说，我需要在 Windows 系统之外执行此操作。我可以假设重启不会有问題，任务也能顺利完成吗？再次感谢 adamW

me wrote on 2015-09-22 05:44:

我于 2015 年 9 月 22 日 05:44 写道:

UEFI boot: how does that shit actually work, then?

UEFI 启动: 那玩意儿到底是怎么工作的?

[Christopher Price](#) wrote on 2015-10-05 09:37:

[Christopher Price](#) 于 2015 年 10 月 5 日 09:37 写道:

Regarding Secure Boot: "If you don't like this, don't buy one. But also don't buy an iDevice, or an Android device with a locked bootloader " Secure Boot is disliked because it gave legitimacy to the practices of Apple and a handful of Android device makers. Before that, Microsoft had to sign a settlement agreement for its similar practices against Be, and others in the 1990s. Since you wrote this article, Microsoft has given OEMs the ability to revoke the ability to disable Secure Boot on Windows 10 systems. This can make it impossible to tell which machines, in a retail setting, for an average consumer, can/can't run Linux. Average consumers aren't going to spend hours researching each PC to know its Linux viability. And worse, they may learn after the fact (once interested in Linux) that they "bought the wrong one." Now the entire PC market is at risk of slowly being locked to the OS that shipped with it. You can argue that slippery slope is counterbalanced by a rise of IoT devices, and Android becoming the most popular OS today. Others will argue that it has ensured Linux on the desktop will be owned by Google. Either way, the critics of Secure Boot do have a sound argument... one time has proven. The UEFI specification could be modified to require a user setting to disable it. And OS vendors could require Secure Boot to boot their kernel. That would mean Windows 10 in a datacenter could remain secure, while home PCs remain open. The question is, if the UEFI group has the guts to stay open... or not.

关于安全启动: “如果你不喜欢, 就别买。但也不要买苹果设备, 或者带有锁定引导程序的安卓设备。” 安全启动之所以不受欢迎, 是因为它使苹果和少数安卓设备制造商的做法合法化。在此之前, 微软在上世纪 90 年代就因其类似做法与 Be 和其他公司签署了和解协议。自从你写这篇文章以来, 微软已经允许 OEM 厂商撤销在 Windows 10 系统上禁用安全启动的功能。这使得普通消费者在零售环境中根本无法判断哪些机器可以运行 Linux。普通消费者不会花几个小时研究每台电脑是否支持 Linux。更糟糕的是, 他们可能事后 (一旦对 Linux 感兴趣) 才发现自己“买错了”。现在, 整个 PC 市场都面临着被预装操作系统逐渐锁定的风险。有人可能会说, 物联网设备的兴起以及安卓成为当今最流行的操作系统, 在一定程度上抵消了这种滑坡效应。另一些人则认为, 这确保了桌面 Linux 系统将由谷歌掌控。无论如何, 安全启动的批评者确实有其道理.....一次成功就证明了这一点。UEFI 规范可以修改, 要求用户设置才能禁用安全启动。操作系统厂商也可以要求使用安全启动来启动他们的内核。这意味着数据中心的 Windows 10 可以保持安全, 而家用 PC 则可以保持开放。问题是, UEFI 组织是否有勇气保持开放.....或者没有。

Computer Scientist wrote on 2018-08-20 20:13:

计算机科学家于 2018-08-20 20:13 写道:

Any computer that is Windows certified is required (by virtue of that certification) to *not* being locked into a specific OS, or even being locked into the Secure Boot keys from Microsoft. So no, that's a completely misleading and wrong point to make.

任何通过 Windows 认证的计算机 (根据认证本身的要求) 都不得被锁定在特定的操作系统中, 甚至不得被锁定在微软的安全启动密钥中。所以, 这种说法完全是误导性的, 也是错误的。

[Christopher Price](#) wrote on 2018-08-20 20:33:

[Christopher Price](#) 于 2018 年 8 月 20 日 20:33 写道:

My comment was written three years ago, when there was serious concern (with valid reasons) that the Other OS key would be pulled from Windows to enforce secure boot. That didn't come to pass, as most of the industry that wanted that control has switched to Windows on ARM. Still, it is a valid thing to be concerned about with Windows certification - it all hinges on being able to bypass or use an Other OS key that many feel could be revoked at the next security FUD event. I don't think it will happen today - but it's always possible, and the FOSS community should stay vigilant toward.

我的评论写于三年前, 当时人们非常担心 (而且理由充分) Windows 会取消“其他操作系统密钥”功能以强制执行安全启动。这种情况最终并未发生, 因为大多数需要这项控制权的厂商都已转向 ARM 架构的 Windows 系统。尽管如此, Windows 认证仍然值得关注——一切都取决于能否绕过或使用“其他操作系统密钥”, 而许多人认为这项密钥可能会在下一次安全恐慌事件中被撤销。我认为这种情况今天不会发生, 但并非完全不可能, 开源软件社区应该对此保持警惕。

[Gordon Myers](#) wrote on 2015-10-07 15:56:

[Gordon Myers](#) 于 2015 年 10 月 7 日 15:56 写道:

I just want to say that I, too, found this article immensely helpful and instructive, and am so grateful that you wrote it.

我只想说, 我也觉得这篇文章非常有帮助, 很有启发性, 非常感谢你写了这篇文章。

Joe wrote on 2015-10-11 14:16:

Joe 于 2015 年 10 月 11 日 14:16 写道:

Hi. I would like the UEFI specification to require something else, besides the the ability to use it without Secure Boot. I would like the CSM mode to be an essential feature of a UEFI implementation.

您好。我希望 UEFI 规范除了要求能够在不使用安全启动的情况下运行之外，还能包含其他功能。我希望 CSM 模式成为 UEFI 实现的一项基本特性。

[Patty P](#) wrote on 2015-10-17 02:41:

[Patty P](#) 于 2015-10-17 02:41 写道:

I liked the part where he said UEFI

我喜欢他说 UEFI 的那部分。

[Patty P](#) wrote on 2015-10-17 02:42:

[Patty P](#) 于 2015-10-17 02:42 写道:

(Amazing article by the way)

(顺便说一句，文章写得真棒！)

Kelly K wrote on 2015-10-18 07:17:

Kelly K 于 2015-10-18 07:17 写道:

I'm a self-pronounced ignorant computer nerd (wanna-be) who, while on my way to gathering fragments of information on how to build a mini-itx, just happens to stumble on to this article--I mean essay. I'm more than thrilled that this wasn't going to be short and snappy reading. After having wandered through so many sites of informatively un-selfexplanatory information, I'm more than eager to pull up a chair to the very front of this long, refreshingly elaborative class. I don't know enough about bios or uefis right now to really join in the passionate arguments/complaints/debates about whose computer has what inside and what their anatomy is called. After having read this essay (and still reading it..and will be reading it again..and again), I would like to raise my hand and say-- that this is the BEST, MOST INFORMATIVE, MOST ENTERTAININGLY CONDESCENDING DETAILED thing I've had the pleasure of reading by far! If there's a place I can go to nominate you, AdamW, as the MOST OUTSTANDING bootloading essayist-speaker-writer-instructor OF THE YEAR and year after this, I would be one of your biggest supporters. Thank you.. that's all I wanna comment on right now. And if you'll please excuse me-- I would like to get back to reading this immensely instructional article--I mean essay.

我自诩为电脑小白（或者说是想成为电脑小白），在搜集如何组装 mini-ITX 电脑的零碎信息时，偶然发现了这篇文章——哦不，是这篇长文。我非常高兴这篇文章不是那种短小精悍的速读文章。在浏览了那么多信息量大但又晦涩难懂的网站之后，我迫不及待地想要坐在这篇内容详实、条理清晰的文章的最前面。我现在对 BIOS 和 UEFI 的了解还不够深入，所以暂时无法参与到关于电脑内部配置和各种组件名称的激烈争论中。读完这篇文章（而且还在读.....以后还会再读.....），我得举手说一句——这绝对是我迄今为止读过的最棒、信息量最大、最有趣（带着居高临下的姿态）又最详尽的文章！如果有什么地方可以提名你，AdamW，为年度以及未来年度最杰出的“引导式写作”作家、演讲者、作家兼导师，我绝对会是你最坚定的支持者之一。谢谢.....我现在只想说这些。请允许我先告辞——我想继续阅读这篇极具指导意义的文章——我是说，这篇论文。

Jeremy Fields wrote on 2015-10-23 22:09:

Jeremy Fields 于 2015 年 10 月 23 日 22:09 写道:

Really excellent post. UEFI to me is now a lot more likeable. I'd ask for an quick foray into PXE booting, but the article is already massive. Thanks for writing!

这篇文章写得真好。现在我对 UEFI 的印象好多了。本来想请你简单介绍一下 PXE 启动，但文章已经很长了。感谢你的分享！

pgmer6809 wrote on 2015-10-27 23:39:

pgmer6809 于 2015-10-27 23:39 写道:

Can anyone explain why after all this time the Linux foundation, (or Red Hat, or ...) have not come up with their own key and lobbied the major Mobo makers and PC vendors to include this key alongside the Microsoft key? That way there would be at least one other entity besides MS that could 'sign' bootloaders. (Very well written article BTW)

谁能解释一下，为什么这么久了，Linux 基金会（或者 Red Hat 之类的）还没推出自己的密钥，并游说各大主板厂商和 PC 厂商将这个密钥与微软密钥一起纳入系统？这样一来，除了微软之外，至少还有另一个机构可以对引导加载程序进行“签名”。（顺便说一句，这篇文章写得很好。）

[adamw](#) wrote on 2015-10-28 00:00:

[adamw](#) 于 2015-10-28 00:00 写道:

I wasn't directly involved in any discussions about that, but what I understand second hand is that it's a fairly complex, expensive and thankless task that involves shouldering quite a chunk of implied legal liability. RH can't justify that to its shareholders, and the Linux Foundation probably literally doesn't have the resources.

我没有直接参与任何相关讨论，但据我了解，这是一项相当复杂、昂贵且吃力不讨好的任务，需要承担相当大的隐性法律责任。RH 无法向其股东证明这一点，而 Linux 基金会可能也确实没有足够的资源。

pgmer6809 wrote on 2015-10-27 23:42:

pgmer6809 于 2015-10-27 23:42 写道:

I don't think you approve of this configuration but I run one machine with a GPT disk that boots in BIOS compatibility mode. I use it to test Linux distros. I have about 16 partitions on it (and counting) at the moment, one per distro. No ESP. Grub has the ability to define a 'extra grub code partition' on GPT disks that make all the problems with 'magic space' go away. Thanks to GRUB and OSprober I do not have a problem booting multiple OS's from this one GPT disk. And I can create as many partitions as I want without worrying about Logical, Extended etc. etc.

我觉得您可能不赞同这种配置，但我有一台机器使用 GPT 磁盘，并以 BIOS 兼容模式启动。我用它来测试 Linux 发行版。目前这块磁盘上有大约 16 个分区（还在增加），每个发行版一个分区。没有 ESP 分区。GRUB 可以在 GPT 磁盘上定义一个“额外的 GRUB 代码分区”，从而解决所有“魔法空间”的问题。多亏了 GRUB 和 OSprober，我可以轻松地从此块 GPT 磁盘启动多个操作系统。而且我可以创建任意数量的分区，而不用担心逻辑分区、扩展分区等等问题。

[adamw](#) wrote on 2015-10-27 23:59:

[adamw](#) 于 2015-10-27 23:59 写道:

BIOS on GPT *can* work, it's just not really a great idea unless you actually *need* GPT, i.e. you're working with a very large disk or the four primary partition limit is somehow important to you (though I really can't think of any situation where that limit would be a problem *and* whatever had the problem would be perfectly happy dealing with GPT).

GPT 分区表上的 BIOS 可以工作，但除非你真的需要 GPT 分区表，否则这并不是一个好主意。例如，你正在使用一个非常大的磁盘，或者四个主分区的限制对你来说很重要（尽管我真的想不出任何情况下这个限制会成为问题，而且任何遇到问题的东西都能很好地处理 GPT 分区表）。

We made GPT the default for BIOS installs for one Fedora release, and it was a train wreck; some of the issues were user error (people doing custom installs not understanding about the BIOS boot partition), but we also found that there are quite a few firmwares out there that simply won't work with it, they just will not boot.

我们在 Fedora 的一个版本中将 GPT 设置为 BIOS 安装的默认分区表，结果却是一场灾难；一些问题是用户错误造成的（进行自定义安装的用户不了解 BIOS 启动分区），但我们也发现有很多固件根本无法与 GPT 兼容，它们根本无法启动。

Julian wrote on 2015-11-19 01:39:

Julian 于 2015-11-19 01:39 写道:

I have new PC, which I assembled myself: Asus motherboard, which during startup displays UEFI BIOS, 8 core AMD processor, 250 GB SSHD, extended partition on my 2TB SATA drive and two partitions drives - E and F. I run Win 7 pro 64 and everything was fine. I've did upgrade to Win 10 and everything was fine until Nov Win 10 update. After rebooting my drives E and F disappeared in fact there was no SATA drive at all in device manager ??? Why? Any light in the darkness wise guys? I was able one time restore my machine to pre update state but Windows installed updates and again no HD. I had huge troubles to return to Win 7. For some magic power it finally happened and I run Win 7 again. I am thankful for sharing your knowledge.

我新组装了一台电脑：华硕主板，启动时显示 UEFI BIOS，8 核 AMD 处理器，250GB SSHD 混合硬盘，2TB SATA 硬盘上的扩展分区，以及两个分区——E 盘和 F 盘。我之前运行的是 Windows 7 专业版 64 位系统，一切正常。后来我升级到了 Windows 10，也一直很顺利，直到 11 月份的 Windows 10 更新。重启后，我的 E 盘和 F 盘都消失了，设备管理器里甚至连 SATA 硬盘都找不到了！这是怎么回事？各位大神，有没有人能帮我解答一下？我曾经尝试将系统还原到更新前的状态，但是 Windows 更新后硬盘又不见了。我费了九牛二虎之力才恢复到 Windows 7。最后，不知怎么的，终于成功了，我又能用 Windows 7 了。感谢各位分享知识。

Michael C. wrote on 2016-07-26 16:54:

Michael C. 于 2016 年 7 月 26 日 16:54 写道:

Julian you made me laugh (thanks); I too prefer Win 7 Pro; Short long story, my Toshiba crashed, after installing new board and HD, that too crashed after one month usage, techy friend advised it was an inherent problem of Toshiba units with AMD CPU (?); I purchased new unit with Win 10 installed (nobody wanted to sell me new unit with Win 7 of any kind)sooo, I purchased complete full blown Win 7 Pro install disks for new PC's, I yelled, bitched, almost threw the unit under my car until another friend said it's new PC with newer motherboard; I finally accessed the UEFI, disabled it and finally installed my Win 7 Pro; I'm still leery of the MS updates, but now I'm armed, just running out of knowledge.

朱利安，你逗乐我了（谢谢）；我也更喜欢 Win 7 专业版；简单来说，我的东芝电脑坏了，换了新主板和硬盘之后，一个月后也坏了，懂技术的朋友说这是东芝电脑配 AMD CPU 的通病（？）；我买了一台预装了 Win 10 的新电脑（没人愿意卖给我任何预装 Win 7 的新电脑），所以，我买了一整套 Win 7 专业版的安装光盘，我当时又吼又骂，差点把电脑扔到车底下，直到另一个朋友说这是新电脑，主板也更新了；我终于进入了 UEFI，把它禁用，然后成功安装了 Win 7 专业版；我仍然对微软的更新心存疑虑，但现在我已经做好了准备，只是知识储备不足了。

sixpack wrote on 2015-12-01 17:09:

sixpack 于 2015-12-01 17:09 写道:

Very interesting article, though it does take a couple of reads - at least for me it does!! To go on a tangent, MS insisting on the x86 chip architecture (plus now ARM) is perhaps more harmful to innovation and development than it's o/s requirements. There is little room for new chip architectures which could

- and are likely to - provide significant benefits to the rather old x86 one. Now, there would be extra costs for MS but then they aren't poor and don't give out dividends so I'm less sympathetic to them. Also Intel has benefited from the scale of production that this restriction enables which in turn should benefit the consumer (prices) although prices are very high for chips it seems. I ran out of knowledge a while back so I'll stop here!

这篇文章很有意思，不过确实需要读几遍才能理解——至少对我来说是这样！顺便提一下，微软坚持使用 x86 芯片架构（现在又加上 ARM）对创新和发展的阻碍，或许比其操作系统要求本身更大。这样一来，几乎没有空间容纳那些能够——而且很可能——为略显老旧的 x86 架构带来显著优势的新芯片架构。当然，这会增加微软的成本，但他们财力雄厚，而且也不分红，所以我对他们没什么好同情的。此外，英特尔也从这种限制带来的规模化生产中获益，这反过来应该会让消费者受益（价格），尽管芯片价格似乎仍然很高。我之前就没什么可说的了，所以就打住吧！

Tim wrote on 2015-12-08 10:04:

Tim 于 2015-12-08 10:04 写道：

Very interesting how the author has not made a peep since Microsoft has changed their position on the disable secure boot feature. SO like its some big surprise secure boot now is no longer required either by the UEFI or MS sert to allow secure boot to have a disable feature. You can damn well be sure MS further pressures or offers some sort of kick back for the feature to be locked. So what happens when a manf under pressure from MS you know the trillion dollar company locks its secure boot with only the MS license in the key ring? That's right no other OS can be loaded unless you hack the firmware. At the very least the Secure boot should have been open to allow user entered keys but that too is not a mandate. All the heat about Secure boot from those of us who understand it was never about is basic workings. Those are all very good the issue was how MS was using its market share to bully requirements. Your artilce reads like MS is some sort of jonny be good misunderstood good guy. Please. MS had originally tried to have the secureboot locked with only its key from the get go under its cert but it got so much backlash from other OS manf such as RedHat and customers as well as HW manf its relented and made itself look good by including the disbale feature in its mandate. It has always wanted that locked. If you are going to aruge this is not true give one good reason why it would not be in MS best interested to have a locked secureboot? There are none plain and simple. Not to mention look what we have now with Windows 10. All home and even paid pro users have become nothing but beta tester group for the big dollar Enterprise group. Now they have laxed their testing to barely alpha level before release. See how it fails in the wild then patch the patch and once its solid send it to enterprise. That is what you get when you accept MS giving you an OS as a service. Really they mean service them. They make their money from OEM cert for the home and pro so you have no weight to your voice at all anymore other than to stop using them.

很有意思的是，自从微软改变了对禁用安全启动功能的立场后，作者竟然一直保持沉默。好像现在 UEFI 和 MS SERT 都不再要求禁用安全启动功能似的。你完全可以肯定，微软会进一步施压，或者提供某种回扣来阻止这项功能的禁用。那么，当一家制造商在微软——这家市值万亿美元的公司——的压力下，将安全启动功能锁定，只允许使用微软的授权码时，会发生什么呢？没错，除非破解固件，否则无法加载其他操作系统。至少，安全启动功能应该允许用户输入密钥，但这也不是强制性的。我们这些了解安全启动的人，对它的批评从来不是针对其基本工作原理的。这些基本原理都很好，问题在于微软如何利用其市场份额来胁迫厂商制定规则。你的文章读起来好像微软是个被误解的好人似的。拜托。微软最初试图从一开始就使用其证书和密钥锁定安全启动，但由于遭到红帽等其他操作系统厂商、客户以及硬件厂商的强烈反对，最终不得不妥协，并在其授权中加入禁用功能以挽回颜面。微软一直都想锁定安全启动。如果你要反驳说微软的做法不对，请举出一个合理的理由，说明为什么锁定安全启动不符合微软的最佳利益？根本没有这样的理由。更不用说看看我们现在在 Windows 10 上看到的情况了。所有家庭用户甚至付费专业用户都沦为了大型企业用户的测试员。现在，他们在发布前甚至放松了测试，几乎只进行了初步测试。看看它在实际应用中表现如何，然后打补丁修复漏洞，一旦稳定下来就发布到企业版。这就是你接受微软“操作系统即服务”的后果。实际上，他们说的“服务”是指给企业用户提供操作系统。他们通过家庭和专业领域的 OEM 认证赚钱，所以除了停止使用他们的产品之外，你再没有任何发言权了。

[adamw](#) wrote on 2016-02-18 22:59:

[adamw](#) 于 2016-02-18 22:59 写道：

I didn't hear anything about Microsoft "changing their position". I haven't been paying super close attention to UEFI stuff lately. And, er, you know I *work* for Red Hat, right?

我没听说微软“改变立场”的事。最近我也没太关注 UEFI 方面的事情。还有，呃，你知道我在红帽工作，对吧？

Rogério Hilbert wrote on 2015-12-22 15:01:

Rogério Hilbert 于 2015 年 12 月 22 日 15:01 写道：

I was wondering about the beginning of the article in the terminology section when it says: "Please don't ever say 'UEFI BIOS'". I've been seeing quite a lot that terminology everywhere, including the title of the UEFI Utility Interfaces, like below: <http://rog.asus.com/wp-content/uploads/2012/07/11.png> It doesn't seem to be wrong, does it?

我在术语部分看到文章开头说“请不要使用‘UEFI BIOS’”时感到疑惑。我到处都能看到这个术语，包括 UEFI 实用程序界面的标题，例如下面这个：<http://rog.asus.com/wp-content/uploads/2012/07/11.png> 看起来好像没什么问题，对吧？

[adamw](#) wrote on 2016-02-18 22:58:

[adamw](#) 于 2016-02-18 22:58 写道：

It's still wrong, but at this point I'm feeling distinctly Canute-like. I still maintain that UEFI and BIOS are types of PC firmware. Saying "UEFI BIOS" is like saying "Linux Windows" or something.

这仍然是错的，但此时此刻，我感觉自己就像克努特一样固执己见。我仍然坚持认为 UEFI 和 BIOS 都是电脑固件的类型。说“UEFI BIOS”就像说“Linux Windows”一样荒谬。

Pugly wrote on 2016-02-20 00:03:

Pugly 于 2016 年 2 月 20 日 00:03 写道：

In 2015 i helped my father in law buy a laptop. We bought one under the premise i would nuke the pre-installed win8 and go with win7. Turned out the oem had removed the ability to turn off secure boot from the firmware. There was no way to tell this from either the internet or the packaging. The online help from the oem had screenshots showing how you could simply turn off secure boot from the firmware menu when infact this menu item infact no longer existed. There should be a clear warning/notice in the product description if secure boot is forever on. It limits the device in a way that was not typical of pcs up till now.

2015 年，我帮岳父买了一台笔记本电脑。我们买的前提是我会把预装的 Win8 系统卸载掉，换成 Win7。结果发现，厂商竟然在固件里取消了关闭安全启动的功能。无论是网上还是包装盒上，都找不到任何相关信息。厂商提供的在线帮助里有截图显示如何从固件菜单里关闭安全启动，但实际上这个菜单选项已经不存在了。如果安全启动是永久开启的，产品说明里应该有明确的警告/提示。这大大限制了设备的功能，而这在以前的电脑上是很罕见的。

Jacob Burroughs wrote on 2016-03-10 02:07:

Jacob Burroughs 于 2016 年 3 月 10 日 02:07 写道：

I enjoyed reading this article. Having read this article after spending several hours one night making secure boot, Windows, and Fedora (later switched to Ubuntu because reasons, but not really relevant to this comment) off a single laptop SSD, I will say that might life would have been a lot easier had I followed basically any of those last points. That said, where would be the fun in doing things the easy way? I therefore want to point out rEFInd, which I don't know if you are familiar with. It offers a nice interface for UEFI multi-booting, and supports using Secure Booth through shim, and chain-loading another bootloader and/or detecting Linux kernels. Once setup once, it is actually nice and painless to use, but I spent way too long getting it set up. Of course, the second time I had to do it, it was much easier since I had done it once, but that doesn't make it any more user-friendly. Also, on that note, I particularly liked your line "Or you can read up on how to configure your own chain of trust and sign your kernels and kernel modules and leave Secure Boot turned on, which will make you feel like an uber geek and be slightly more secure." since that more or less reflects my feelings, even if I am currently not building my own kernel. (Is there a good reason I might someday want to experiment with doing that?)

我很喜欢这篇文章。读完这篇文章之前，我花了几个小时，在一个笔记本电脑的固态硬盘上安装了安全启动、Windows 和 Fedora（后来因为一些原因换成了 Ubuntu，但这和本文无关）。如果当初我按照文章最后提到的几点去做，那肯定会轻松很多。话虽如此，走捷径还有什么乐趣可言呢？因此，我想推荐一下 rEFInd，不知道你是否熟悉它。它为 UEFI 多系统启动提供了一个友好的界面，支持通过 shim 使用安全启动，以及链式加载其他引导程序和/或检测 Linux 内核。一旦设置好，它用起来确实很方便，但我花了太多时间才把它设置好。当然，第二次设置的时候就容易多了，因为已经设置过一次了，但这并不意味着它更人性化。另外，说到这儿，我特别喜欢你说的“或者你可以研究一下如何配置自己的信任链，如何对内核和内核模块进行签名，以及如何启用安全启动，这会让你觉得自己像个超级极客，而且安全性也会更高一些。”这句话，因为这或多或少地表达了我的感受，即使我目前并没有自己编译内核。（我以后会不会想尝试一下呢？）

Srj wrote on 2016-03-18 02:07:

Srj 于 2016-03-18 02:07 写道：

When I tried efibootmgr in installed ubuntu OS along with Win10 in UEFI mode, a message appears stating that "efivar is missing" and all. They also suggest me to trick madprob . So please let me know what should be done if efibootmgr shows some troubles.

当我在安装了 Ubuntu 和 Win10 的 UEFI 模式下尝试使用 efibootmgr 时，出现一条消息提示“缺少 efivar”。他们还建议我使用 madprob 来解决这个问题。所以，如果 efibootmgr 出现问题，我应该怎么办？

LINSEY TIRADO wrote on 2016-05-11 05:46:

LINSEY TIRADO 于 2016 年 5 月 11 日 05:46 写道：

Helpful writing , With reference to which , if one requires to merge some PDF files , my boss ran across article here <https://disqus.com/by/rest04plate/>.

这篇文章很有帮助，如果需要合并一些 PDF 文件，我的老板在这里找到了一篇文章：<https://disqus.com/by/rest04plate/>。

Fred J. Tydeman wrote on 2016-05-20 04:31:

Fred J. Tydeman 于 2016 年 5 月 20 日 04:31 写道：

On a Fedora 23 system, as root, I enter: efibootmgr -t 15 and the timeout changes as expected. But, when I enter any of: efibootmgr -b 0001 -L FedoraSSD efibootmgr -b 1 -L FedoraSSD efibootmgr -v -b 1 --label FedoraSSD nothing happens. I expected the name on the boot menu to change.

在 Fedora 23 系统上，以 root 用户身份运行命令 `efibootmgr -t 15`，超时时间如预期发生变化。但是，当我运行以下任何命令时：`efibootmgr -b 0001 -L FedoraSSD`、`efibootmgr -b 1 -L FedoraSSD` 或 `efibootmgr -v -b 1 --label FedoraSSD`，没有任何反应。我预期启动菜单中的名称会发生变化。

andreino wrote on 2016-05-23 12:17:

andreino 于 2016-05-23 12:17 写道:

Hello, state that I am newbie, so have a lot of patience, I installed arch sun an external drive gpt-UEFI I used systemd-boot and efibootmgr and I added the entries, and everything works. Now I have a problem on the same disk I installed another distro archmate, and I can not figure out if it is possible to start the distro dall'entries I created and if I may kindly indicate the steps. thank you

您好，我是个新手，所以请多多包涵。我在一个外置硬盘上安装了 Arch Linux（GPT UEFI 分区表），使用了 systemd-boot 和 efibootmgr，并添加了相应的启动项，一切正常。现在我在同一块硬盘上安装了另一个发行版 Archmate，遇到了问题。我不知道是否可以启动我之前创建的发行版启动项，能否请您指点一下具体步骤？谢谢！

andreino wrote on 2016-05-23 12:19:

andreino 于 2016-05-23 12:19 写道:

```
[nuovo@pippo ~]$ ls -l /boot totale 26836 drwxr-xr-x 4 root root 4096 22 mag 17.54 EFI -rwxr-xr-x 1 root root 18629478 23 mag 10.35 initramfs-linux-fallback.img -rwxr-xr-x 1 root root 3678073 23 mag 10.35 initramfs-linux.img -rwxr-xr-x 1 root root 692736 18 gen 18.37 intel-ucode.img drwxr-xr-x 3 root root 4096 22 mag 16.52 loader -rwxr-xr-x 1 root root 4463456 11 mag 22.23 vmlinuz-linux [nuovo@pippo ~]$ ls -l /boot/EFI/ totale 8 drwxr-xr-x 2 root root 4096 22 mag 17.54 Boot drwxr-xr-x 2 root root 4096 22 mag 17.54 systemd [nuovo@pippo ~]$ ls -l /boot/loader/entries/ totale 4 -rwxr-xr-x 1 root root 180 22 mag 17.42 arch.conf [nuovo@pippo ~]$ paeted -l bash: paeted: comando non trovato [nuovo@pippo ~]$ parted -l [nuovo@pippo ~]$ sudo parted -l
Modello: ATA TOSHIBA MQ01ABD0 (scsi) Disco /dev/sda: 750GB Dimensione del settore (logica/fisica): 512B/4096B Tabella delle partizioni: gpt
Flag del disco: Numero Inizio Fine Dimensione File system Nome Flag 1 1049kB 420MB 419MB ntfs Basic data partition nascosta, diag 2 420MB 693MB 273MB fat32 EFI system partition avvio, esp 3 693MB 827MB 134MB Microsoft reserved partition msftres 4 827MB 652GB 651GB ntfs Basic data partition msftdata 5 652GB 729GB 77,3GB ntfs Basic data partition msftdata 6 729GB 730GB 895MB ntfs nascosta, diag 7 730GB 731GB 510MB ntfs nascosta, diag 8 731GB 731GB 472MB ntfs nascosta, diag 9 731GB 750GB 18,8GB ntfs Basic data partition nascosta, msftdata
Modello: TOSHIBA MK5075GSX (scsi) Disco /dev/sdb: 500GB Dimensione del settore (logica/fisica): 512B/512B Tabella delle partizioni: gpt
Flag del disco: Numero Inizio Fine Dimensione File system Nome Flag 1 1049kB 634MB 633MB fat32 avvio, esp 2 634MB 19,6GB 19,0GB ext4 3 19,6GB 381GB 361GB ntfs Basic data partition msftdata 4 381GB 467GB 86,0GB ntfs msftdata 5 467GB 480GB 12,8GB ext4 [nuovo@pippo ~]$ sdc1 archlinux sdc5 archmate
```

```
[nuovo@pippo ~]$ ls -l /boot totale 26836 drwxr-xr-x 4 root root 4096 22 mag 17.54 EFI -rwxr-xr-x 1 root root 18629478 23 mag 10.35 initramfs-linux-fallback.img -rwxr-xr-x 1 root root 3678073 23 mag 10.35 initramfs-linux.img -rwxr-xr-x 1 root 根 692736 18 gen 18.37 intel-ucode.img drwxr-xr-x 3 root 根 4096 22 mag 16.52 loader -rwxr-xr-x 1 root 根 4463456 11 弹匣 22.23 vmlinuz-linux [nuovo@pippo ~]$ ls -l /boot/EFI/ 总计 8 drwxr-xr-x 2 root root 4096 22 mag 17.54 Boot drwxr-xr-x 2 root root 4096 22 mag 17.54 systemd [nuovo@pippo ~]$ ls -l /boot/loader/entries/ 总计 4 -rwxr-xr-x 1 root root 180 22 mag 17.42 arch.conf [nuovo@pippo ~]$ paeted -l bash: paeted: comando non trovato [nuovo@pippo ~]$ pared -l [nuovo@pippo ~]$ sudo parted -l
Modello: ATA TOSHIBA MQ01ABD0 (scsi) Disco /dev/sda: 750GB Dimensione del settore (logica/fisica): 512B/4096B Tabella delle Partizioni: gpt
Flag del disco: Numero Inizio Fine Dimensione 文件系统 Nome Flag 1 1049kB 420MB 419MB ntfs 基本数据分区 nascosta, diag 2 420MB 693MB 273MB fat32 EFI 系统分区 avvio, esp 3 693MB 827MB 134MB Microsoft 保留分区 msftres 4 827MB 652GB 651GB ntfs 基本数据分区 msftdata 5 652GB 729GB 77.3GB ntfs 基本数据分区 msftdata 6 729GB 730GB 895MB ntfs nascosta, diag 7 730GB 731GB 510MB ntfs nascosta, diag 8 731GB 731GB 472MB ntfs nascosta, diag 9 731GB 750GB 18.8GB ntfs 基本数据分区 nascosta, msftdata
型号: TOSHIBA MK5075GSX (scsi) 磁盘 /dev/sdb: 500GB 分区尺寸 (逻辑/物理): 512B/512B 分区表: gpt 标志 disco: Numero Inizio Fine Dimensione 文件系统 Nome Flag 1 1049kB 634MB 633MB fat32 avvio, esp 2 634MB 19,6GB 19,0GB ext4 3 19,6GB 381GB 361GB ntfs 基本数据分区 msftdata 4 381GB 467GB 86,0GB ntfs msftdata 5 467GB 480GB 12,8GB ext4 [nuovo@pippo ~]$ sdc1 archlinux sdc5 archmate
```

andreino wrote on 2016-05-26 15:46:

andreino 于 2016-05-26 15:46 写道:

i solved thank 我解决了，谢谢

Joshua wrote on 2016-07-18 17:04:

Joshua 于 2016 年 7 月 18 日 17:04 写道:

It's long and a bit wordy, but good stuff. The advice is sound. Thanks!

虽然篇幅较长，文字也有些冗长，但内容很好。建议很中肯。谢谢！

Dhruv wrote on 2016-07-22 22:03:

Dhruv 于 2016-07-22 22:03 写道:

I came across this thread while searching for a solution to the problem am facing right now. When I turn on my laptop Dell Inspiron 15 win 8.1, the OS is not loading and the system is stuck at the Aptio setup utility page.Can anyone help me fix this..?? P.S - I am a total noob whwn it comes to computers.

我在寻找解决当前问题的方案时偶然发现了这个帖子。我的戴尔 Inspiron 15 笔记本电脑（Windows 8.1 系统）开机后，操作系统无法启动，一直卡在 Aptio 设置实用程序页面。请问有人能帮我解决这个问题吗？PS：我对电脑一窍不通。

poke49152 wrote on 2016-07-24 10:30:

poke49152 于 2016-07-24 10:30 写道:

UEFI - why to do it simple if you can mess it with old system so that nothing works than anymore. I had issue on linux, when system didnt recognized a disk, then there was some legacy or something like this setting, and it worked by mistake, but in next version probably wont. Also we se theres problem with standard backup . So no matter how many payed bullshit is put in this site, nobody cantr escape from simple fact - UEFI is just another corporate scam to prevet you of using older software and hardware, and to by their new bullshittn they sell now regulary every month, usualy every version more stupid and more unusable and with less concern of users privacy.

UEFI——既然能搞砸旧系统，让所有东西都无法正常工作，为什么还要搞得这么简单呢？我在 Linux 上就遇到过这个问题，系统无法识别磁盘，后来发现是某个遗留的设置出了问题，侥幸能用，但下一个版本可能就不行了。而且我们发现标准备份也存在问题。所以不管这个网站上充斥着多少付费的垃圾信息，没人能逃脱一个简单的事实——UEFI 不过是企业为了阻止你使用旧软件和硬件而耍的又一个骗局，目的是让你购买他们每月定期推出的新产品，而且通常每个版本都更愚蠢、更难用，对用户隐私也更加漠不关心。

amazing wrote on 2016-07-24 16:52:

amazing 于 2016-07-24 16:52 写道：

I stumbled upon this blog and I just wanted to inform you how pompous and ignorant you are: "You do not have a 'UEFI BIOS'. No-one has a 'UEFI BIOS'. Please don't ever say 'UEFI BIOS'" You barely know what firmware is. You have a very weak grasp of EE and chips. And yet you don't have any problem making these kinds of grandiose statements. That's amazing. I don't have much EE knowledge (but certainly more than you) and I've worked closely developing software drivers for chips. If ASUS calls its firmware "UEFI BIOS" then it is called UEFI BIOS. If you think otherwise then you're basically saying that Microsoft doesn't know what Windows is and it is really called dos. Or linux doesn't know what it is talking about and there's no such thing as "linux kernel" - it's always linux. The fact that you can't make the distinction doesn't make your words fact. In fact here's an Intel glossary mentioning an "EFI BIOS": <http://www.intel.com/content/www/us/en/support/topics/glossary.html> And the definition from that page of BIOS/SMBIOS/System BIOS: "BIOS □ Basic Input/Output System is the firmware for a computer whose main function is to identify and initialize various motherboard components as well as to load and transfer control to a small program that then loads the operating system." Also a cursory search on google shows both ASUS and GigaByte say things like "UEFI BIOS" - both online and in manuals. Basically the absolute opposite of what you were certain of is correct. Now I don't think you're a bad guy or that it's bad to be wrong. But you should have some humility when speaking about things you don't know anything about. And I don't even care what Wikipedia or the UEFI organization say because it doesn't matter. If the chip and board manufacturer/integrators say that it is X, then it is X. End of story.

我偶然发现了这个博客，只想告诉你你有多么自大无知：“你没有‘UEFI BIOS’。没人有‘UEFI BIOS’。请永远不要再提‘UEFI BIOS’这个词。”你连固件是什么都搞不清楚。你对电子工程和芯片的理解非常肤浅。然而，你却能毫不费力地发表这种夸夸其谈的言论。真是令人匪夷所思。我的电子工程知识也不多（但肯定比你多），而且我曾参与过芯片软件驱动程序的开发。如果华硕称其固件为“UEFI BIOS”，那就是 UEFI BIOS。如果你不这么认为，那就相当于你在说微软不知道 Windows 是什么，它实际上叫 DOS。或者 Linux 不知道自己在说什么，根本不存在“Linux 内核”这种东西——它永远都是 Linux。你分不清这些概念并不代表你的话就是事实。事实上，英特尔的术语表里提到了“EFI BIOS”：<http://www.intel.com/content/www/us/en/support/topics/glossary.html> 该页面对 BIOS/SMBIOS/系统 BIOS 的定义是：“BIOS——基本输入/输出系统，是计算机的固件，其主要功能是识别和初始化各种主板组件，以及加载并将控制权移交给一个小型程序，该程序随后加载操作系统。”另外，在谷歌上简单搜索一下就会发现，华硕和技嘉都使用过“UEFI BIOS”之类的术语——无论是在网站上还是在手册中。基本上，这与你之前确信的完全相反。我并不认为你是个坏人，或者犯错有什么错。但是，在谈论你一无所知的事情时，你应该保持谦逊。而且，我根本不在乎维基百科或 UEFI 组织怎么说，因为那并不重要。如果芯片和电路板制造商/集成商说它是 X，那就是 X。就这么简单。

Somalia Pirate wrote on 2016-08-05 03:46:

索马里海盗于 2016 年 8 月 5 日 03:46 写道：

In the "UEFI native booting: how it actually works – boot manager entries" section where you say "What does these entries actually mean, though?" should be "What do these entries actually mean, though?"

在“UEFI 原生启动：它的实际工作原理 - 启动管理器条目”部分，您提到的“这些条目究竟是什么意思？”应该改为“这些条目究竟是什么意思？”

Chris wrote on 2016-08-24 16:03:

Chris 于 2016 年 8 月 24 日 16:03 写道：

Adam, Thanks. Great! article. Now I understand how UEFI works better. Maybe I can help out here. First. Any idea of UEFI BIOS is dreck. Plain and simple. No matter who says what. As one said above here BIOS was not the reason for memory limit here. The real reason was the hardware addressing limits caused by the 32-bit word registers used in the memory addresses. Take a simple read in the MS A+ textbook and you will see the initial limit was dealt with in the concept of HighMEM. I was surprised to see the Intel actually developed this as can be seen on this site; https://en.wikipedia.org/wiki/Unified_Extensible_Firmware_Interface. It was done to address the constraints against multi-core, multi-thread CPU's. UEFI REPLACES BIOS, but the compatibilty has to remain. These are just the correct facts of the case. Right now I am busy changing over my Lenovo Windows 8 laptop to Gentoo Linux. The incessant crashes and involuntary OneKey Recoveries finally got to me. I stumbled over your site to make sure I do not destroy the LT. Funny thing. The MBR always baffled me since I never twigged to the unformatted space in front of the first partition where the rest of it was written. Conditions now. My Gentoo livecd is iso'd, but boots up ONLY when UEFI goes AFTER Legacy Boot. Otherwise, with Secure Boot Disabled, it still boots into Windows. I want to strip Win 8 completely, last update was more than a year ago when the forced 8.1 update crashed everything. No time for back ups. Not even manual. Took complete control. Much as people said here, I do not want to be involved in Win 10. fdisk -l show it is a gpt disc. /dev/sda1 1G Win recovery /dev/sda2 260M EFI System /dev/sda3 1G Lenovo boot /dev/sda4 128M MS reserved /dev/sda5 884G MS data /dev/sda6 25G MS data /dev/sda7 20G Win recovery Questions. It seems I can keep /sda2 for future use as you said here. Correct? (I think I wll partition manually). With Linux cp I sould also be able to save /sda3. Previously I wanted to dual install Windows and Linux (on BIOS XP), but

the reigning wisdom was that you had to install Windows first then Linux ot=r get wioped out. Is this still the situation with UEFI? Thanks again, this is very helpful.

Adam, 谢谢。文章写得真棒！现在我对 UEFI 的工作原理有了更深的理解。也许我能帮上忙。首先，任何关于 UEFI BIOS 的说法都是垃圾。就这么简单。不管别人怎么说。正如上面有人提到的，BIOS 并不是造成内存限制的原因。真正的原因是硬件寻址的限制，这是由于内存地址中使用的 32 位字寄存器造成的。随便翻翻微软 A+教材，你就会发现最初的限制是通过 HighMEM 的概念来解决的。我很惊讶地发现英特尔竟然开发了这项技术，你可以在这个网站上看到：https://en.wikipedia.org/wiki/Unified_Extensible_Firmware_Interface。这样做是为了解决多核多线程 CPU 的限制。UEFI 取代了 BIOS，但兼容性必须保持。这些都是事实。我现在正忙着把我的联想 Windows 8 笔记本电脑换成 Gentoo Linux 系统。无休止的崩溃和强制一键恢复最终让我崩溃了。我偶然发现了你们的网站，以确保我的 LT 系统不会彻底报废。说来也怪，MBR 一直让我困惑，因为我从未意识到第一个分区前面那块未格式化的空间才是其余数据写入的地方。现在的情况是这样的：我的 Gentoo LiveCD 是 ISO 镜像，但只有在 UEFI 启动顺序排在 Legacy 启动之后才能启动。否则，即使禁用安全启动，它仍然会启动到 Windows。我想彻底卸载 Win 8，上次更新还是在一年多以前，强制升级到 8.1 导致所有系统崩溃。没时间备份了，连手动备份都没做。我完全掌控了系统。正如这里其他人所说，我不想再碰 Win 10 了。fdisk -l 显示它是 GPT 磁盘。 /dev/sda1 1G Windows 恢复 /dev/sda2 260M EFI 系统 /dev/sda3 1G Lenovo 启动 /dev/sda4 128M MS 保留 /dev/sda5 884G MS 数据 /dev/sda6 25G MS 数据 /dev/sda7 20G Windows 恢复 问题：正如您所说，我似乎可以保留 /sda2 以备将来使用，对吗？（我想我会手动分区）。使用 Linux cp 命令，我应该也可以保留 /sda3。之前我想在 BIOS XP 系统上双安装 Windows 和 Linux，但普遍的说法是必须先安装 Windows，否则 Linux 会被覆盖。UEFI 系统下情况是否仍然如此？ 再次感谢，这很有帮助。

[Tim Gatton](#) wrote on 2016-08-30 13:33:

[Tim Gatton](#) 于 2016 年 8 月 30 日 13:33 写道：

Thanks for the UEFI path explanation ... been trying for 2 days via the DELL web site (my OEM) and was getting no where.

感谢您对 UEFI 路径的解释.....我已经通过戴尔网站（我的 OEM 厂商）尝试了两天，但毫无进展。

OK wrote on 2016-09-13 11:14:

OK 于 2016-09-13 11:14 写道：

Such a douchey way of writing an article.

这种写文章的方式真是令人作呕。

[adamw](#) wrote on 2016-09-13 23:47:

[adamw](#) 于 2016-09-13 23:47 写道：

Such a douchey way of writing a comment.

这种评论方式真是太没品了。

Ef wrote on 2016-09-29 01:55:

Ef 于 2016-09-29 01:55 写道：

I too believe that words should be explicit rather than customary. Just because many use a phrase like "UEFI BIOS" doesn't excuse that it is misleading and confusing if you don't already know what is being referenced. My pet peeve is "dark matter". Two things about dark matter, it isn't dark (it's clear) and it isn't matter (it has a mass property but no other form characteristics of matter). But I learned to no longer be upset when someone says dark matter. I'll never say 'dark matter' or 'UEFI BIOS'. When I hear them, it tells me that the speaker or writer is lazy, misinformed or doesn't actually understand what they are talking about. Thanks for the UEFI article. Very helpful. Although I still can't see how to stop Windows from changing the EFI settings every time it runs. I figured out how to turn off 'recoveryenabled'. But somehow the Windows Boot Loader always gets reinserted first in the fwbootmgr settings. Maybe it's the firmware doing it. I seemed to have learned that from this article.

我也认为用词应该明确，而不是墨守成规。仅仅因为很多人都用“UEFI BIOS”这样的短语，并不能掩盖它在不了解其含义的情况下会造成误导和困惑的事实。我最反感的就是“暗物质”这个词。关于暗物质，有两点需要说明：它并不暗（它是透明的），而且它不是物质（它有质量，但没有物质的其他形态特征）。但我已经学会了不再对别人说“暗物质”感到恼火。我永远不会说“暗物质”或“UEFI BIOS”。每当我听到这些词，我都会觉得说话者或作者很懒惰、信息不足，或者根本不理解他们在说什么。感谢您提供的关于 UEFI 的文章，非常有用。虽然我仍然不知道如何阻止 Windows 每次启动时都更改 EFI 设置。我已经找到了关闭“recoveryenabled”的方法，但不知何故，Windows 启动加载程序总是会被重新插入到 fwbootmgr 设置中。也许是固件的问题。我似乎是从这篇文章中学到这一点的。

Profff wrote on 2016-10-03 20:07:

Profff 于 2016-10-03 20:07 写道：

why does my laptop bott to windows 10 quicker via a UEFI boot and GPT partition?

为什么我的笔记本电脑通过 UEFI 启动和 GPT 分区启动到 Windows 10 的速度更快？

Tom Hale wrote on 2016-10-14 07:06:

Tom Hale 于 2016 年 10 月 14 日 07:06 写道：

Great article. One thing to fix: "what it'll really likely do add is configure"

文章写得很好。有一点需要修改：“它真正可能添加的功能是配置”。

[adamw](#) wrote on 2016-10-26 00:58:

[adamw](#) 于 2016-10-26 00:58 写道:

Thanks Tom, fixed! 谢谢汤姆，问题解决了！

HAABackwater Bumpkin wrote on 2016-10-24 06:54:

HAABackwater Bumpkin 于 2016-10-24 06:54 写道:

Thanks AdamW! I'm also a pedandict podunk and while this whole computer business is completely beyond my simplistic nature and etc , you've truly explained things in a way me and that other pleb could grasp :) I have a couple other minor clarifications and I'd really appreciate it if you had another moment to offer some of your information. 1.) please tell me if I'm understanding correctly that the manufacturer has mislabeled the updated firmware as bios on a newer Dell Inspiron and whether it does me any good to attempt to "re flash" it again and again. I've done this many times and this blasted malware just returns. 2.) do I also hear you say that the MALICIOUS code is contained within the firmware and/or the media I'm try to use to fix the problems, and which would be the better scenario and more likely for a guy like me to actually pull off? I've heard it is possible to modify an ISO or image file but I hadn't considered that you might ever attempt to fix the firmware itself. 3) (last one) is this a situation that is better dealt with while an operating system is installed on a permanent disk or better handled with the disk removed using a live cd? Please excuse me my backwards and ordinary ways and let me assure you that not in all my years of blindly meandering through YouTube tutorials that this is a wonderful explanation of how UEFI differs from bios and what it actually does. Maybe you should keep at this writing stuff especially because it's a real treat to read. 🍷

谢谢 AdamW！我也是个喜欢钻研电脑的菜鸟，虽然这套电脑技术对我来说完全是天方夜谭，但你解释得真透彻，我和其他菜鸟都能理解 :) 我还有几个小问题想请教一下，如果你能抽出点时间解答一下就太好了。1) 请问我理解得对吗？戴尔 Inspiron 的新款机型上，厂商把更新后的固件错误地标记为 BIOS，反复“刷新”固件有用吗？我已经试过很多次了，但该死的恶意软件还是会再次出现。2) 你是不是说恶意代码包含在固件和/或我用来修复问题的介质中？哪种情况更容易操作，也更适合像我这样的菜鸟？我听说可以修改 ISO 或镜像文件，但我没想到还可以尝试修复固件本身。3) (最后一个问题) 这种情况是在操作系统安装到永久磁盘上时处理得更好，还是在移除磁盘并使用 Live CD 处理得更好？请原谅我笨拙的表达方式，而且我可以向您保证，在我多年漫无目的地浏览 YouTube 教程的经历中，这绝对是对 UEFI 与 BIOS 的区别以及它实际作用的最佳解释。也许您应该继续写这类文章，因为读起来真的是一种享受。🍷

[adamw](#) wrote on 2016-10-26 00:57:

[adamw](#) 于 2016-10-26 00:57 写道:

1) Yeah, it's not uncommon for vendors to refer to a UEFI firmware as 'a BIOS' or something like that. I am in a bit of a minority with my insistence that the colloquial usage of 'BIOS' to mean 'anything that's kind of a system firmware' is stupid and wrong, but I still think I'm right. :)

1) 是的，厂商把 UEFI 固件称为“BIOS”之类的说法并不罕见。我坚持认为用“BIOS”来指代“任何系统固件”这种说法既愚蠢又错误，我知道这可能属于少数派，但我仍然认为我的观点是对的。:)

2) I'm not quite sure what you mean by 'malicious code' here. This post is a general explainer about how UEFI works, it's not about fixing some kind of malware issue. It sounds like you're dealing with some kind of malware issue but you didn't actually explain what it is, so I can't really answer your question, I'm afraid.

2) 我不太明白你说的“恶意代码”是什么意思。这篇文章是对 UEFI 工作原理的一般性解释，并不是关于修复恶意软件问题的。听起来你似乎遇到了某种恶意软件问题，但你没有具体说明是什么，所以恐怕我无法回答你的问题。

3) Seems like a ditto to #2.

3) 似乎与第 2 点相同。

Azix wrote on 2016-10-27 13:24:

Azix 于 2016-10-27 13:24 写道:

So UEFI is a kind of BIOS?

所以 UEFI 是一种 BIOS?

[adamw](#) wrote on 2016-11-02 01:21:

[adamw](#) 于 2016-11-02 01:21 写道:

Fry meme 炸薯条梗

ryan wrote on 2016-11-01 11:02:

ryan 于 2016 年 11 月 1 日 11:02 写道:

link to uefi spec requires login, fyi

UEFI 规范链接需要登录，仅供参考。

Vis wrote on 2016-11-02 20:28:

Vis 于 2016-11-02 20:28 写道:

Good read. Have one question. I understand every firmware could have their own implementation, but what is the expected or general behavior if there are no boot entries in NVRAM, but the ESP have kernel directories and bootloaders present. I tried this experiment in an ubuntu system. Created 2 directories inside the boot partition. EFI/Ubuntu/grubx64.efi EFI/fedora/grubx64.efi Deleted the boot entries from NVRAM and did a power cycle. When the system came back up, could see Ubuntu entries in efibootmgr -v and not fedora. Am curios to know how figured out one entry and not another. (There is no fedora entry in grub.cfg and only Ubuntu, but i believe that is immaterial to create the boot entry) Thanks.

这篇文章写得很好。我有一个问题。我知道每个固件都有自己的实现方式，但如果 NVRAM 中没有启动项，而 ESP 分区里有内核目录和引导加载程序，那么预期行为或一般行为是什么？我在 Ubuntu 系统上做了这个实验。我在启动分区里创建了两个目录：EFI/Ubuntu/grubx64.efi 和 EFI/fedora/grubx64.efi。然后我删除了 NVRAM 中的启动项并重启了系统。系统重启后，我在 `efibootmgr -v` 命令中看到了 Ubuntu 的启动项，但没有看到 Fedora 的。我很好奇为什么会识别出一个启动项而不是另一个。（grub.cfg 文件中只有 Ubuntu 的启动项，没有 Fedora 的，但我认为这与创建启动项无关。）谢谢。

Vis wrote on 2016-11-02 21:02:

Vis 于 2016-11-02 21:02 写道:

I could get the firmware create the fedora entry, by having a shim.efi file inside \EFI\fedora. When i created another directory say vis (\EFI\vis\), with a bootloader file grubx64.efi and shimx64.efi, firmware doesnt create though. What file does firmware expect to create a boot entry.

我通过在 \EFI\fedora 目录下放置一个 shim.efi 文件，成功让固件创建了 Fedora 启动项。但是，当我创建另一个目录，例如 vis (\EFI\vis\), 并在其中放置了引导加载程序文件 grubx64.efi 和 shimx64.efi 时，固件却无法创建启动项。请问固件需要什么文件才能创建启动项？

[adamw](#) wrote on 2016-11-03 19:37:

[adamw](#) 于 2016-11-03 19:37 写道:

The only thing the UEFI spec *requires* the firmware to implement is the fallback path, which is basically the thing where there's a designated location where the firmware is supposed to look on the first ESP it finds on a disk. For an x86_64 system, the fallback path is \EFI\BOOT\BOOTx64.EFI , as explained above.

UEFI 规范唯一要求固件实现的功能是回退路径，它本质上是一个指定位置，固件在磁盘上找到第一个 ESP 分区时，应该从该位置开始查找。对于 x86_64 系统，回退路径是 \EFI\BOOT\BOOTx64.EFI，如上所述。

Now, what I'm betting happened in your case is this. Fedora has a trick where when we install, we place a bootloader in the fallback path location . If that bootloader gets loaded, it boots Fedora normally, *and re-creates the 'regular' Fedora EFI boot manager entry*. So if you install Fedora alone to a UEFI system, delete the Fedora boot menu entry, then reboot, you should see the system boot to Fedora, then on the next reboot, and 'efibootmgr' will show the 'Fedora' entry again.

现在，我猜测你遇到的情况是这样的。Fedora 有一个小技巧，安装时会在备用路径位置放置一个引导加载程序。如果这个引导加载程序被加载，Fedora 就能正常启动，并重新创建“常规”的 Fedora EFI 启动管理器条目。所以，如果你在 UEFI 系统上单独安装 Fedora，删除 Fedora 启动菜单条目，然后重启，你应该会看到系统启动到 Fedora，然后在下次重启时，“efibootmgr”会再次显示“Fedora”条目。

I'm betting Ubuntu has implemented the same trick, and you installed Ubuntu after Fedora, so it was Ubuntu's fallback loader that wound up getting run, not Fedora's. So Ubuntu did the same thing I described, and recreated its boot manager entry.

我估计 Ubuntu 也用了同样的技巧，而且你是在 Fedora 之后安装的 Ubuntu，所以最后运行的是 Ubuntu 的备用启动器，而不是 Fedora 的。因此，Ubuntu 也做了我之前描述的事情，重新创建了它的启动管理器条目。

BTW, there is an open question I'm supposed to be figuring out here (it's come up a few times recently), which is whether the UEFI spec actually requires the firmware to try fallback booting from attached hard disks / SSDs if no boot manager entries are present, or if it only *requires* the firmware to try fallback booting from removable devices. I still need to look that up again. But most firmwares *do* in fact try fallback boot from attached hard disks / SSDs if no other boot entry is present or works.

顺便说一下，这里有个悬而未决的问题我需要弄清楚（最近出现过几次），那就是 UEFI 规范是否真的要求固件在没有启动管理器条目的情况下尝试从连接的硬盘/固态硬盘启动，还是只要求固件尝试从可移动设备启动。我还需要再查一下。但大多数固件实际上都会在没有其他启动条目或启动失败的情况下尝试从连接的硬盘/固态硬盘启动。

Vis wrote on 2016-11-04 07:15:

Vis 于 2016-11-04 07:15 写道:

In my case, i am not quite sure, it is booting because of fall back path. I believe Firmware is doing something different. Have shown the sequence below. Assuming that with the fallback bootloader being present, why is it not detecting my directory in ESP(say vis, which has the same contents) and detects only ubuntu and fedora boot loaders. Does firmware creates entries of known/registered vendor. Below is the sequence: Before reboot (I have deleted all boot entries using sudo efibootmgr -b # -B /mnt/1/EFI\$ sudo efibootmgr -v | grep -e Ubuntu -e Fedora -e vis -e Vis /mnt/1/EFI\$ Below are my ESP files. /mnt/1/EFI\$ ls -ltr total 12 drwxr-xr-x 2 root root 4096 Nov 2 20:37 ubuntu drwxr-xr-x 2 root root 4096 Nov 2 20:54 fedora drwxr-xr-x 2 root root 4096 Nov 2 21:36 vis /mnt/1/EFI\$ ls -ltr ubuntu/ total 0 -rwxr-xr-x 1 root root 0 Nov 2 20:37 grubx64.efi /mnt/1/EFI\$ ls -ltr fedora/ total 408 -rwxr-xr-x 1 root root 135680 Oct 14 02:48 bootx64.efi -rwxr-xr-x 1 root root 135680 Nov 2 00:14 grubx64.efi -rwxr-xr-x 1 root root 135680 Nov 2 20:54 shim.efi /mnt/1/EFI\$ ls -ltr vis/ total 408 -rwxr-xr-x 1 root root 135680 Nov 2 00:15 shimx64.efi -rwxr-xr-x 1 root root 135680 Nov 2 07:23 grubx64.efi -rwxr-xr-x 1 root root 135680 Nov 2 21:36 shim.efi sudo reboot (/dev/pts/0) at 7:00 ... The system is going down for reboot NOW! Once the system is back up, i see boot entries for Ubuntu and Fedora and not for vis. (I have 3 boot drives. One we are looking at is 965970eb-d8d1-4d73-aa1e-d0c7bad6f948 (/dev/sda1) /mnt/1/EFI\$ ls -ltr /dev/disk/by-partuuid/965970eb-d8d1-4d73-aa1e-d0c7bad6f948 lrwxrwxrwx 1 root root 10 Nov 4 07:02 /dev/disk/by-partuuid/965970eb-d8d1-4d73-aa1e-d0c7bad6f948 -> ../../sda1 /mnt/1/EFI\$ sudo efibootmgr -v | grep -e Ubuntu -e Fedora -e vis -e Vis Boot0010* Fedora HD(1,GPT,965970eb-d8d1-4d73-aa1e-d0c7bad6f948,0x100,0x1ff00)/File(\EFI\FEDORA\SHIM.EFI)..BO Boot0011* Ubuntu HD(1,GPT,965970eb-d8d1-4d73-aa1e-d0c7bad6f948,0x100,0x1ff00)/File(\EFI\UBUNTU\GRUBX64.EFI)..BO Boot0012* Ubuntu HD(1,GPT,efd05c7c-c77b-475f-876e-12a585bf9a15,0x100,0x1ff00)/File(\EFI\Ubuntu\grubx64.efi)..BO Boot0013* Ubuntu HD(1,GPT,caafb2f8-a788-4210-988b-f7cb8ab46954,0x100,0x1ff00)/File(\EFI\Ubuntu\grubx64.efi)..BO /mnt/1/EFI\$ ls -ltr total 12 drwxr-xr-x 2 root root 4096 Nov 2 20:37 ubuntu drwxr-xr-x 2 root root 4096 Nov 2 20:54 fedora drwxr-xr-x 2 root root 4096 Nov 2 21:36 vis There is no BOOT under EFI. Curious to see the above behavior.

就我而言，我不太确定，它之所以能启动是因为使用了备用路径。我认为固件执行的操作有所不同。以下是启动过程。假设备用引导加载程序存在，为什么它检测不到 ESP 目录（例如 vis 目录，其内容与主目录相同），而只检测到 Ubuntu 和 Fedora 的引导加载程序？固件是否会创建已知/已注册厂商的条目？ 以下是顺序： 重启前（我已经使用 sudo efibootmgr -b # -B 删除了所有启动项） /mnt/1/EFI\$ sudo efibootmgr -v | /mnt/1/EFI\$ grep -e Ubuntu -e Fedora -e vis -e Vis /mnt/1/EFI\$ 以下是我的 ESP 文件。 /mnt/1/EFI\$ ls -ltr 总共12 drwxr-xr-x 2 root root 4096 11 月 2 日 20:37 ubuntu drwxr-xr-x 2 root root 4096 11 月 2 日 20:54 fedora drwxr-xr-x 2 root root 4096 11 月 2 日 21:36 vis /mnt/1/EFI\$ ls -ltr ubuntu/ 总计 0 -rwxr-xr-x 1 root root 0 11 月 2 日 20:37 grubx64.efi /mnt/1/EFI\$ ls -ltr fedora/ 总计 408 -rwxr-xr-x 1 root root 135680 10 月 14 日 02:48 bootx64.efi -rwxr-xr-x 1 root root 135680 十一月 2 00:14 grubx64.efi -rwxr-xr-x 1 root root 135680 11 月 2 日 20:54 shim.efi /mnt/1/EFI\$ ls -ltr vis/ 总计 408 -rwxr-xr-x 1 root root 135680 11 月 2 日 00:15 shimx64.efi -rwxr-xr-x 1 root root 135680 十一月 2 07:23 grubx64.efi -rwxr-xr-x 1 root root 135680 11 月 2 日 21:36 shim.efi sudo 重启 (/dev/pts/0) at 7:00 ... 系统即将重启！系统恢复后，我看到了 Ubuntu 和 Fedora 的启动项，但没有看到 vis 的启动项。（我有 3 个启动盘。我们正在查看的其中一个是 965970eb-d8d1-4d73-aa1e-d0c7bad6f948 (/dev/sda1)） /mnt/1/EFI\$ ls -ltr /dev/disk/by-partuuid/965970eb-d8d1-4d73-aa1e-d0c7bad6f948 lrwxrwxrwx 1 root root 10 11 月 4 07:02 /dev/disk/by-partuuid/965970eb-d8d1-4d73-aa1e-d0c7bad6f948 -> ../../sda1 /mnt/1/EFI\$ sudo efibootmgr -v | /mnt/1/EFI\$ grep -e Ubuntu -e Fedora -e vis -e Vis Boot0010* Fedora HD(1,GPT,965970eb-d8d1-4d73-aa1e-d0c7bad6f948,0x100,0x1ff00)/File(\EFI\FEDORA\SHIM.EFI)..BO Boot0011* Ubuntu HD(1,GPT,965970eb-d8d1-4d73-aa1e-d0c7bad6f948,0x100,0x1ff00)/File(\EFI\UBUNTU\GRUBX64.EFI)..BO Boot0012* Ubuntu HD(1,GPT,efd05c7c-c77b-475f-876e-12a585bf9a15,0x100,0x1ff00)/File(\EFI\Ubuntu\grubx64.efi)..BO Boot0013* Ubuntu HD(1,GPT,caafb2f8-a788-4210-988b-f7cb8ab46954,0x100,0x1ff00)/File(\EFI\Ubuntu\grubx64.efi)..BO /mnt/1/EFI\$ ls -ltr 总共12 drwxr-xr-x 2 root root 4096 11 月 2 日 20:37 ubuntu drwxr-xr-x 2 root root 4096 11 月 2 日 20:54 fedora drwxr-xr-x 2 root root 4096 11 月 2 日 21:36 vis EFI 下没有 BOOT 选项。很好奇上述行为究竟是如何发生的。

Irvine Specter wrote on 2016-11-12 02:55:

Irvine Specter 于 2016 年 11 月 12 日 02:55 写道：

As a beneficiary of your graphomania, thanks for your work. I hope to have enough time to read all the useful things you write!

作为您写作热情的受益者，感谢您的付出。我希望有足够的时间阅读您所有精彩的文章！

Gary Gapinski wrote on 2016-11-12 04:09:

Gary Gapinski 于 2016 年 11 月 12 日 04:09 写道：

Nicely written. I learned recommendation #5 the hard way after much puzzlement when trying to get a VirtualBox kernel module loaded with Secure Boot enabled.

写得很好。我之前在启用安全启动的情况下加载 VirtualBox 内核模块时，遇到了很多困惑，最终才明白第五条建议的道理。

Ravi wrote on 2016-11-12 12:55:

Ravi 于 2016 年 11 月 12 日 12:55 写道：

Your article is a great! After reading rEFInd docs and installing it I was able to grasp idea about how UEFI works but was still looking for more info. I am using Linux on a MacBook Pro and needless to say things don't work perfectly, though still good. One of the problems I face is slowness in UEFI's firmware. You press a key it takes solid 10 seconds to go to different menu. GRUB's timeout of 8 seconds takes more than a minute. Hard to explain but it has been observed by many people. Link: [http://apple.stackexchange.com/questions/245918/osx-startup-manager-boot-drive-selection-screen-slow-unresponsive-in-macbook#comment306041_245918]. And its not just that grub is slow, whole UEFI environment is slow. MacBook's boot menu, rEFInd

menu and GRUB's menu. And sometimes it vanishes. How to troubleshoot this?

您的文章太棒了！阅读并安装 rEFInd 文档后，我对 UEFI 的工作原理有了初步了解，但仍然想了解更多信息。我在 MacBook Pro 上使用 Linux 系统，虽然目前还不错，但不得不说，它并非完美无缺。我遇到的一个问题是 UEFI 固件运行缓慢。按下一个键后，需要整整 10 秒钟才能进入不同的菜单。GRUB 的 8 秒超时时间，我却要等一分钟以上。这很难解释，但很多人都遇到过这个问题。链接：
[http://apple.stackexchange.com/questions/245918/osx-startup-manager-boot-drive-selection-screen-slow-un-responsive-in-macbook#comment306041_245918]。而且不仅仅是 GRUB 慢，整个 UEFI 环境都很慢。MacBook 的启动菜单、rEFInd 菜单和 GRUB 菜单都存在这个问题。有时，菜单甚至会消失。如何排查这个问题？

Ravi wrote on 2016-11-12 13:02:

Ravi 于 2016 年 11 月 12 日 13:02 写道：

Thanks to your patient explanation, I was able to confidently change boot order using efibootmgr.

感谢您的耐心解释，我才能自信地使用 efibootmgr 更改启动顺序。

payu wrote on 2016-11-12 22:41:

payu 于 2016-11-12 22:41 写道：

One following question: Is the GSM locked in certain carrier also related to secure boot? For example, I have a phone, unlocking the secure boot and installing a custom android system, can it possibly to use another carrier if the phone was originally locked to certain carrier?

还有一个问题：GSM 网络锁定是否与安全启动有关？例如，我有一部手机，解锁安全启动并安装了自定义安卓系统后，如果这部手机原本被锁定到某个运营商，我还能使用其他运营商的网络吗？

Oyvind Overby wrote on 2016-11-13 10:29:

奥温德·奥弗比 于 2016-11-13 10:29 写道：

It would be nice to know why one should consider changing from the old BIOS/MBR thing to UEFI, assuming the scenario where I already have Win 10 installed on my PC the old fasion BIOS/MBR way. What is there to benefit? Boot Speed? Overall increased average HDD speed? Better protection against, or ability to recover from bad sectors or other intermittent HDD defects? Less HDD space wasted due to more felxible block sizes (in case of zillions of small files)? Improved read-ahead? Delayed write? An security improvents (apart from secure-boot)? Nobody should have to read 40 pages of tech stuff and still wonder; whatever for? Why? What is the benefit?

我想知道，如果我的电脑上已经安装了 Win 10 系统，并且使用的是传统的 BIOS/MBR 模式，那么为什么要考虑从旧的 BIOS/MBR 切换到 UEFI 呢？这样做有什么好处？启动速度？硬盘平均速度的提升？对坏扇区或其他间歇性硬盘故障的更好保护或恢复能力？由于更灵活的块大小（在存储大量小文件的情况下）而减少硬盘空间的浪费？预读性能的提升？延迟写入？安全性的提升（除了安全启动之外）？没人应该读了 40 页的技术文档后仍然一头雾水：到底是为了什么？为什么？有什么好处？

Ravi wrote on 2016-11-14 06:58:

Ravi 于 2016 年 11 月 14 日 06:58 写道：

If its working for you no need to change. You never had to read how BIOS boot works since its an old method with lot of tools, troubleshooters and tutorials to help you. But its still a hack. 512 bytes is too little of a space to work. You still need "magic" to load your bootloader, which loads the kernel. Read the article again. UEFI takes a break and provides a clean implementation which can be replicated. Kernels can boot directly using it.

如果它对你有效，就无需更改。你以前从未了解过 BIOS 启动的工作原理，因为它是一种古老的方法，而且有很多工具、故障排除程序和教程可以帮助你。但它仍然是一种变通方法。512 字节的空间太小了，无法正常工作。你仍然需要一些“技巧”来加载引导加载程序，而引导加载程序又会加载内核。请再次阅读文章。UEFI 提供了一种更简洁的实现方式，可以进行复现。内核可以直接使用它启动。

[Daniel Scott](#) wrote on 2019-02-03 00:41:

[Daniel Scott](#) 于 2019 年 2 月 3 日 00:41 写道：

Oyvind Overby: One of the chief "practical" benefits of UEFI is that it will allow you to boot an operating system that is installed on a partition larger than 2TB. BIOS/MBR cannot do this.

Oyvind Overby：UEFI 的主要“实用”优势之一是，它允许你启动安装在大于 2TB 分区上的操作系统。BIOS/MBR 无法做到这一点。

[adamw](#) wrote on 2019-02-06 18:30:

[adamw](#) 于 2019-02-06 18:30 写道：

Well, that's a benefit of GPT, strictly speaking. BIOS + GPT can do it (with a special partition, and if the BIOS in question doesn't choke on GPT). Anaconda supports doing this.

严格来说，这是 GPT 的一个优势。BIOS + GPT 可以实现这一点（需要创建一个特殊的分区，并且前提是相应的 BIOS 不会对 GPT 造成兼容性问题）。Anaconda 支持这种做法。

[Overmind](#) wrote on 2016-11-15 12:10:

[Overmind](#) 于 2016-11-15 12:10 写道：

I disagree with the statement that 'MBR itself is not big enough'. it does not have to be. The bootloader is not and should not act like an operating system. The bootloader is a simple small file that if needed can just load a bigger file. No problem here whatsoever.

我不同意“MBR 本身不够大”的说法。它不需要那么大。引导加载程序不是也不应该像操作系统那样运行。引导加载程序只是一个简单的、小的文件，如果需要，它可以加载一个更大的文件。这里完全没有问题。

kaleemullah360 wrote on 2016-11-17 07:28:

kaleemullah360 于 2016-11-17 07:28 写道：

this tutorial need some pictures, screenshots. rest is good. you must add some pictures.

本教程需要一些图片和截图。其他部分都很好。你必须添加一些图片。

UrezD wrote on 2017-02-09 06:35:

UrezD 于 2017-02-09 06:35 写道：

Thanks for the read. Question though, I wrote down the wrong password to my admin UEFI. If I completely reimage the EFI partition, will that reset the UEFI boot loader?

感谢阅读。不过我有个问题，我不小心把管理员 UEFI 密码记错了。如果我完全重装 EFI 分区，会重置 UEFI 引导程序吗？

Indeximal wrote on 2017-05-20 12:13:

Indeximal 于 2017-05-20 12:13 写道：

Thank you! This was amazingly easy and interesting to read.

谢谢！这篇文章读起来既轻松又有趣。

Ajay wrote on 2017-06-01 18:58:

Ajay 于 2017-06-01 18:58 写道：

Well, when we say BIOS, we refer a program that starts the computer system. It is of course stored as firmware. It initializes hardware and invokes booting process. UEFI does the same what traditional BIOS programs (distributed by Award, American Megatrends etc.) have been doing. So, Computers keep evolving. After 8 bit 16 bit machines we have have 32 bit and 64 bit machines. It is still computer. Saying UEFI is not a BIOS program is like saying like 16-bit machines were not computers.

我们说的 BIOS，指的是启动计算机系统的程序。它当然是以固件的形式存储的。它负责初始化硬件并启动系统。UEFI 的功能与传统的 BIOS 程序（例如 Award、American Megatrends 等公司提供的程序）相同。所以，计算机一直在不断发展。从 8 位、16 位计算机发展到 32 位、64 位计算机，它们仍然是计算机。说 UEFI 不是 BIOS 程序，就如同说 16 位计算机不是计算机一样。

Computer Scientist wrote on 2018-08-20 20:01:

计算机科学家于 2018 年 8 月 20 日 20:01 写道：

You still misunderstand the term BIOS, which makes your analogy flawed. BIOS is a specific type of firmware, BIOS is not a generic way to refer to a program that starts the commuter. You may be using it that way, but you would be wrong to do so. Firmware is the program that starts the computer, BIOS is a specific implementation of that. To call every program that starts a computer a BIOS is just plain wrong, and shows limited experience with computers beyond your average IBM PC compatible machine.

你仍然误解了 BIOS 这个术语，这使得你的类比存在缺陷。BIOS 是一种特定类型的固件，它并非指启动计算机的通用程序。你或许是这样理解的，但这是不正确的。固件是启动计算机的程序，而 BIOS 是固件的一种特定实现。将所有启动计算机的程序都称为 BIOS 是完全错误的，这表明你对计算机的了解仅限于普通的 IBM PC 兼容机。

R. Subudhi wrote on 2017-06-04 15:48:

R. Subudhi 于 2017 年 6 月 4 日 15:48 写道：

Good article, informative, well-written and easily understood, especially by novices like me. All my confusions regarding MBR boot and UEFI boot are now gone. However, it would be more useful if a paragraph or two could be added bow a uefi boot system handles devices attached to the system, since at least two of my devices have stopped functioning after I switched over to UEFI booting.

这篇文章写得很好，内容丰富，文笔流畅，通俗易懂，尤其适合像我这样的新手。我之前对 MBR 启动和 UEFI 启动的疑惑都解开了。不过，如果能再加一两段关于 UEFI 启动系统如何处理连接到系统的设备的说明就更好了，因为切换到 UEFI 启动后，至少有两个设备无法正常工作了。

Dinitri wrote on 2017-06-18 18:27:

Dinitri 于 2017-06-18 18:27 写道：

Great explanation with so much to absorb. (love your sense of humor too--much lacking in the tech world!) One thought; the UEFI standard can/does look for extensions or devices which are obviously not bootable (such as a "video adapter"...lol), then this forward thinking allows for possible numerous future doors to insert a layer within the 'UEFI' structure AND is that not defeating the original motive/reason to codifying the booting firmware in the first place? In simple terms: UEFI is getting rid of the possibility of having an UNDERLYING root program unknowing placed below the booting

instructions WHILE creating an obvious future problem of being corrupted by a real 'bootable video adapter' lol not so funny then lol tc thanks for your efforts and clarity. nice job..Dinitri

解释得真好，信息量很大。（我也很喜欢你的幽默感——科技界现在很缺乏这种幽默感！）我有个想法：UEFI 标准会查找那些明显无法启动的扩展或设备（比如“视频适配器”……哈哈），那么这种前瞻性的设计就可能在未来给 UEFI 结构内部插入一层，但这岂不是违背了最初对启动固件进行编码的初衷吗？简单来说：UEFI 消除了启动指令下方放置底层 root 程序的可能性，但同时也制造了一个显而易见的问题：它可能会因为真正的“可启动视频适配器”而损坏。哈哈，这样看来就没那么好笑了。谢谢你的努力和清晰的解释。干得漂亮！——Dinitri

Ariel wrote on 2017-06-22 12:55:

Ariel 于 2017-06-22 12:55 写道：

Thank you friend. Great explanation.

谢谢朋友，解释得很清楚。

AwokeKnowing wrote on 2017-06-27 19:54:

AwokeKnowing 于 2017-06-27 19:54 写道：

Thank you. I've been putting off studying this precisely because of all the vague "features/sales explanations". This article was just what I needed as I built my new system with a mix of old and new drives and several OS. As Einstein says "simple as possible, but no simpler".

谢谢。我一直拖着没研究这方面的内容，正是因为那些含糊不清的“功能/销售说明”。这篇文章正是我需要的，因为我正在组装一台新系统，里面混用了新旧硬盘和几个操作系统。正如爱因斯坦所说，“尽可能简单，但不能过于简单”。

Richard H wrote on 2017-07-09 22:06:

Richard H 于 2017-07-09 22:06 写道：

A few times I've removed a secondary drive, and for some reason it messed up my bootloader.....I'm curious what might have happened there as it wasn't the OS drive, I figured that one wouldn't have anything to do with the computer booting.

我曾几次移除辅助硬盘，不知何故都搞砸了我的引导加载程序……我很好奇到底发生了什么，因为那不是系统盘，我以为系统盘跟电脑启动没什么关系。

CHRIS wrote on 2017-07-21 16:50:

CHRIS 于 2017-07-21 16:50 写道：

Indication of TOOLS for managing UEFI file wood be very welcome . General concepts are fine but practical management is better . Thank you any way for enlighntning this VERY DARK subject ...especially the SECURE BOOT part JCD

非常欢迎提供用于管理 UEFI 文件的工具。一般概念固然好，但实际操作更佳。无论如何，感谢您阐明这个非常晦涩的主题……特别是安全启动部分，JCD

JM C wrote on 2017-07-30 03:04:

JM C 于 2017-07-30 03:04 写道：

cluster disks that are connected to shared SCSI or Fibre Channel buses that are used by the Cluster service. Poor mortals like me cannot understand the above.

集群磁盘连接到集群服务使用的共享 SCSI 或光纤通道总线。像我这样的凡人无法理解以上内容。

D E Davis wrote on 2017-08-28 17:27:

DE Davis 于 2017-08-28 17:27 写道：

Very informative, with but two small gripes: 1. In your explanation you've veered off onto Lunix, and at 86 years of age I may never know Lunix. 2. You didn't list DOS/Windows tools for examining the UEFI code.

内容非常详实，只有两点小小的不足：1. 你在解释中提到了 Linux，而我已 86 岁高龄，可能永远也无法了解 Linux。2. 你没有列出用于检查 UEFI 代码的 DOS/Windows 工具。

Raman Katoch wrote on 2017-09-03 08:52:

拉曼·卡托奇于 2017-09-03 08:52 写道：

Too informative to digest ! But superbly written !

信息量太大，难以消化！但文笔极佳！

Old Sarge wrote on 2017-09-04 11:54:

老萨奇于 2017 年 9 月 4 日 11:54 写道：

The BIOS settings allow me to set a boot order, but the UEFI shows the hard drive as master and is greyed out. I am retired IT, but a simpler article with step-by-step bullets would be better.

BIOS 设置允许我设置启动顺序，但 UEFI 显示硬盘为主启动盘，且该选项呈灰色不可用状态。我已退休，从事 IT 工作，但如果能提供一篇步骤更清晰、更简单的文章就更好了。

tshilscorn wrote on 2017-09-04 20:28:

tshilscorn 于 2017-09-04 20:28 写道：

I had windows boot manager controlling my bios on Samsung NP300E5X then after too many unsuccessful attempts trying to reset bios to default I decided to reset the cmos battery. However since I reset the cmos battery, the windows boot manager was wiped out and now the computer is stuck at bios with no drive detected nor any other boot option. When I start the computer it first restart, then automatically goes to bios with only two blank options, boot menu and app menu. Please help me fix this either hardware or software.

我之前用 Windows 启动管理器控制三星 NP300E5X 的 BIOS，但多次尝试将 BIOS 重置为默认设置都失败后，我决定重置 CMOS 电池。然而，重置 CMOS 电池后，Windows 启动管理器被清除，现在电脑卡在 BIOS 界面，检测不到任何硬盘，也没有其他启动选项。开机后，电脑会先重启，然后自动进入 BIOS，但只有启动菜单和应用程序菜单两个空白选项。请帮我解决这个问题，无论是硬件问题还是软件问题。

Shaun Clarke wrote on 2017-09-26 12:11:

Shaun Clarke 于 2017 年 9 月 26 日 12:11 写道：

Just wanted to thank the author for the write up. Very informative and I certainly learned a lot, even though I'm a Windows guy :). One thing that has had me curious, and actually led to me researching UEFI in more detail is why UEFI uses a FAT (compatible) file system, as FAT is a proprietary file system by Microsoft. Do UEFI implementations have to pay royalties as a result?

非常感谢作者的文章。内容非常丰富，即使我是个 Windows 用户，也从中受益匪浅。其中有一点让我很好奇，也促使我更深入地研究 UEFI：为什么 UEFI 会使用 FAT（兼容）文件系统？毕竟 FAT 是微软的专有文件系统。UEFI 的实现是否需要为此支付版权费？

[SteveSi](#) wrote on 2017-09-26 12:38:

[SteveSi](#) 于 2017-09-26 12:38 写道：

<http://download.microsoft.com/download/1/6/1/161ba512-40e2-4cc9-843a-923143f3456c/fatgen103.doc> Microsoft granted rights specifically for UEFI.

<http://download.microsoft.com/download/1/6/1/161ba512-40e2-4cc9-843a-923143f3456c/fatgen103.doc> 微软专门授予了 UEFI 权限。

Nikolay Kuchumov wrote on 2017-11-12 19:52:

尼古拉·库楚莫夫于 2017-11-12 19:52 写道：

Fag 烟

Dieter Pfeuffer wrote on 2017-11-14 09:07:

Dieter Pfeuffer 于 2017 年 11 月 14 日 09:07 写道：

I have had some issues with UEFI boot and GPT partitioned disks while setting up a multi boot system (Windows 10, OpenSUSE, Kali Linux). I searched the internet and read some forum posts and howtos to solve my problems. But there are really a lot of “half true info” and erroneous info out there. At last I found this very good article. Thanks a lot to share your expert knowledge!

我在设置多系统启动（Windows 10、OpenSUSE、Kali Linux）时遇到了一些 UEFI 启动和 GPT 分区磁盘的问题。我上网搜索，阅读了一些论坛帖子和教程来解决问题。但是网上充斥着大量“半真半假”的信息和错误信息。最终，我找到了这篇非常好的文章。非常感谢您分享您的专业知识！

Chas B wrote on 2018-09-06 16:09:

Chas B 于 2018-09-06 16:09 写道：

How does the UEFI firmware find the ESP partition location on disk ? Where does it get the disk partition information ? Apologies if you mentioned this already.

UEFI 固件是如何找到磁盘上 ESP 分区位置的？它从哪里获取磁盘分区信息？如果您之前已经提到过这一点，请见谅。

[adamw](#) wrote on 2018-09-06 18:05:

[adamw](#) 于 2018-09-06 18:05 写道：

Simple - it reads the partition table, like anything else would. :)

很简单——它读取分区表，就像其他程序一样。:)

Chas B wrote on 2018-09-06 21:37:

Chas B 于 2018-09-06 21:37 写道:

So how does the UEFI firmware know where the partition table is ? I know that for legacy BIOS it is always in the first disk sector (MBR) And also does it look for MBR partition table and GPT partition table ?

那么 UEFI 固件是如何知道分区表位置的呢？我知道对于传统 BIOS 来说，分区表总是位于磁盘的第一个扇区（MBR）。另外，它会同时查找 MBR 分区表和 GPT 分区表吗？

[adamw](#) wrote on 2018-09-06 21:53:

[adamw](#) 于 2018-09-06 21:53 写道:

MBR and GPT both live in defined positions on the disk. The primary GPT from LBA 1 to LBA 33, the secondary GPT (basically a backup) at the end of the disk, in the last 33 blocks. See https://en.wikipedia.org/wiki/GUID_Partition_Table .

MBR 和 GPT 都位于磁盘上的特定位置。主 GPT 从 LBA 1 到 LBA 33，辅助 GPT（本质上是一个备份）位于磁盘末尾，即最后 33 个块。参见 https://en.wikipedia.org/wiki/GUID_Partition_Table。

UEFI firmwares *can* read both MBR and GPT, usually, and many BIOS firmwares can read and boot from GPT. Typically, though, UEFI is and should be used with GPT. The Fedora installer actually refuses to let you do a UEFI install to an MBR-labelled disk, we just don't want to support that.

UEFI 固件通常可以读取 MBR 和 GPT 分区表，许多 BIOS 固件也可以读取 GPT 分区表并从中启动。不过，UEFI 通常（也应该）与 GPT 分区表一起使用。Fedora 安装程序实际上不允许将 UEFI 安装到 MBR 分区表的磁盘上，我们只是不想支持这种安装方式。

Chas B wrote on 2018-09-06 22:14:

Chas B 于 2018-09-06 22:14 写道:

Thanks Adam, you have provided some good information and advice here, I found my way to here to this site while trying to fix an incurable dual boot system which I think is to do with a mix of BIOS and UEFI installations.

谢谢 Adam，你在这里提供了一些很好的信息和建议。我在尝试修复一个无法解决的双启动系统时找到了这个网站，我认为这与 BIOS 和 UEFI 安装的混合有关。

Stanford Mayfield wrote on 2018-09-10 15:37:

斯坦福·梅菲尔德于 2018 年 9 月 10 日 15:37 写道:

good read!! 值得一读!

Guy wrote on 2018-10-12 10:31:

Guy 于 2018-10-12 10:31 写道:

Does UEFI offer anything around making startup of systems more resilient? e.g. On an embedded system I might have two possible boot media. If one is completely unresponsive then boot order, just like in BIOS, should let me move to the next. But if the media is present and the issue is later in the boot process (e.g. Linux kernel panic and fails to load) - is there any facility in UEFI that allows me to detect that and boot from the alternate media the next time around? e.g. boot sequence first writes to somewhere in UEFI and alters the boot order - if it starts up fine it then undoes that change but if not then next boot it will use the fallback media? I'm also looking at whether the hardware watchdog can be accessed on the system so that I can force that reboot on the first boot hanging (not much help changing the boot order if I can't also ensure the system will cycle as well). Any ideas? thanks

UEFI 是否提供了提高系统启动稳定性的功能？例如，在嵌入式系统中，我可能有两个启动介质。如果其中一个完全无响应，那么启动顺序设置应该像 BIOS 一样，让我切换到下一个介质。但是，如果介质存在，而问题发生在启动过程的后期（例如 Linux 内核崩溃导致加载失败），UEFI 是否有任何机制可以检测到这种情况，并在下次启动时使用备用介质？例如，启动顺序设置可以先写入 UEFI 的某个位置并更改启动顺序——如果启动正常，则撤销该更改；如果启动失败，则下次启动时使用备用介质？我还在研究是否可以访问系统的硬件看门狗，以便在第一次启动卡住时强制重启（如果不能确保系统重启，更改启动顺序就没什么意义了）。有什么想法吗？谢谢！

James Up de Graff wrote on 2018-10-26 04:54:

James Up de Graff 于 2018 年 10 月 26 日 04:54 写道:

It will be well when the BIOS firmware is totally abandoned due to occasional incompatibility with GPI disks, etc. I used to like Veritas System Recovery for backup. It used to work well with BIOS based computers; however, with the GPI disks it will no longer function in some instances. For example, I have a two year old Dell 8910 desktop computer that is BIOS based. The OS is installed on a GPI SSD, which works with almost no difficulties. When I backup a system image to a GPI external disk, there are no problems. However, when I try to restore the system image, the restore fails. The error message states that with a BIOS based computer the system image can only be restored to an MBR formatted disk. As a consequence I can no longer use Veritas System

Recovery for backups of a system image. If anyone knows how to avoid this failure, I would be pleased to hear.

BIOS 固件彻底被弃用是件好事，因为它偶尔会与 GPI 磁盘等设备不兼容。我以前很喜欢用 Veritas System Recovery 做备份。它以前在基于 BIOS 的电脑上运行良好；但是，在某些情况下，它无法在 GPI 磁盘上运行。例如，我有一台两年前的戴尔 8910 台式机，它是基于 BIOS 的。操作系统安装在 GPI SSD 上，运行几乎没有任何问题。当我将系统映像备份到 GPI 外置磁盘时，一切正常。但是，当我尝试恢复系统映像时，恢复失败。错误信息显示，对于基于 BIOS 的电脑，系统映像只能恢复到 MBR 格式的磁盘。因此，我无法再使用 Veritas System Recovery 来备份系统映像。如果有人知道如何避免这种情况，请不吝赐教。

[Steve Si](#) wrote on 2018-10-26 07:57:

[Steve Si](#) 于 2018 年 10 月 26 日 07:57 写道：

Try this: Convert the ISO of the Veritas Recovery CD to a .imgPTN file using the Easy2Boot MPI Tool Pack. Copy the .imgPTN file to an Easy2Boot USB drive. MBR-boot to the .imgPTN file (or select it using SWITCH_E2B.exe under Windows). MBR-boot to the CSM menu an choose the Clover boot option which will boot the UEFI-version of Veritas System Recovery. www.easy2boot.com

试试这个方法：使用 Easy2Boot MPI 工具包将 Veritas Recovery CD 的 ISO 镜像转换为 .imgPTN 文件。将 .imgPTN 文件复制到 Easy2Boot USB 驱动器。使用 MBR 引导程序启动到 .imgPTN 文件（或者在 Windows 系统下使用 SWITCH_E2B.exe 选择该文件）。使用 MBR 引导程序启动到 CSM 菜单，然后选择 Clover 引导选项，这将启动 UEFI 版本的 Veritas System Recovery。 www.easy2boot.com

jamesup wrote on 2018-10-26 16:38:

jamesup 于 2018-10-26 16:38 写道：

Steve, Thank you for your recommendation. I will give that suggestion a try and let you know if there is success.

史蒂夫，谢谢你的建议。我会试试这个方法，如果成功了我会告诉你。

Will Matthews wrote on 2018-10-26 11:22:

Will Matthews 于 2018 年 10 月 26 日 11:22 写道：

Adam, This is excellent. Many thanks.

亚当，这太棒了，非常感谢。

Frank L Y Speaking wrote on 2018-11-02 20:44:

Frank LY Speaking 于 2018 年 11 月 2 日 20:44 写道：

The word "actually" appears 32 times in this article. If you omitted ALL of those occurances it would not change the meaning in this slightest. I often wonder if people who insist on interjecting that word are playing a variation of Simon Says. If you don't say "actually", does it mean that you are lying?

这篇文章里“实际上”这个词出现了32次。就算把这32次全删掉，意思也不会有丝毫改变。我常常在想，那些非要插上这个词的人是不是在玩“西蒙说”的变体游戏。如果不说“实际上”，是不是就意味着你在撒谎？

[adamw](#) wrote on 2018-11-02 21:20:

[adamw](#) 于 2018-11-02 21:20 写道：

That's funny, I don't recall advertising for an editor. No, I'm not hiring. Buh bye.

真奇怪，我不记得我发过招聘编辑的广告。不，我不招人。再见。

Night Eagle wrote on 2018-11-09 03:19:

Night Eagle 于 2018-11-09 03:19 写道：

Wow, my head is spinning. I have learned a great deal reading all this. Unfortunately not enough to understand why on my HP laptop I can't boot from my DVD/cd in anything but "Legacy" mode thus when I do I cant use the UEFI to reinstall windows. When I boot from the DVD in UEFI the install disk cant be read. I think my computer is haunted.

哇，我头都大了。读了这么多，我学到了很多。可惜的是，我还是不明白为什么我的惠普笔记本电脑只能以“传统”模式从 DVD/CD 启动，所以即使以“传统”模式启动，也无法使用 UEFI 重新安装 Windows。而当我尝试以 UEFI 模式从 DVD 启动时，安装光盘又无法读取。我觉得我的电脑闹鬼了。

Night Eagle wrote on 2018-11-09 03:38:

Night Eagle 于 2018-11-09 03:38 写道：

It has to do with my install disk, doesn't it?

这跟我的安装光盘有关，对吧？

[adamw](#) wrote on 2018-11-09 06:52:

[adamw](#) 于 2018-11-09 06:52 写道：

It's possible, yes. The disc you're trying to boot could simply not be set up to be UEFI bootable at all. An image can legacy bootable, UEFI bootable, or both.

是的，有可能。您尝试启动的磁盘可能根本没有设置为 UEFI 启动。镜像文件可以是传统启动盘、UEFI 启动盘，或者两者兼具。

ranran wrote on 2018-11-26 21:58:

ranran 于 2018-11-26 21:58 写道：

Hello Adam, I hope you can help me, May I please ask a question about adding boot entry for MBR shadow (used with sedutils) ? The problem: I don't have the "shadow mbr" entry in the bios. I can find it with uefi shell, but can't add it as entry using "bcfg" because bcfg is not supported in this BIOS..... (Is it the same in your issue) ? This is the entry as seen by uefi shell; fs0:\EFI\boot\bootx64.efi So, since I don't have bcfg, I wanted to add it with efibootmgr, but it didn't boot when I pressed on it (in bios), so I thought that maybe I didn't add it as required. Do you know how I can find it with efibootmgr (just as we find it with ls in uefi shell) ? I tried the following options but both have no result on pressing this new entry. efibootmgr -c -d /dev/sda -p 1 -L "mypba" -l "\EFI\boot\bootx64.efi" efibootmgr -c -d /dev/sda -p 2 -L "mypba" -l "\EFI\boot\bootx64.efi" Thank you for any idea you'll have !!!!!!!!!!!!!!! Ran

你好 Adam，希望你能帮帮我。请问一下关于添加 MBR 影子启动项（使用 sedutils）的问题？问题是：我的 BIOS 里没有“shadow mbr”启动项。我可以用 UEFI shell 找到它，但是无法使用“bcfg”命令添加，因为这个 BIOS 不支持 bcfg.....（你的问题也是这样吗？）这是 UEFI shell 显示的启动项：fs0:\EFI\boot\bootx64.efi。由于没有 bcfg，我想用 efibootmgr 添加，但是按下后（在 BIOS 里）系统没有启动，所以我怀疑是不是添加的方式不对。你知道怎么用 efibootmgr 找到它吗（就像在 UEFI shell 里用 ls 命令查找一样）？我尝试了以下方法，但是都无法添加这个新启动项。efibootmgr -c -d /dev/sda -p 1 -L "mypba" -l "\EFI\boot\bootx64.efi" efibootmgr -c -d /dev/sda -p 2 -L "mypba" -l "\EFI\boot\bootx64.efi" 非常感谢您的任何建议！

Northpass Kid wrote on 2018-12-05 00:21:

Northpass Kid 于 2018-12-05 00:21 写道：

The post cleared up a couple of details about UEFI and reassured me that what I did know was valid. It does need a little fact-check though. In a Mac, the ESP as it is created by the Disk Utility is a 200G FAT32 filesystem. I don't recall that it was ever HFS+.

这篇文章澄清了关于 UEFI 的一些细节，也让我确信我之前了解的知识是正确的。不过，它确实需要一些事实核查。在 Mac 系统中，磁盘工具创建的 ESP 分区是一个 200GB 的 FAT32 文件系统。我不记得它曾经是 HFS+ 格式的。

[adamw](#) wrote on 2018-12-05 07:51:

[adamw](#) 于 2018-12-05 07:51 写道：

I didn't say the ESP is HFS+ on Macs, though. I said "Apple, for instance, ships at least some Macs with their bootloaders in an HFS+ partition." That is, there was some point at which Apple was shipping Macs set up to run an EFI bootloader from an HFS+ partition.

我并没有说 Mac 上的 ESP 分区是 HFS+ 格式。我说的是“例如，苹果公司至少在某些 Mac 电脑上出厂时，其引导加载程序就位于 HFS+ 分区中。”也就是说，苹果公司曾经在某个时期出厂的 Mac 电脑上设置了从 HFS+ 分区运行 EFI 引导加载程序的功能。

I don't think I have the references any more - this post is over four years old now, after all - but I'm reasonably sure that was the case at some point, as IIRC we had to teach anaconda to recognize it.

我想我已经找不到相关的参考资料了——毕竟这篇文章已经超过四年了——但我相当肯定在某个时候确实如此，因为如果我没记错的话，我们当时不得不教 Anaconda 来识别它。

Northpass Kid wrote on 2018-12-06 06:52:

Northpass Kid 于 2018-12-06 06:52 写道：

OK, I understand your distinction. And yes, I seem to remember that Apple took liberties by having the bootloader NOT in the ESP. The ESP was left empty but the partition had to be there by GPT standard. Maybe it can be blamed on Apple being the early adopter of UEFI.

好的，我明白你的意思了。是的，我记得苹果公司确实采取了一些特殊做法，没有把引导加载程序放在 ESP 分区里。ESP 分区是空的，但按照 GPT 标准，这个分区必须存在。这或许要归咎于苹果公司是 UEFI 的早期采用者。

tech4him wrote on 2019-01-03 18:17:

tech4him 于 2019-01-03 18:17 写道：

Looks like you can actually download the spec directly now, without having to register :). <http://www.uefi.org/specifications>

现在似乎可以直接下载规范文件了，无需注册哦 :)。 <http://www.uefi.org/specifications>

davit wrote on 2019-01-06 21:19:

戴维特于 2019-01-06 21:19 写道：

thanks. enjoyed a lot

谢谢。非常享受。

Dee wrote on 2019-01-09 07:50:

Dee 于 2019 年 1 月 9 日 07:50 写道:

I really enjoyed this explanation, especially the tone of acerbic disdain for those who, inaccurately, use "BIOS" to refer to any system firmware when they should use the generic "firmware" or specify UEFI, ePROM, etc. I am currently in training to be a Certified Dell Client Technician and the material is incredibly poorly written & inaccurate. The module I am on (which led me here via a search for the details of GPT vs. MBR partitioning) specifically points out that the hardware being discussed is only compatible with UEFI, and then instructs the technician to install the most current "BIOS" firmware. Your irritation echoed my own. One question: As someone who went through the early days of running Windows on the internet, I was always terrified of malicious software designed to infect the MBR or even hide in BIOS, I wondered if UEFI, since it allows for the OS to directly access and interact with firmware, might present a larger attack surface for malicious code. An unexpected "benefit" of the MBR was that it was very difficult to alter (partially since it was hap-hazard and didn't work very well). With UEFI, the firmware and boot instructions are (if I understand) stored in a specific location and backed up in another easily found location; the OS is able to interact directly with it and it uses only a CRC check to verify the validity of the boot instructions -- wouldn't it be much harder to secure in a Windows environment? Just wondering.

我非常喜欢这篇解释，尤其是其中对那些错误地将“BIOS”指代所有系统固件的人的尖锐批评，他们应该使用通用的“固件”或具体说明是 UEFI、ePROM 等等。我目前正在接受戴尔认证客户端技术员的培训，但培训材料写得非常糟糕，而且错误百出。我正在学习的模块（我是通过搜索 GPT 和 MBR 分区细节找到这里的）明确指出，所讨论的硬件仅兼容 UEFI，然后指示技术人员安装最新的“BIOS”固件。你的不满正是我所感同身受的。我还有一个问题：作为一个经历过早期 Windows 互联网时代的人，我一直很担心恶意软件会感染 MBR 甚至隐藏在 BIOS 中。我想知道，UEFI 既然允许操作系统直接访问固件并与之交互，是否会为恶意代码提供更大的攻击面。MBR 的一个意想不到的“好处”是它很难被篡改（部分原因是它本身设计随意且运行不佳）。而 UEFI 中，固件和启动指令（如果我理解正确的话）存储在一个特定位置，并在另一个容易找到的位置进行备份；操作系统可以直接与它交互，并且仅使用 CRC 校验来验证启动指令的有效性——在 Windows 环境下，这难道不会更难保证安全吗？我只是好奇而已。

Piotr wrote on 2019-01-24 14:26:

Piotr 于 2019-01-24 14:26 写道:

Hi Dee, I have got the same feeling as UEFI systems are more attractive to access boot instructions from OS level. Maybe that's why Secure Boot has been implemented to allow only signed drives and signatures to allow in the booting process.

嗨，Dee，我也有同感，UEFI 系统似乎更倾向于从操作系统层面访问启动指令。也许这就是为什么安全启动机制的存在，它只允许经过签名的驱动器和签名才能启动。

Russell wrote on 2019-01-13 21:55:

Russell 于 2019 年 1 月 13 日 21:55 写道:

Thankyou Adam for a very useful and informative post. I actually enjoyed reading through! Discovered this while trying to give new life to my 2011 MacBook Pro with a dual boot setup (Ubuntu). Your comment around Mac firmware interface and getting a pretty icon to show was funny to me. I've got the icon (after much effort) but I can not for the life of me get the entry to show as "Ubuntu" just keep getting "EFI Boot".

亚当，非常感谢你这篇实用又内容丰富的帖子！我读得很开心！我是在尝试给我的 2011 款 MacBook Pro 安装双系统（Ubuntu）时发现这篇文章的。你关于 Mac 固件界面以及如何显示漂亮图标的评价让我觉得很有意思。我费了好大劲才弄到图标，但是无论如何都无法让启动项显示为“Ubuntu”，一直显示的是“EFI 启动”。

koko wrote on 2019-02-07 19:52:

科科于 2019-02-07 19:52 写道:

UEFI is a peace of shit... or rather to say vendors locking it's abilities

UEFI 简直就是一坨屎.....或者更确切地说，是厂商限制了它的功能。

Pedro Costa wrote on 2019-02-14 19:51:

Pedro Costa 于 2019 年 2 月 14 日 19:51 写道:

Incredibly useful and well written article! Congratulations and thank you for sharing your knowledge on such acronym-heavy subject in a simple way to understand. Keep it up! Cheers

这篇文章真是太实用、太精彩了！恭喜你，感谢你用如此浅显易懂的方式分享了你在这个充斥着大量缩略语的主题的知识。继续加油！祝好！

Desert Tripper wrote on 2019-03-06 10:32:

Desert Tripper 于 2019 年 3 月 6 日 10:32 写道:

Great article, but why the unnecessary jab at Macs?

文章写得很好，但为什么要无端地抨击 Mac 电脑呢？

[adamw](#) wrote on 2019-04-15 21:52:

[adamw](#) 于 2019-04-15 21:52 写道:

Because it's my blog and I'll jab at anyone I like! Macs have historically been a medium-sized pain in the ass to anyone trying to maintain an operating system that isn't macOS which tries to work on them. So we like to work the frustration out somewhere.

因为这是我的博客，我想吐槽谁就吐槽谁！Mac 电脑一直以来都让那些试图在 Mac 上运行非 macOS 操作系统的人头疼不已。所以我们喜欢找个地方发泄一下这种挫败感。

Porfy wrote on 2019-05-06 15:51:

Porfy 于 2019-05-06 15:51 写道:

I'm new to UEFI. I've done many dual-boot installations, but all based on BIOS/MBR. I basically tried to do UEFI when I got a new laptop that is dual-disc. It came only with the HDD, and with my curiosity piqued, I bought the NVMe SSD then re-installed Windows using UEFI firmware, as this is supposed to bring out the speed and all intended for the NVMe SSD. Did I understand that correctly? At any rate, I managed to get this particular machine to do dual-boot. Secondly, I have older laptops, like 3-4 years old, and some have the mSATA SSD slot. Some I am able to turn off Legacy support and install Windows under UEFI firmware. No problem with that. My question is on those machines that I am not able to turn off Legacy/CSM. I have tried and I am still able to install Windows using UEFI firmware (yes, my installer is UEFI-enabled created using Rufus). Is this okay, or am I headed in the wrong direction? I am asking as once I am done with Windows installation, I will be attempting to do dual-boot: install Ubuntu Linux (UEFI mode), and finally, Grub Customizer, which I found in previous installs is able to read and use the EFI boot manager. I hope my short explanation is clear. Let me know please on my queries. Thank you!

我是 UEFI 新手。我之前做过很多双启动安装，但都是基于 BIOS/MBR 的。我最近入手了一台双硬盘笔记本电脑，想尝试一下 UEFI 系统。这台电脑出厂时只有一块机械硬盘，出于好奇，我又买了一块 NVMe 固态硬盘，然后用 UEFI 固件重新安装了 Windows 系统，因为据说这样可以充分发挥 NVMe 固态硬盘的性能。我的理解对吗？总之，我成功地在这台电脑上实现了双启动。另外，我还有一些比较老的笔记本电脑，大概用了三四年了，有些有 mSATA 固态硬盘插槽。有些电脑我可以关闭 Legacy 支持，然后用 UEFI 固件安装 Windows 系统，这没问题。我的问题是，对于那些无法关闭 Legacy/CSM 支持的电脑，我尝试过，但仍然能够用 UEFI 固件安装 Windows 系统（是的，我的安装程序是用 Rufus 创建的，支持 UEFI）。这样做可以吗？还是我走错了方向？我这么问是因为，Windows 安装完成后，我打算尝试双系统启动：安装 Ubuntu Linux（UEFI 模式），最后安装 Grub Customizer。我之前安装过其他系统，发现 Grub Customizer 可以读取和使用 EFI 启动管理器。希望我的解释足够清楚。请不吝赐教。谢谢！

[adamw](#) wrote on 2019-05-29 23:32:

[adamw](#) 于 2019-05-29 23:32 写道:

Hi Porfy! I am not sure about this whole NVMe SSD thing, that's new to me. But to answer your main question: you don't need to turn CSM support off entirely to do a native UEFI boot, in theory. If the firmware is actually a UEFI firmware, there *ought* to be a way in there for you to do a UEFI native boot of your OS install medium. But because the firmware UIs aren't standardized at all and vary a lot between systems, that's about as specific as I can get. Good luck!

嗨，Porfy！我对 NVMe SSD 不太了解，这对我来说也是个新领域。不过，回答你的主要问题：理论上，你不需要完全关闭 CSM 支持就能实现原生 UEFI 启动。如果固件确实是 UEFI 固件，那么*应该*有办法让你从操作系统安装介质中实现原生 UEFI 启动。但由于固件的用户界面完全没有统一的标准，而且不同系统之间的差异很大，所以我只能提供这么多具体的信息了。祝你好运！

[Nick Brown](#) wrote on 2019-06-03 10:23:

[Nick Brown](#) 于 2019 年 6 月 3 日 10:23 写道:

Great article. Reminds me of the days when the Internet was full of stuff like this - actual detailed, text-intensive, informative articles rather than click-bait and ads.

好文章。让我想起了互联网上曾经充斥着这类文章的时代——真正详尽、文字丰富、信息量大的文章，而不是充斥着标题党和广告。

Paul Bonneau wrote on 2019-07-02 02:22:

保罗·博诺于 2019-07-02 02:22 写道:

I am somewhat amused at the recommendations, e.g. "If you can possibly manage it, have one OS per computer." Here I was reading all about the wonderfulness of UEFI allowing me to flexibly and powerfully boot anything at all - and then the number one recommendation is to hobble yourself the same way you'd do with a BIOS setup, just to have a fair chance the thing will work. Yes, I know you and Adam are two distinct people, but you have to admit this is pretty funny.

我对那些建议感到有些好笑，比如“如果可以的话，每台电脑最好只装一个操作系统”。我之前一直在看 UEFI 的各种优点，说它能让我灵活强大地启动任何程序——结果第一条建议居然是像用 BIOS 一样给自己设限，就为了保证系统能正常运行。我知道你和 Adam 是两个不同的人，但你不得不承认这确实挺搞笑的。

[adamw](#) wrote on 2019-10-18 18:25:

[adamw](#) 于 2019-10-18 18:25 写道:

"Yes, I know you and Adam are two distinct people" Er...are we? I was pretty sure I was just me. :P I mean, that's my practical recommendation for how to have a simple and happy life, born of painful experience. It applies to BIOS and UEFI (and every other case I've encountered, honestly). UEFI makes a

game *effort* to improve things, and really does improve some things, and if it had all been implemented in the best way possible by everybody would *really* make multibooting less of a pain. But in the sad reality we live in, it gets implemented by firmware engineers and OS developers, so it is of course awful just like everything else...

“是的，我知道你和亚当是两个人。”呃.....是吗？我一直以为只有我自己呢。:PI 我的意思是，这是我从痛苦的经历中总结出的，关于如何拥有简单快乐生活的实用建议。它适用于 BIOS 和 UEFI（说实话，也适用于我遇到的所有其他情况）。UEFI 确实在努力改进，也确实在某些方面有所改进，如果所有人都以最佳方式实现它，那么多系统启动的麻烦程度将会大大降低。但在我们所处的悲哀现实中，它是由固件工程师和操作系统开发人员实现的，所以它当然和其他所有东西一样糟糕.....

Kt wrote on 2019-07-31 11:35:

Kt 于 2019-07-31 11:35 写道：

Far too many words without much technical content

文字太多，技术内容却很少。

[adamw](#) wrote on 2019-10-18 18:15:

[adamw](#) 于 2019-10-18 18:15 写道：

In that case I suggest you try just reading every third word instead. Free tip! You're welcome!

既然如此，我建议你试试每隔三个字读一个。免费小贴士！不用谢！

MAC wrote on 2019-08-01 23:12:

MAC 于 2019-08-01 23:12 写道：

Very simple explanation. Was able to understand without any difficulties as I have had experience with BIOS installation. Keep on indulging the Grahpomania with more expert/simplified stuff like this. Great stuff. Many thanks.

解释得非常简单明了。因为我之前安装过 BIOS，所以理解起来毫不费力。请继续为图形爱好者们带来更多这样专业又简明的教程。太棒了！非常感谢。

readfull wrote on 2019-08-02 09:27:

readfull 于 2019-08-02 09:27 写道：

Thank you very very much

非常非常感谢！

n3p wrote on 2019-10-02 00:54:

n3p 于 2019-10-02 00:54 写道：

3 Fedoras (30,31,rawhide) + Debian SID + Ubuntu 19.10 +Ubuntu 19.04 + Windows 10 ALL OF THEM WITH SECURE BOOT in 2 disks (SDD [UEFI] + HDD).

3 个 Fedora 系统（30、31、rawhide）+ Debian SID + Ubuntu 19.10 + Ubuntu 19.04 + Windows 10，全部都启用了安全启动，安装在 2 个磁盘上（SSD [UEFI] + HDD）。

Johny wrote on 2019-10-18 01:34:

Johny 于 2019 年 10 月 18 日 01:34 写道：

What described in this article is not 100% true. I just booted from my windows efi partition and I checked my system disk, it does have an efi partition but the disk is MBR format, not GPT So even if you have a MBR partition table, you can still boot from an efi partition in that table, this follows the specs of uefi that you could boot from anywhere that contains efi boot loader. I don't think uefi exclusively asking for GPT disk when booting

本文所述内容并非完全正确。我刚刚从 Windows EFI 分区启动，并检查了系统磁盘，发现它确实有一个 EFI 分区，但磁盘格式为 MBR，而非 GPT。因此，即使分区表为 MBR，仍然可以从该分区表中的 EFI 分区启动，这符合 UEFI 的规范，即可以从任何包含 EFI 引导加载程序的分区启动。我不认为 UEFI 在启动时会专门要求使用 GPT 磁盘。

[adamw](#) wrote on 2019-10-18 18:16:

[adamw](#) 于 2019-10-18 18:16 写道：

Quote from article: "For the sake of sanity, I am going to assume the use of disks with a GPT partition table, and EFI FAT32 EFI system partitions. Depending on how deep you're going to dive into this stuff you may find out that it's not strictly speaking the case that you can always assume you'll be dealing with GPT disks and EFI FAT32 ESPs when dealing with UEFI native boot, but the UEFI specification is quite strongly tied to GPT disks and EFI FAT32 ESPs, and this is what you'll be dealing with in 99% of cases. Unless you're dealing with Macs, and quite frankly, screw Macs."

文章摘录：“为了方便起见，我假设使用的是 GPT 分区表磁盘和 EFI FAT32 系统分区。根据你深入研究的程度，你可能会发现，严格来说，在处理 UEFI 原生启动时，并非总是可以假设你处理的是 GPT 磁盘和 EFI FAT32 ESP 分区。但 UEFI 规范与 GPT 磁盘和 EFI FAT32 ESP 分区紧密相关，99% 的情况下你都会遇到这种情况。除非你处理的是 Mac，说实话，Mac 什么的都见鬼去吧。”

Mark Dinges wrote on 2019-10-21 13:16:

Mark Dinges 于 2019 年 10 月 21 日 13:16 写道:

I thought you might find this interesting. From <https://pcsupport.lenovo.com/us/en/products/laptops-and-netbooks/thinkpad-edge-laptops/thinkpad-e595-type-20nf/downloads/DS539418> Description This package updates the UEFI BIOS (including system program and Embedded Controller program) stored in the Lenovo computer to fix problems, add new functions, or expand functions as noted below.

我想你可能会对这个感兴趣。来自 <https://pcsupport.lenovo.com/us/en/products/laptops-and-netbooks/thinkpad-edge-laptops/thinkpad-e595-type-20nf/downloads/DS539418> 描述: 此软件包更新联想电脑中存储的 UEFI BIOS (包括系统程序和嵌入式控制器程序), 以修复问题、添加新功能或扩展功能, 如下所述。

Alexander wrote on 2019-10-24 09:18:

Alexander 于 2019 年 10 月 24 日 09:18 写道:

Gawd... What the Secure Boot has to do with ATi/NVIDIA proprietary drivers or your own kernels??? I didn't get that comletely. Doesn't Secure Boot just define that partition too boot from is encrypted with some known to firmware key??

我的天.....安全启动跟 ATi/NVIDIA 的专有驱动程序或者你自己的内核有什么关系??? 我没完全明白。安全启动不就是把启动分区用固件密钥加密了吗?

[adamw](#) wrote on 2019-12-11 02:04:

[adamw](#) 于 2019-12-11 02:04 写道:

No, it's not about encryption. It's about signing. Rather than "encrypted", Secure Boot says "all the bits involved in booting the system are *signed* with a known-to-firmware key". The relationship to proprietary drivers and your own kernels is pretty simple: those things are part of the boot chain. So if you enable Secure Boot, they must be signed appropriately. If not, the system won't boot or the driver won't load.

不, 这与加密无关, 而是与签名有关。安全启动并非“加密”, 而是“所有参与系统启动的数据都使用固件已知的密钥进行*签名*”。它与专有驱动程序和您自己的内核之间的关系非常简单: 它们都是启动链的一部分。因此, 如果您启用安全启动, 它们必须经过适当的签名。否则, 系统将无法启动或驱动程序将无法加载。

tram wrote on 2019-11-11 20:56:

tram 于 2019-11-11 20:56 写道:

Your article is fantastic and really helped me understand why I went through 2 days-of-pain trying to get various gaming operating systems installed on different hardware recently!

你的文章太棒了, 真的帮我理解了为什么我最近花了整整两天时间才在不同的硬件上安装好各种游戏操作系统!

[Digi](#) wrote on 2019-12-16 13:43:

[Digi](#) 于 2019-12-16 13:43 写道:

Great explanation with so much to absorb. Thanks

讲解非常精彩, 信息量很大。谢谢。

Harper Pekay wrote on 2019-12-17 02:35:

Harper Pekay 于 2019 年 12 月 17 日 02:35 写道:

This helped me connect the dots together. Thanks for expanding the Internet knowledge base!

这帮我把所有线索串联起来了。感谢您拓展了互联网知识库!

Adam Piggott wrote on 2020-02-10 19:09:

Adam Piggott 于 2020 年 2 月 10 日 19:09 写道:

An eminently readable, informative and useful article. Thank you!

这是一篇非常易读、内容丰富且实用的文章。谢谢!

Ludwik wrote on 2020-03-06 13:01:

路德维克于 2020-03-06 13:01 写道:

I was planning to read just the important stuff from the beginning and skip the rest, but the rest was written in so much logical and nice way, I had to do it all :D totally worth it. Great knowledge and tutor skills

我本来打算只看开头重要的部分, 跳过其他内容, 但后面的内容写得逻辑清晰、条理分明, 我忍不住全部读完了 :D 绝对值得! 知识渊博, 辅导技巧一流。

Keith wrote on 2020-03-07 15:05:

Keith 于 2020-03-07 15:05 写道:

Love the article. Purchased new mobo and all the fixens to build system. Forget about issues that drove me here in first place. You must forgive us for saying UEFI BIOS when that's exactly what system says after booting into it.

很喜欢这篇文章。我已经购买了新的主板和所有必要的配件来组装电脑。之前困扰我的那些问题现在都迎刃而解了。请原谅我们说的是 UEFI BIOS，因为系统启动后显示的就是这个。

Thomas Tang wrote on 2020-03-12 06:31:

Thomas Tang 于 2020-03-12 06:31 写道:

This is a very cool blog. Could you give generic solution hints to "This is not a bootable disk error". A fallback boot loader is in EPS in the SSD; UEFI 2.7. Thank you very much.

这是一个很棒的博客。请问您能否提供一些关于“这不是可启动磁盘”错误的通用解决方案提示？备用启动加载程序位于固态硬盘 (SSD) 的 EPS 分区中；系统版本为 UEFI 2.7。非常感谢！

Jo Hutabarat wrote on 2020-03-29 02:47:

乔·胡塔巴拉特于 2020-03-29 02:47 写道:

Pheeew... Thank you for the "human" explanation of the UEFI technical details contained in those various official documents. I now understand why I kept failing to boot from a second installation of Debian 10 (BUSTER) on my external HDD. In addition, this article has given me a fundamentally important expanded view on how I will from now on select my next mobo. Before reading this article, I took matters very simply when it came to choosing a motherboard: Does it support my chosen CPU? Does it have a built-in graphics adapter? What are the ports it has? How many memory slots are there? And so on. But BIOS was a given and never a consideration! I'd take whatever BIOS the mobo's maker had wisely adopted... (hahaha...) Now, I will parse through as much information as I possible can on the UEFI installed on the mobo being considered. Thank you again!

呼.....非常感谢您对各种官方文档中 UEFI 技术细节的“通俗易懂”的解释。我现在终于明白为什么我一直无法从外置硬盘上的第二个 Debian 10 (BUSTER) 系统启动了。此外，这篇文章让我对如何选择下一块主板有了更全面的认识。在阅读这篇文章之前，我选择主板的方式非常简单：它是否支持我选择的 CPU？它是否集成显卡？它有哪些接口？有多少个内存插槽？等等。但 BIOS 对我来说是理所当然的，我从来不考虑！我会选择主板厂商采用的任何 BIOS.....（哈哈.....）现在，我会尽可能多地研究我正在考虑的主板上安装的 UEFI 信息。再次感谢！